GlobalGeoTree Dataset Analysis

Exploring source bias, temporal trends, and regional patterns in a global tree observation dataset

GlobalGeoTree team

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1 Introduction

> *Authors: Nataliia (source bias & Germany analysis), Anna (data structure), Camille (worldwide distribution), Santiago (US distribution)*

> *— Nataliia*

This report analyses the GlobalGeoTree dataset, a global collection of approximately 6 million georeferenced tree observations covering species, taxonomic groups, and geographical locations of trees collected by diverse groups across the world. The goal is to understand how the dataset was assembled, what biases it contains, and how tree recording patterns vary across time, geography, and data sources.

GlobalGeoTree aggregates tree observations from a large number of independent sources: citizen-science platforms, institutional herbarium and museum collections, systematic field inventories, compiled floristic atlases, and various research and conservation projects. The loaded dataset covers observations from 2015 to 2024.

Because GlobalGeoTree is assembled from many independent sources, it does not reflect a systematic global sampling design — it reflects where people happened to record trees. The result is a dataset with strong geographic, taxonomic, and temporal biases that are source-driven rather than biologically meaningful. Understanding these biases is a prerequisite for any downstream analysis. To make the source structure interpretable, each observation's free-text source name is classified into one of seven groups using keyword matching (see the Methodology section for details and limitations).

The report is organised in three parts. **Part I** examines global patterns: which sources dominate the dataset and how they shape what we see, how observations have changed over time, and how taxonomic composition and species richness vary across the world. **Part II** zooms into Germany as a whole and then Berlin in detail. Baden-Württemberg and NRW appear only through selected density and NUTS3 district maps that provide visual context for interpreting Berlin. **Part**

III examines the United States, the most-represented country in the dataset, analysing species composition and geographic patterns at the national level.

Because this is a data-visualisation project, several questions are intentionally shown through more than one visual encoding. These alternatives demonstrate different techniques and emphasise different aspects of the same data — for example, absolute counts versus distribution shape, or geographic coverage versus local concentration.

All charts in this report show **recording intensity** — how many observations were recorded and where — not a direct census of trees. High values can reflect actual tree presence, but they equally reflect where observers and platforms are most active.

> Limitations to keep in mind
>
> The dataset has no systematic global sampling design, so observation patterns must not be interpreted as direct measures of tree abundance. Records redistributed through aggregators such as GBIF may overlap with records attributed to individual platforms; source classification is based on keyword rules and remains uncertain for multilingual or opaque source names; and the `year` field describes when an observation was recorded, not necessarily when it was added to GlobalGeoTree. These limitations apply throughout the report.

Setup

Packages are loaded below; inline comments note their role in the report.

The following variables define the dataset year bounds and how many items appear in rankings and maps.

2 Methodology

This section summarises the analytical choices that the code implements. It provides context for interpreting the results without requiring the reader to examine every code block.

Data scope. The loaded GlobalGeoTree observation file covers observations dated from 2015 to 2024. Rows with missing country codes or species names are removed during loading. Coordinates are converted to numeric values and validated before the continent spatial join; records with missing or out-of-range coordinates, unmatched continents, or open-ocean assignments are excluded from the spatially analysed dataset. The remaining records form the working dataset for the global, Germany, and US analyses.

Source classification. Each observation carries a free-text source field with thousands of distinct values. A keyword-matching function (`classify_source()`) assigns each source name to one of seven groups: iNaturalist, Pl@ntNet, Other citizen-science apps, GBIF (aggregator), Institutional & survey data, Reference & thematic databases, and Other / unknown. The rules are applied in order, and the first match wins. This approach works well for English-language source names and major platforms but can misclassify source names in other languages or those using project-specific abbreviations. The "Other / unknown" group is a catch-all for everything that did not match any rule.

Geographic boundaries. For the Germany analysis, administrative boundaries at the NUTS1 level (Bundesländer) and NUTS3 level (districts) are loaded from GISCO, the geographic information system of the European Commission. Berlin district boundaries come from a separate open-data project. If a local copy of these files exists, it is used directly; otherwise the file is downloaded from the public URL on first render.

Visualisation choices. Maps with individual observation points use reproducible random subsamples for readability and to keep the PDF size manageable (see below). Aggregate charts, counts, richness tables, rarefaction analyses, and source-analysis summaries always use the full relevant data. English common names for species are fetched from the GBIF API and cached locally.

Subsampling strategy. The full dataset contains over 6 million observations. Rendering every row as an individual map point is prohibitively slow and produces visually indistinguishable results compared to a well-chosen subsample. All point maps in this report therefore use reproducible random subsampling (fixed seed 42) via a `sample_up_to()` helper, which returns all rows when the data is smaller than the requested cap. Worldwide non-source point maps draw up to 40,000 observations (per facet where applicable). Source-analysis maps draw up to 50,000 per source group. Density maps (kernel density estimation) also draw up to 50,000 observations, as KDE smoothing is stable well below this threshold. US maps follow the same approach with US-specific caps. Germany maps use smaller region-specific caps, though the German subset is small enough that most caps are not reached and all available observations are displayed. Crucially, all analytical summaries — counts, rankings, richness calculations, and rarefaction analyses — use the complete, unsampled data.

Reading density maps. Several maps in this report use smooth filled density contours (2D kernel density estimation) to highlight where observations are most concentrated. The legend shows contour levels from low to high: darker or more saturated colours indicate areas where more observations are clustered per unit area. These are *relative* density levels within each individual map — the same level number on two different maps does not imply the same absolute observation count. As with all maps in this report, density maps show *recording intensity*, not true tree abundance.

Part I: Global Patterns

3 Data Preparation

3.1 Data Loading and Cleaning

The raw CSV is read with `fread()` for speed. String fields are trimmed, numeric columns are type-cast, and rows with missing country codes or species names are removed. Each observation's free-text source name is classified into one of seven source groups (see Methodology). Continent labels are derived by spatially joining observation coordinates to a world countries shapefile, with manual corrections for Russia (split at 60°E) and Réunion (reassigned to Africa).

Measure	Value
Observation file	data/GlobalGeoTree.csv
Dataset period	2015–2024
Rows analysed	5,991,220
Distinct sources	2,462
Distinct source groups	6
Distinct countries	211
Distinct species	20,747
Germany rows	204,231

The table above confirms that the dataset loaded correctly and gives an overview of its scope. The analysis uses 5,991,220 observations in total.

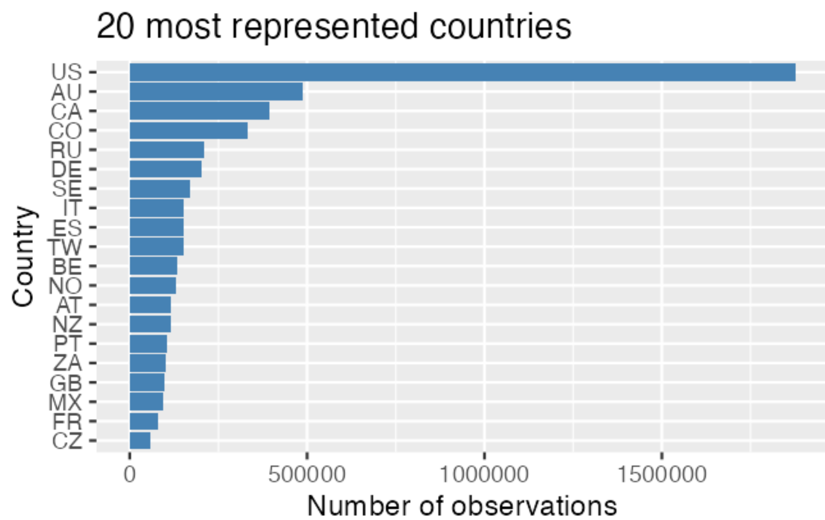
3.2 Dataset Overview

> — *Anna*

Column	Description
`sample_id`	Entry ID
`country_code`	Code of the country (e.g. US for the USA)
`level0`	Broad vegetation category grouping plants by shared ecological and physiological traits (e.g. Evergreen Broadleaf)
`level1_family`	Taxonomic Family — a group of related genera sharing common evolutionary characteristics (e.g. Asphodelaceae)
`level2_genus`	Taxonomic Genus — a group of closely related species (e.g. Aloe)
`level3_species`	Taxonomic Species — the most specific classification level (e.g. Aloe africana)
`location`	Country name
`source`	Source from which the occurrence record was obtained (e.g. iNaturalist Research-grade Observations)
`species_key`	Species identifier
`year`	Year of the observation
`longitude`	Geographic longitude
`latitude`	Geographic latitude

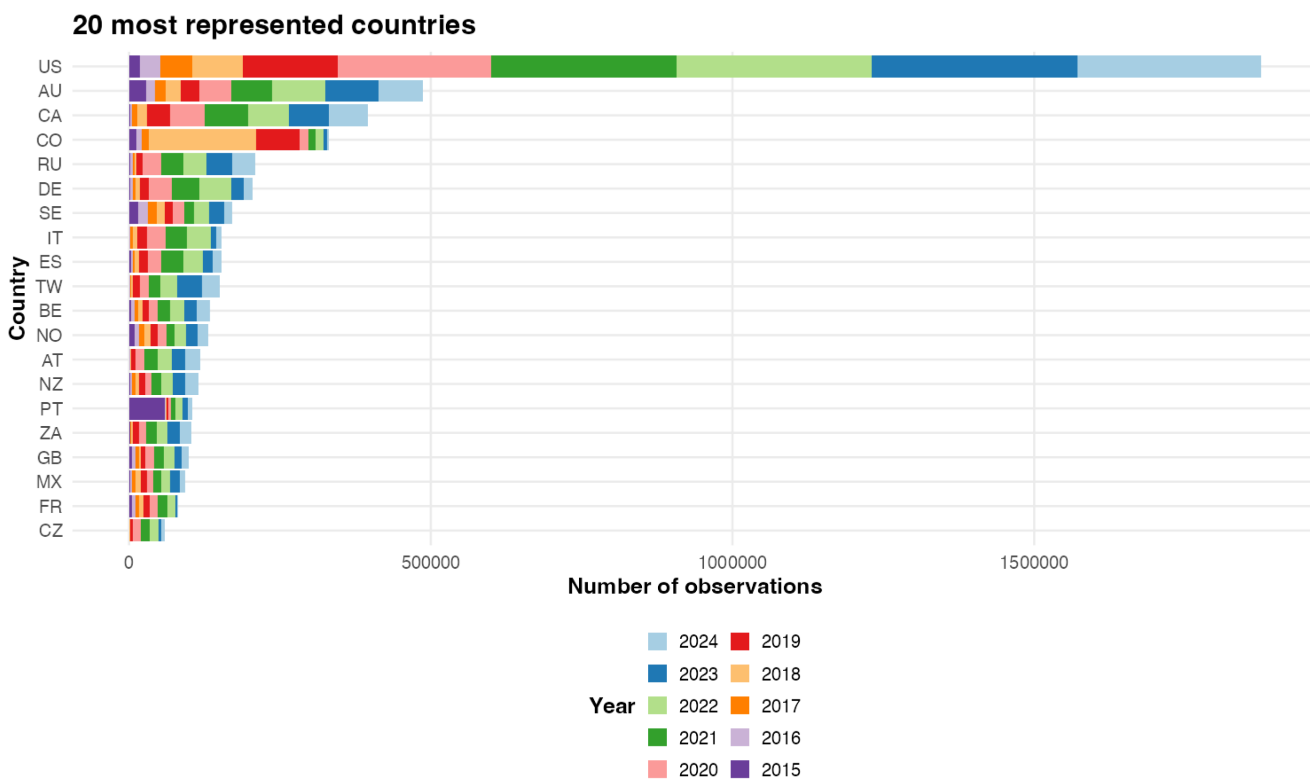
3.3 Distribution of the data across the world

Data distribution across the different countries:



We see that most of the data come from developed western countries, and especially the US that have a very important number of observations compared to the other countries.

Yearly distribution of the data in different countries: is there any evolution of the data collected throughout the years? Do some countries present large imbalance in the data distribution? How could this affect our analysis?



We see on this graph that the observations across the different countries were collected during a period of 10 years and that a slight increase in the number of yearly observations has been seen in the last years, from 2022 to 2024.

4 Source and Platform Bias

— *Nataliia*

4.1 Source Classification

Each observation's source field is matched against keyword rules to produce one of seven groups (ordered from most to least specific):

- **iNaturalist** — the largest citizen-science biodiversity platform worldwide.
- **Pl@ntNet** — a plant-identification app with a stronger presence in French-speaking countries and the Global South.
- **Other citizen-science apps** — regional portals (e.g., Artportalen, iRecord, Naturgucker, NatureMapr).
- **GBIF (aggregator)** — re-distributes records from thousands of organisations; may *overlap* with any other group.
- **Institutional & survey data** — herbarium/museum collections and systematic field campaigns.
- **Reference & thematic databases** — compiled floristic atlases, global plant databases (e.g., BIEN, RAINBIO, Tropicos), Wikidata, and invasive-species tracking.
- **Other / unknown** — everything that did not match any keyword rule.

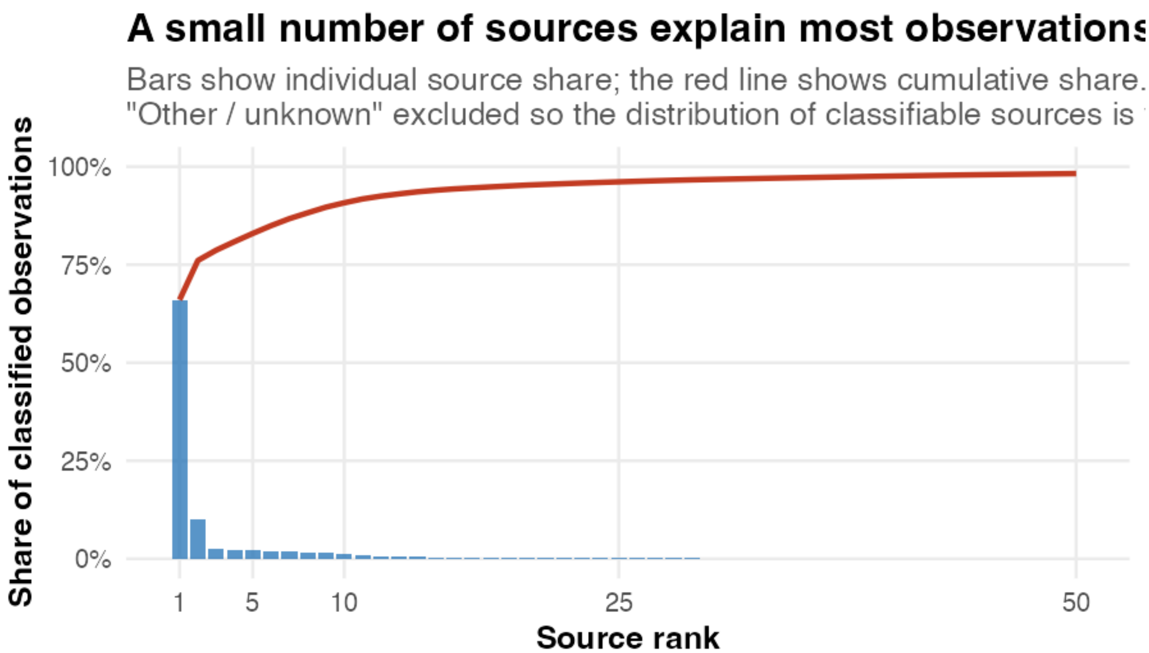
The keyword rules work well for English-language names and major platforms but can misclassify multilingual or opaque source names; the "Other / unknown" group likely includes recognisable sources whose names did not match any rule.

4.1.1 Classification coverage

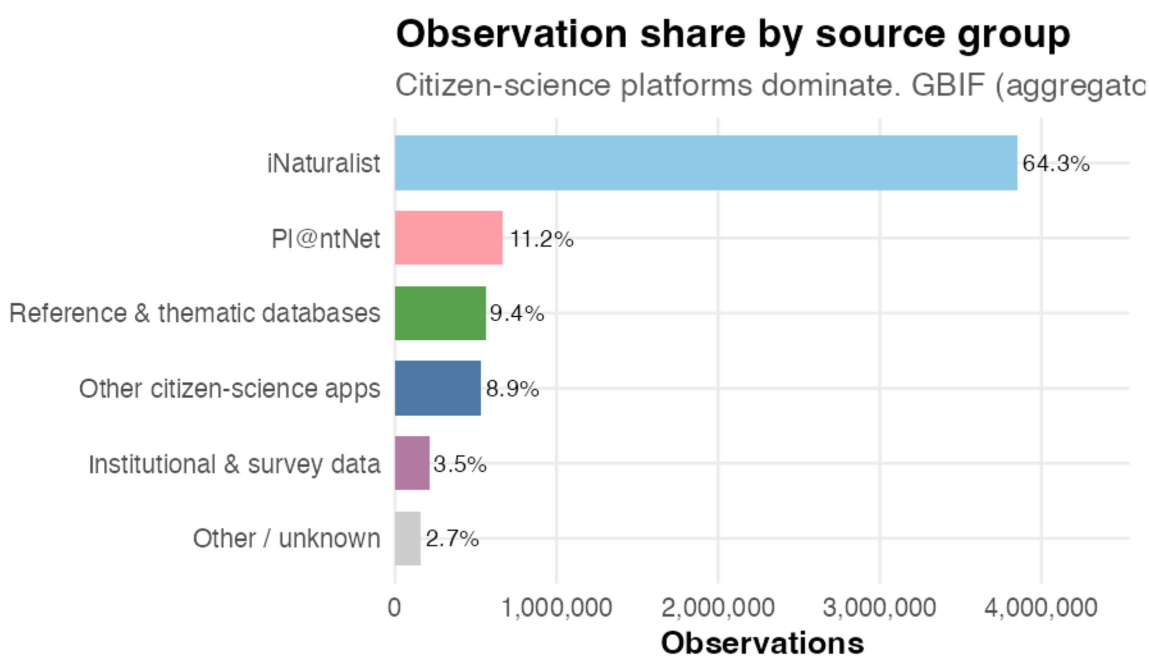
Of all 5,991,220 observations, **97.3%** are assigned to one of the six named groups, and **2.7%** fall into "Other / unknown". The higher the coverage rate, the more reliably the source-group comparisons throughout this report represent the full dataset.

4.2 Source Concentration

iNaturalist alone contributes **64.3%** of all observations; the three citizen-science groups together account for **84.4%**.



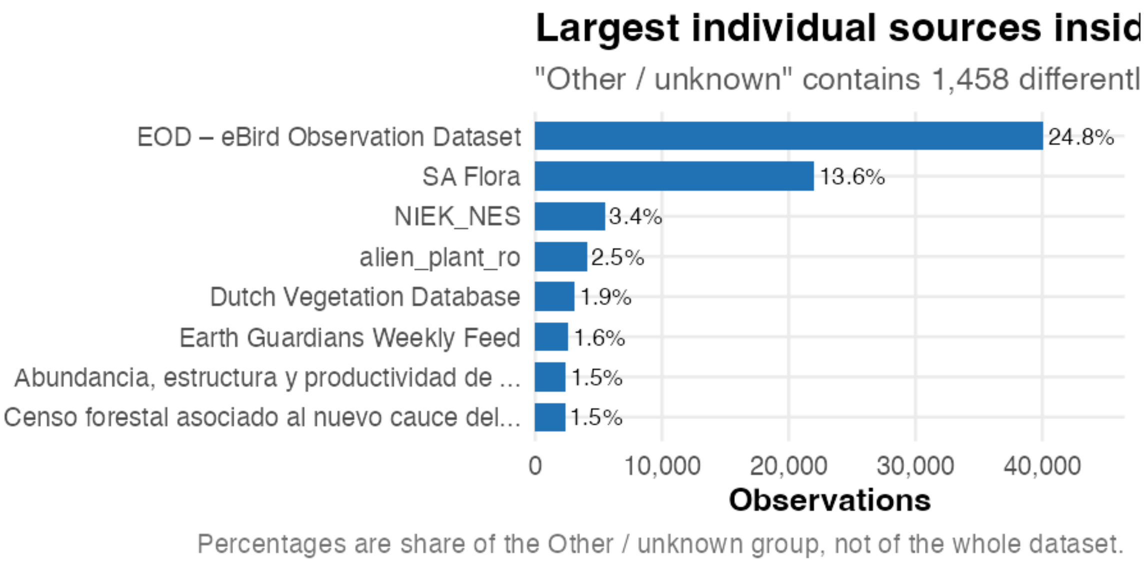
The Pareto chart shows extreme source concentration: a handful of individual sources account for the vast majority of classified observations, and the cumulative share curve flattens quickly. This means that any systematic bias in the top few sources will propagate through the entire dataset.



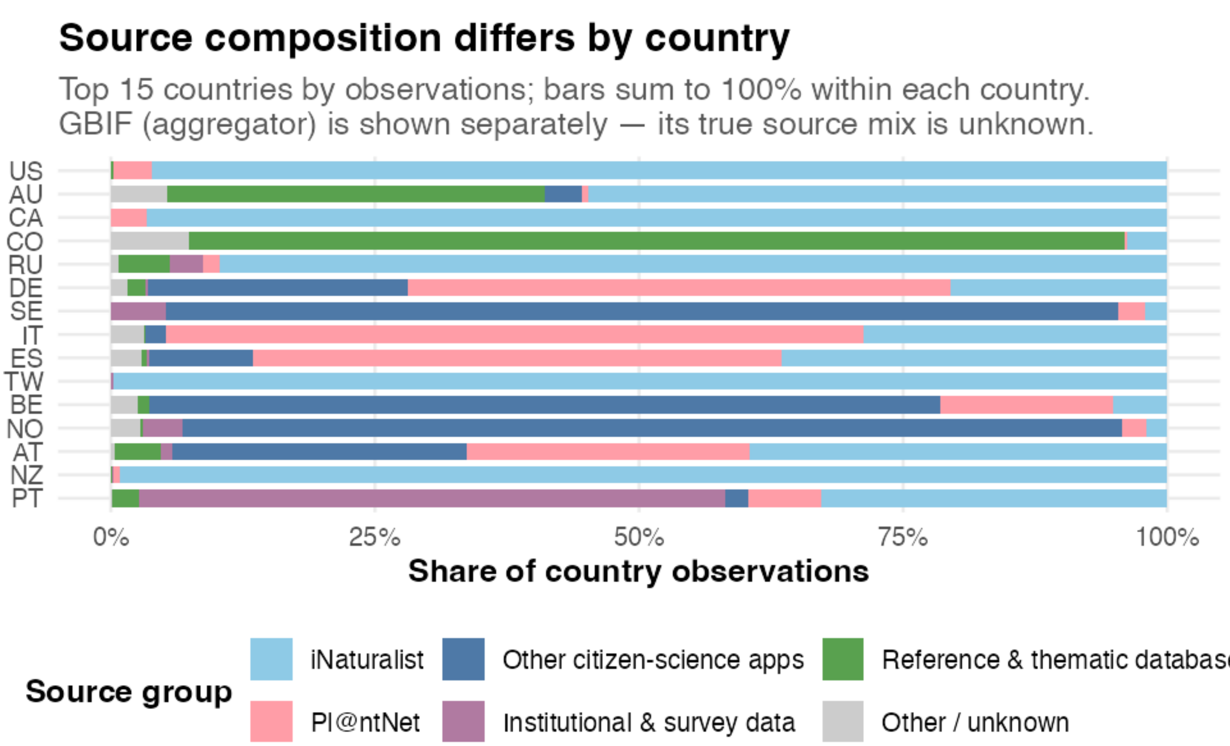
The bar chart confirms that citizen-science platforms collectively dominate the dataset, with iNaturalist as the single largest contributor. Institutional and reference sources, while individually smaller, cover regions and taxa that citizen-science platforms may miss.

4.3 What Is in "Other / Unknown"?

"Other / unknown" contains 1,458 individually named sources that did not match any keyword rule. The chart below shows the largest contributors within this heterogeneous catch-all group.

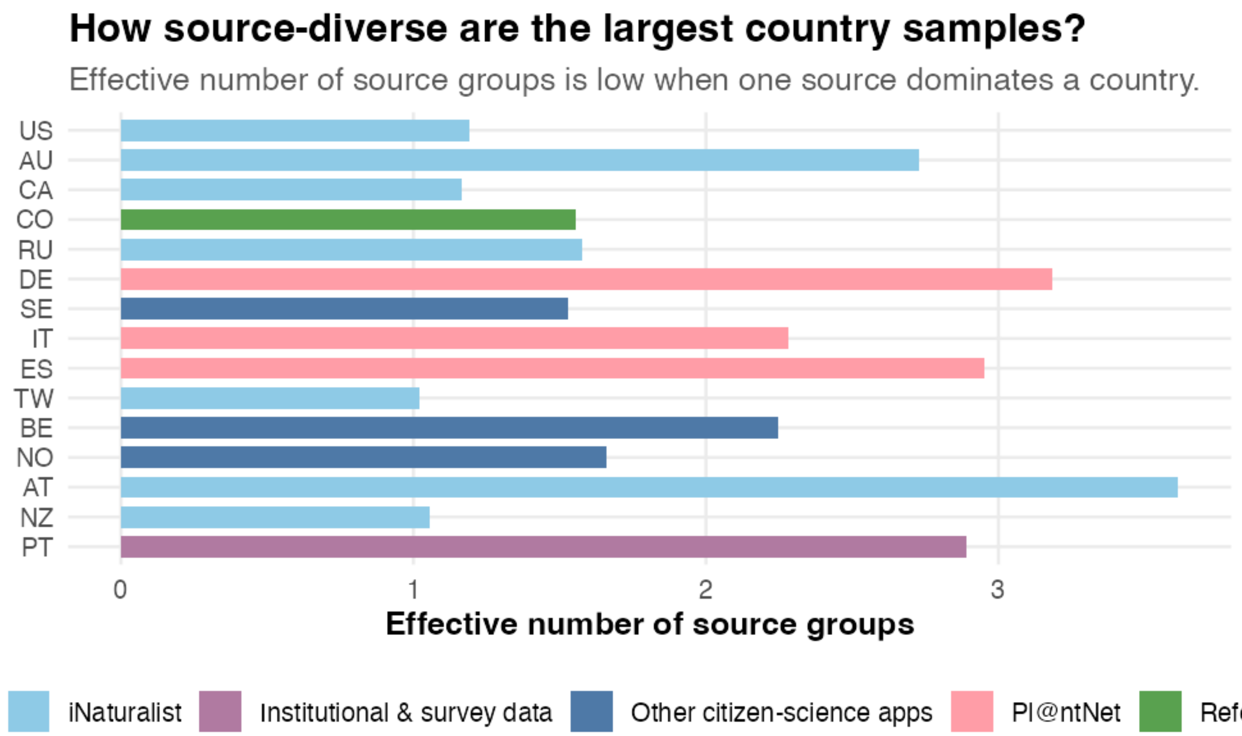


4.4 Countries and Regional Coverage



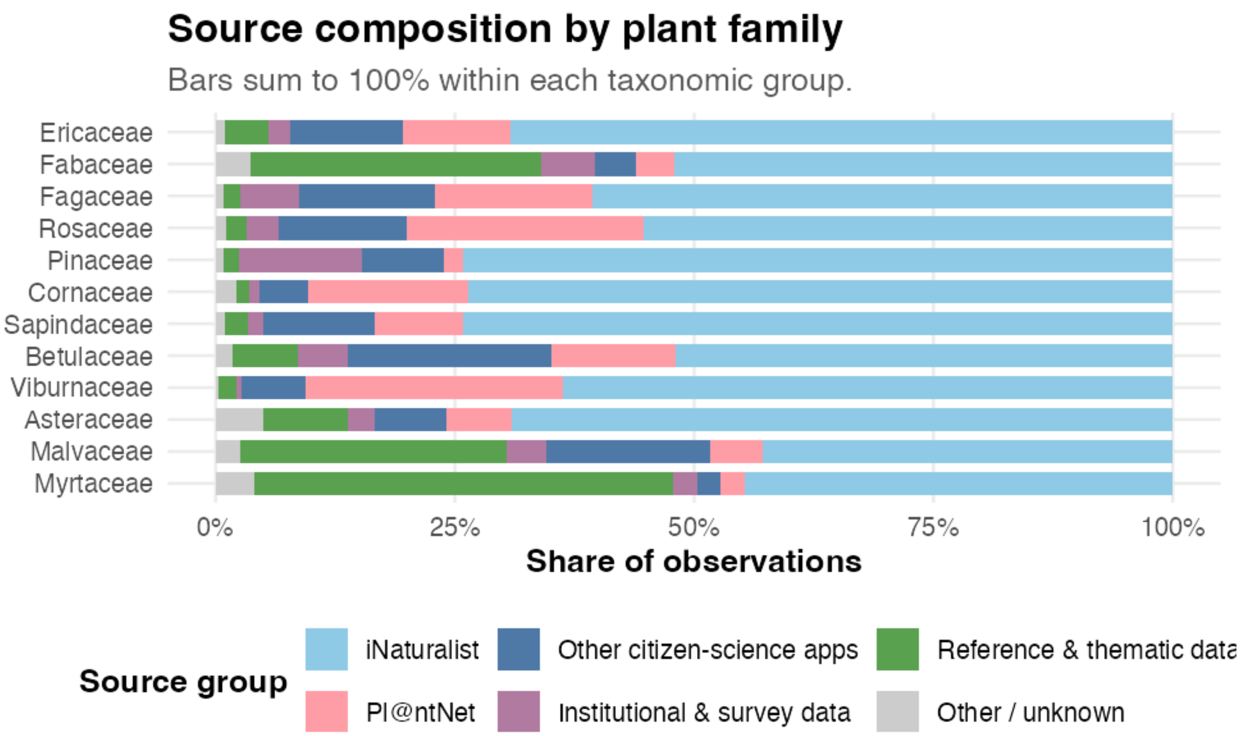
The stacked bar chart reveals that source composition varies sharply by country. Some countries are almost entirely supplied by a single platform (e.g. iNaturalist in the US), while others draw from a more balanced mix. Countries dominated by a single source inherit that platform's geographic and taxonomic biases.

Shannon entropy converted to "effective number of source groups" quantifies source diversity per country (1 = one source dominates).



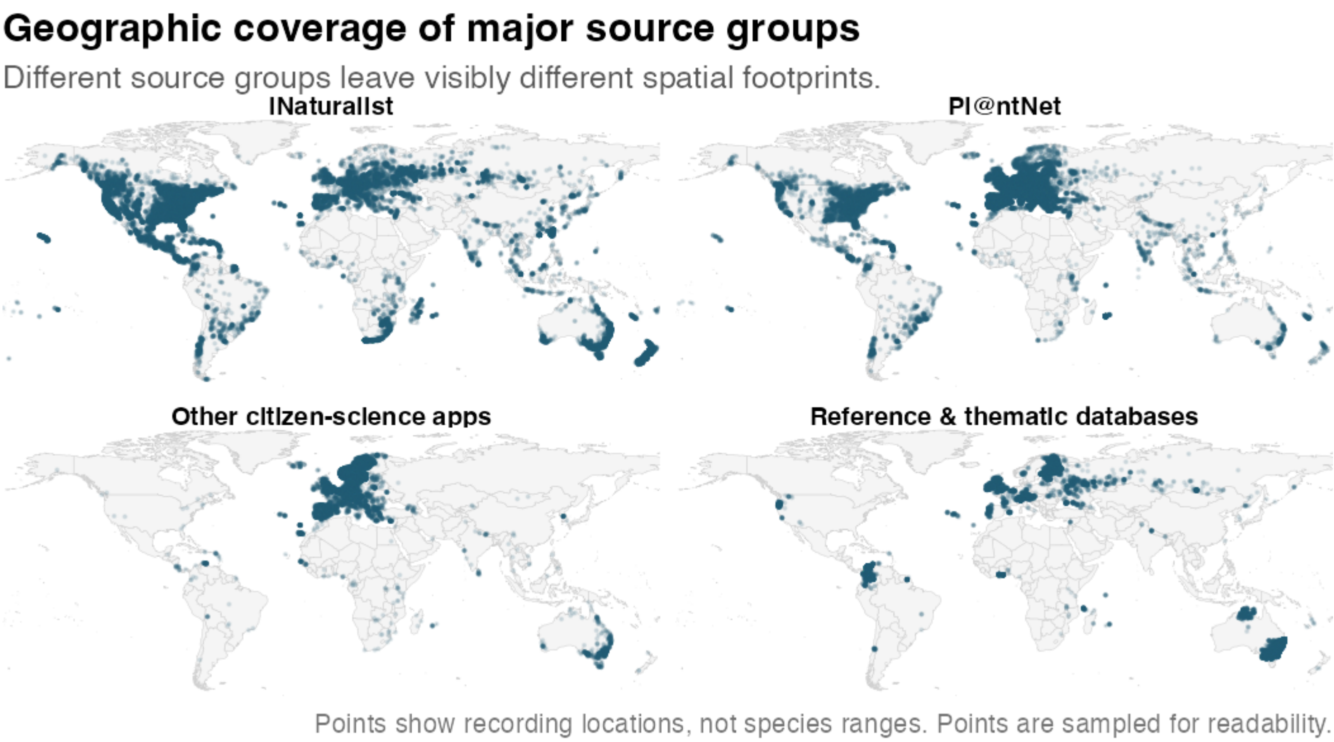
Countries with low effective source diversity (close to 1) are essentially single-source samples, meaning their apparent species composition and geographic coverage are shaped almost entirely by one platform's user base. Countries with higher diversity draw from multiple independent collection efforts, making their data more robust to any single platform's biases.

4.5 Sources and Taxonomic Groups



The source mix is not uniform across plant families. Some families are predominantly recorded through citizen-science apps, while others rely more heavily on institutional surveys or reference databases. Families that are primarily recorded by a single source type may have geographic or temporal gaps that mirror that source's coverage limitations.

4.6 Geographic Footprints of Data Sources



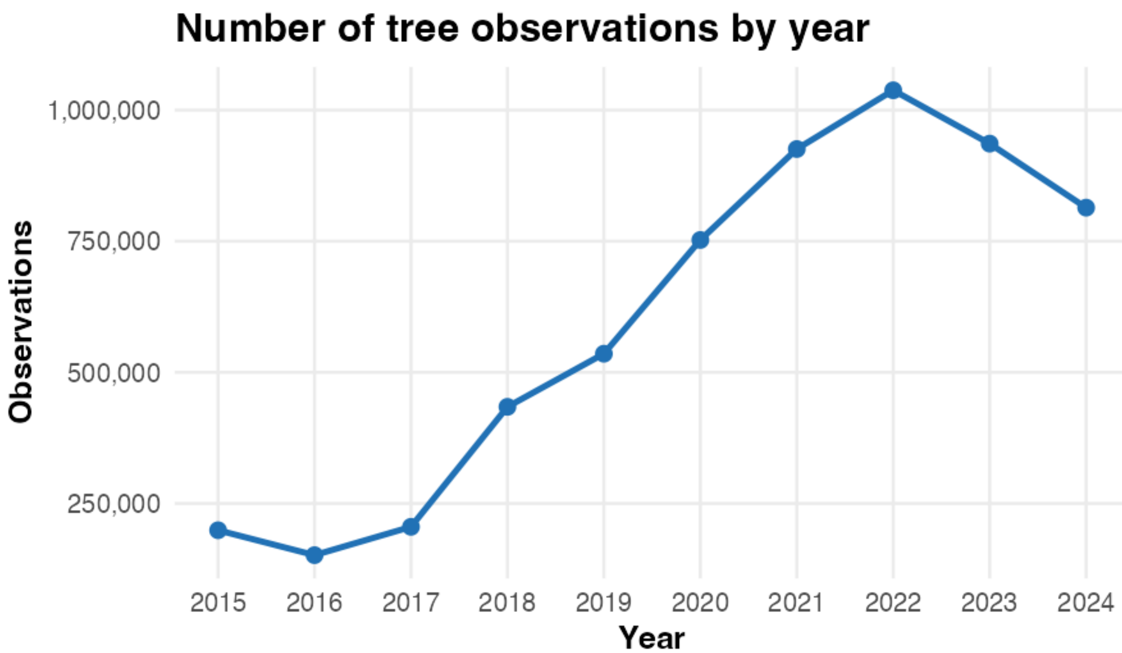
The faceted map reveals strikingly different spatial footprints across source groups. iNaturalist provides the most geographically widespread coverage but is concentrated in North America, Europe, and Oceania. Pl@ntNet complements it with stronger presence in French-speaking Africa and parts of South America. Institutional and survey sources tend to fill in localised regions that citizen-science platforms miss. These differences mean that where the dataset is geographically strong or weak depends directly on which sources are active in a given region.

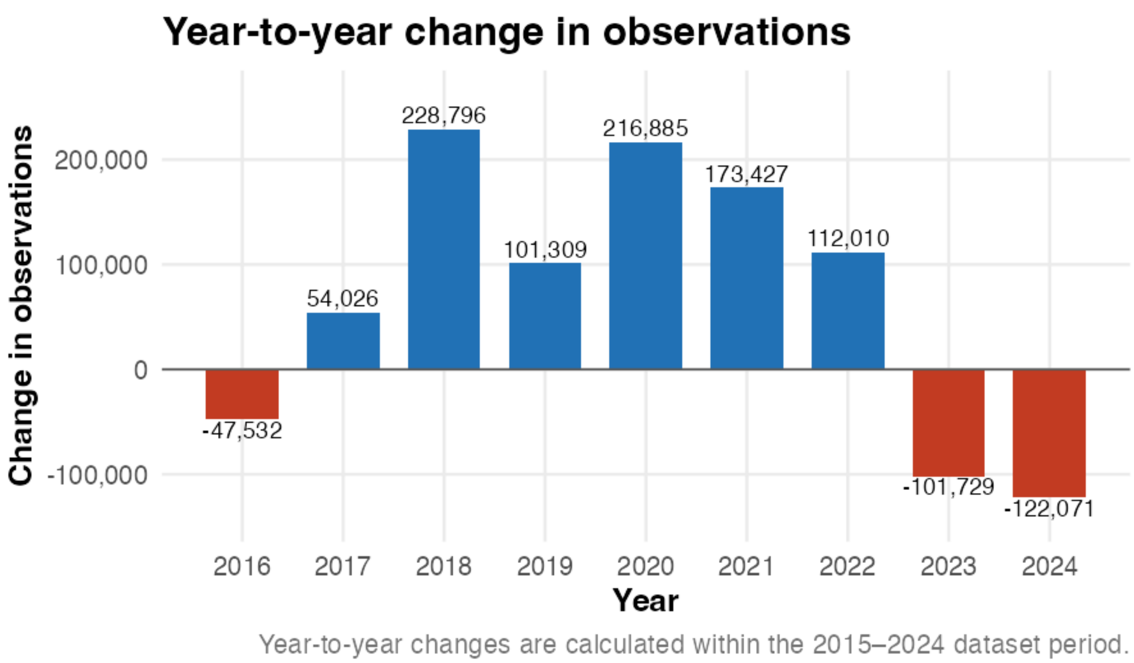
5 Temporal Development

> — *Nataliia*

5.1 Observation Volume Over Time

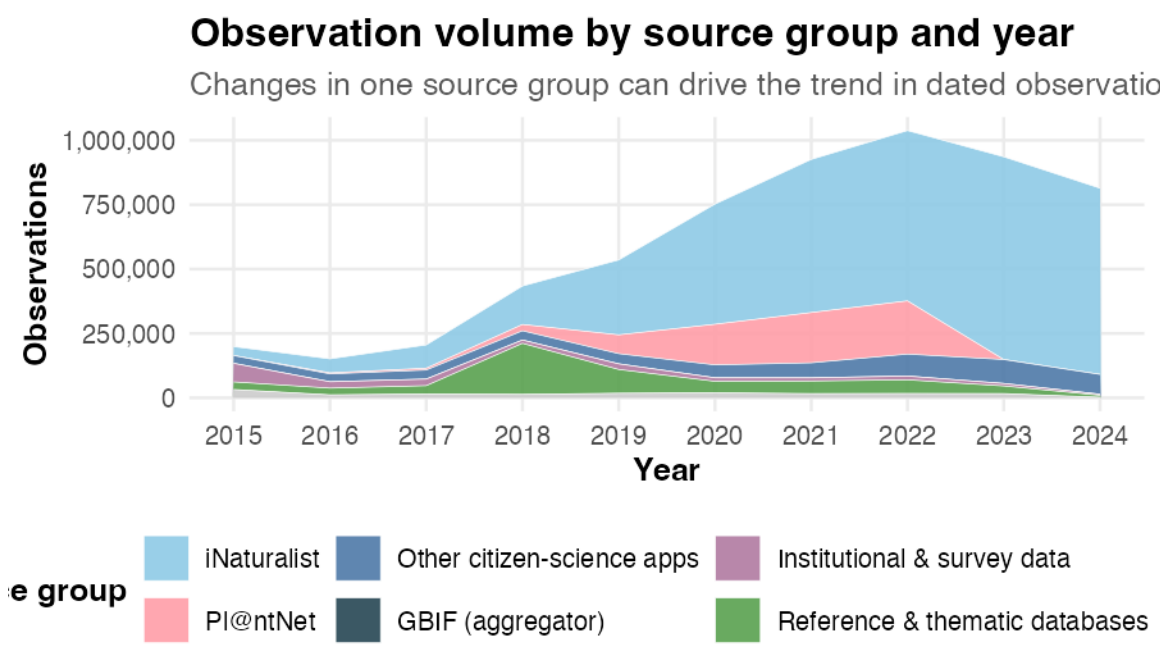
Between 2015 and 2024, observations changed from **198,874** to **813,995**, an overall change of **309.3%**.



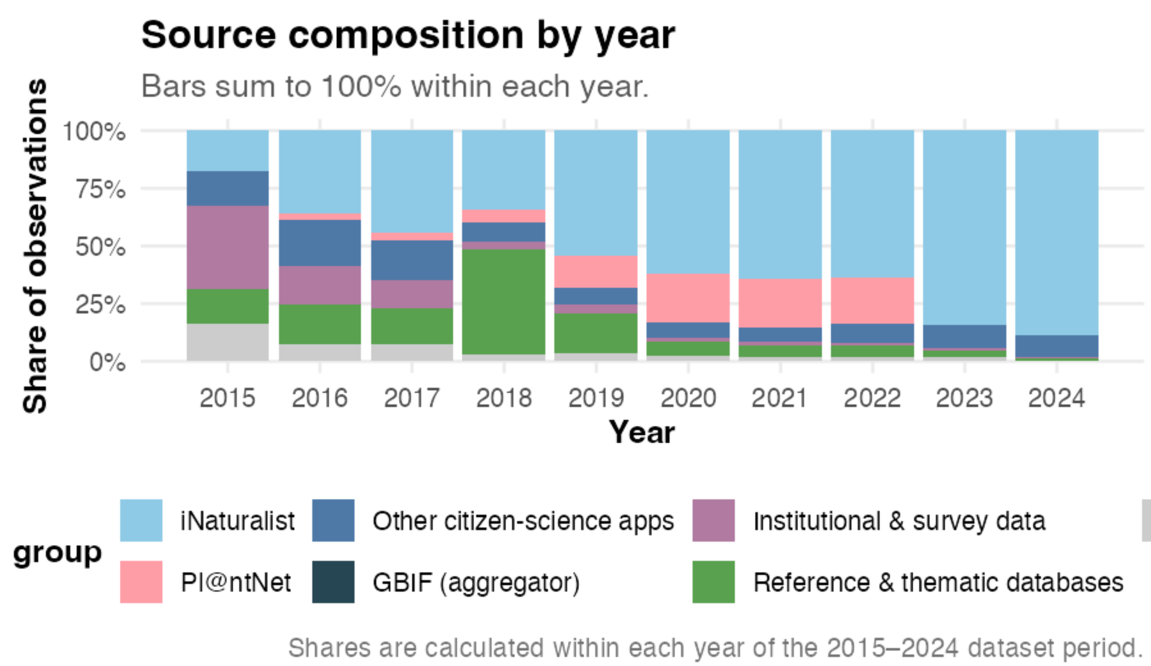


The strongest growth occurs in **2018** (+228,796 observations), and the strongest decline in **2024** (-122,071).

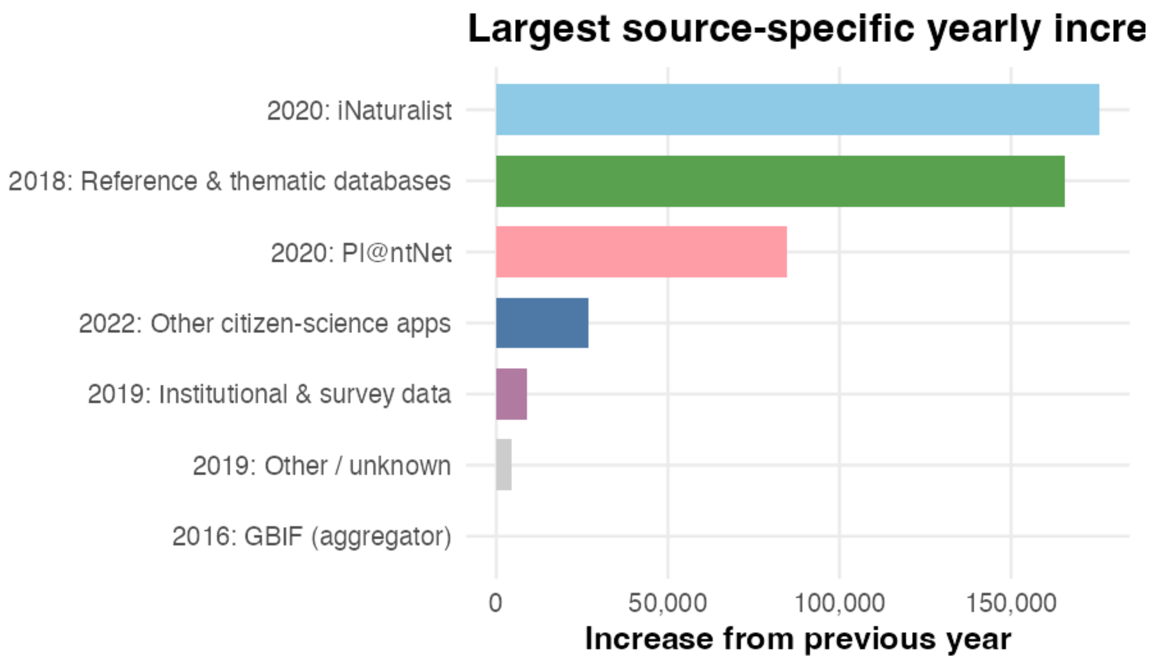
5.2 Which Sources Drive the Temporal Trend?



The stacked area chart shows that the increase in dated observations is largely associated with citizen-science platforms, particularly iNaturalist. When iNaturalist's contribution rises, the total observation count rises accordingly. This pattern is consistent with changing platform activity and coverage, but the observation year does not reveal when records entered GlobalGeoTree.



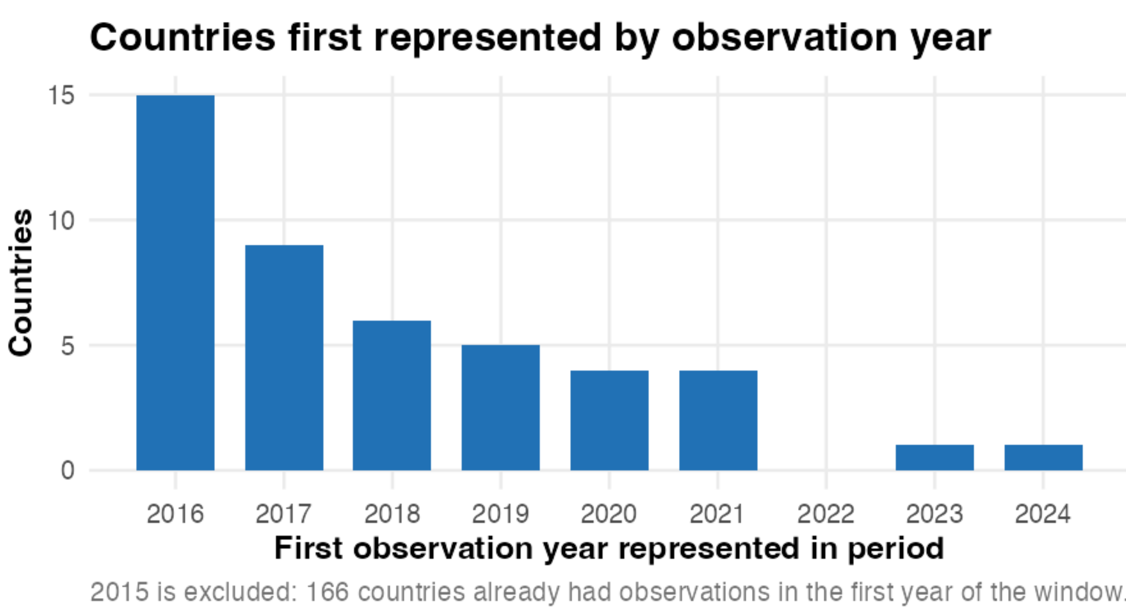
The year-by-year source share view shows whether the source mix is shifting over time. If one source group's share is growing while others shrink, the dataset's overall character is changing — even if total volume increases. A stable composition would suggest that all sources are growing at roughly the same rate.



The chart above highlights the single largest year-over-year jumps for each source group, making it easy to identify which platforms drove the biggest surges in data volume and when those surges occurred.

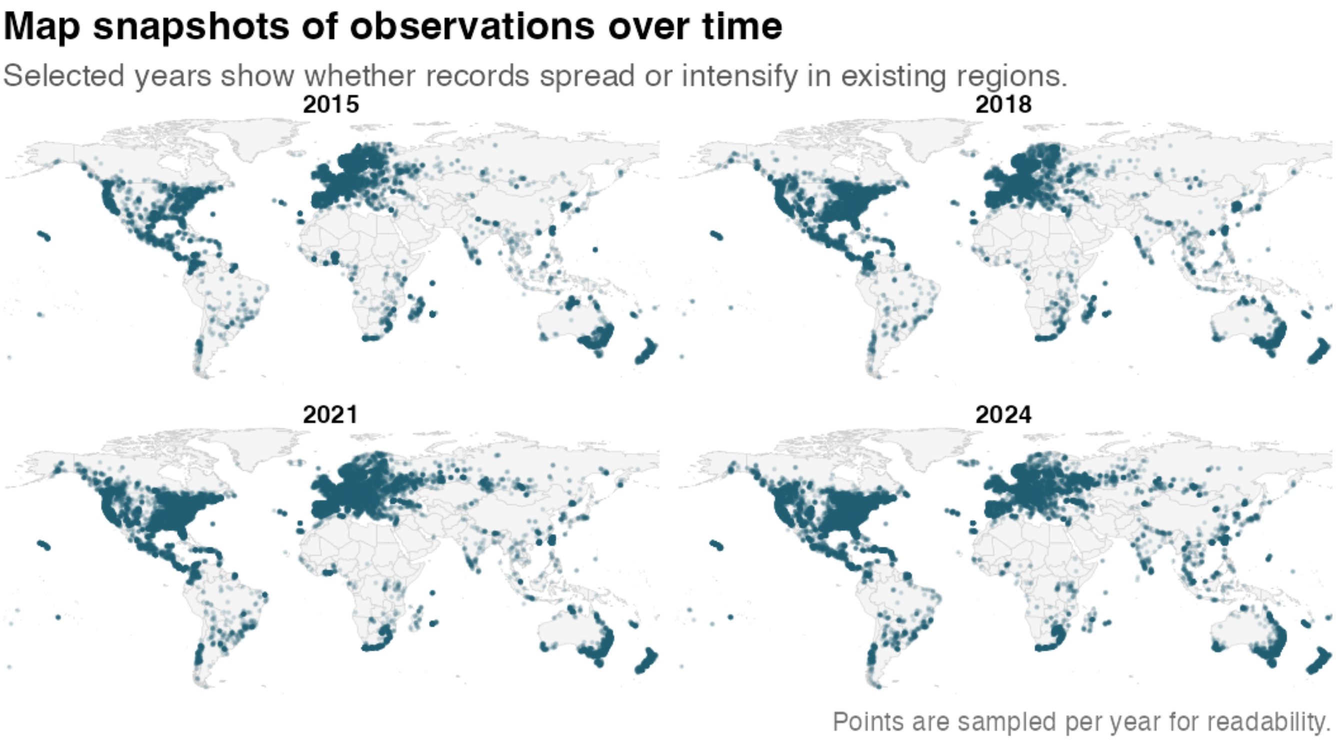
5.3 Geographic Coverage Over Time

This subsection examines how the geographic reach of observations dated from 2015 to 2024 differs over time. It compares the first observation year represented for each country within this period and maps observation locations at selected snapshot years. This describes the dates attached to the records, not when countries or records were added to GlobalGeoTree.



This chart shows when countries first appear among the observations dated within the selected period. A declining count suggests that later observation years add relatively few countries not already represented by earlier-dated records, although it does not reveal when those records were ingested into GlobalGeoTree.

The maps below show observation locations at four snapshot years, revealing whether records spread into new regions or simply become denser in existing ones.



Comparing the four snapshot years shows that the geographic footprint of the dated observations was already broadly established early in the period: North America, Europe, and parts of Oceania and South America are visible from the first panel. Later panels show a clear increase in point density within these same regions, but comparatively little expansion into previously unrepresented areas such as central Africa, interior Asia, or the Amazon basin. Within the observation-year window, the pattern is therefore more consistent with intensification than with geographic expansion.

6 Taxonomic Composition and Geographic Patterns

6.1 Species observed in this dataset

— *Anna*

What are the different tree families referenced in this dataset?

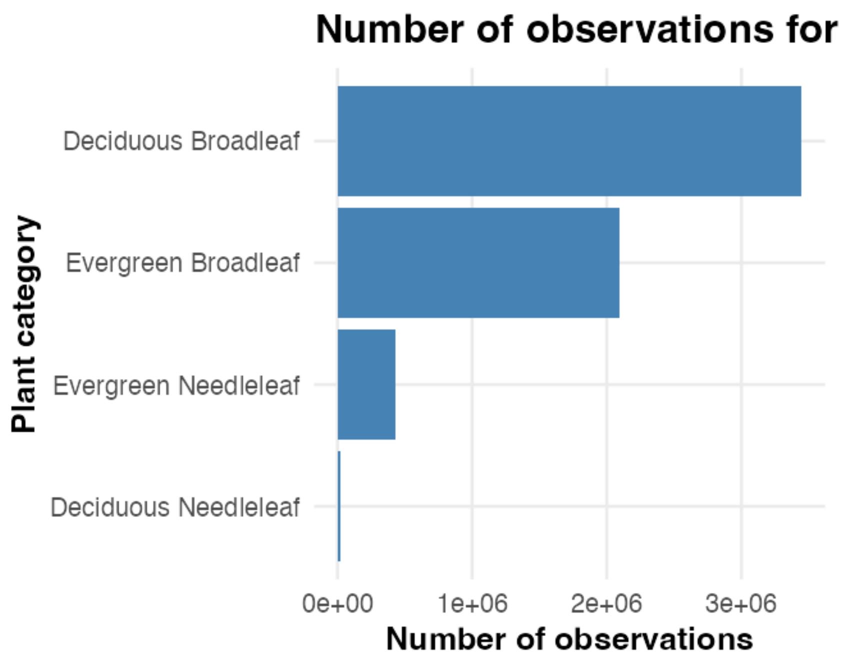


Figure 1: Observation counts for the four foliage categories in GlobalGeoTree.

Definition of each tree category:

- Evergreen Needleleaf: An evergreen needleleaf refers to woody vegetation that retains needle-like or scale-like foliage year-round
- Evergreen Broadleaf: An evergreen broadleaf is a plant or tree that has wide, flat leaves (rather than needles or scales) and retains its foliage year-round
- Deciduous Broadleaf: a diverse group of flowering plants (angiosperms) characterized by flat leaves that are shed annually before winter (Oak, Maple, Beech, Birch, and Poplar/Aspen)
- Deciduous Needleleaf : a unique group of conifers that, unlike their evergreen counterparts, shed all their needles annually in the autumn (rare trees)

Deciduous Broadleaf is the largest foliage category among observations in this dataset. Deciduous Needleleaf accounts for a much smaller share and appears in more geographically restricted areas. These values describe the composition of GlobalGeoTree observations rather than the worldwide abundance of each foliage type.

What are the different taxonomic families represented in this dataset and in which proportion are they represented?

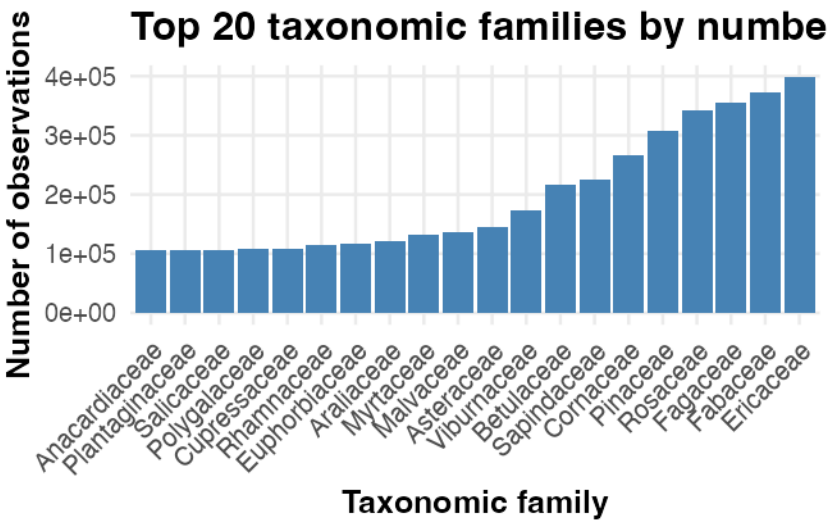


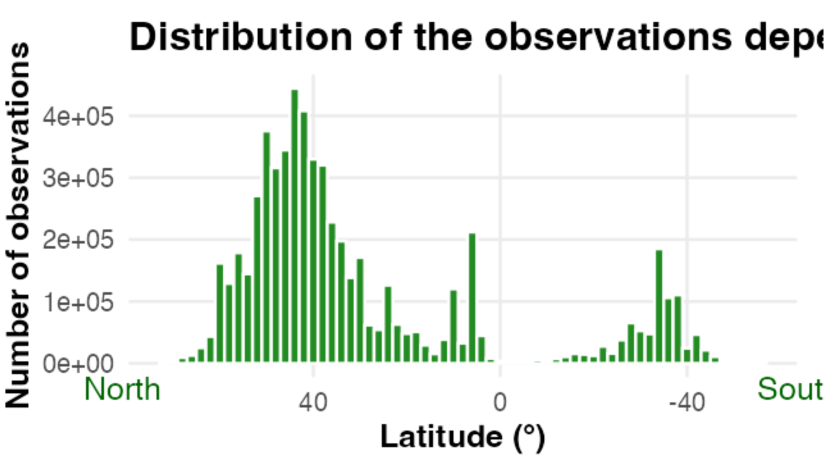
Figure 2: The 20 taxonomic families with the largest observation counts in the dataset.

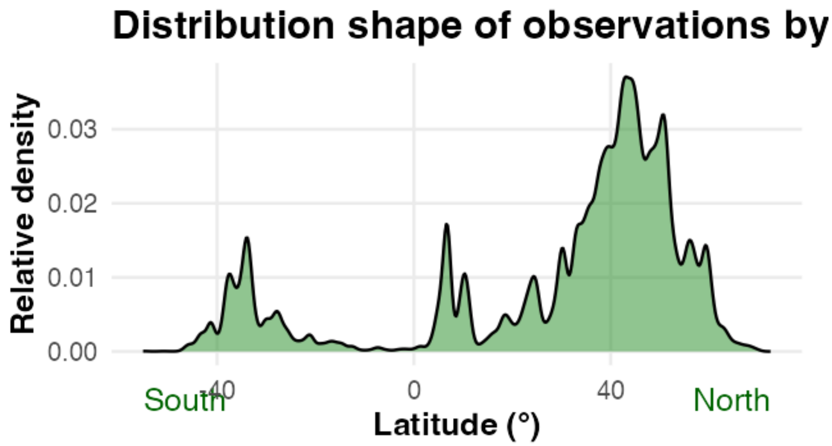
Again, here we see that the most represented taxonomic families in this dataset are also the most represented ones across the world, especially in North America, Europe and the north of Asia, where most of the data come from. However, as the source bias analysis above shows, this apparent representativeness should be interpreted with caution: the dataset reflects where observers and platforms are most active, not a systematic global survey.

6.2 Geographical distribution of the observations

— *Anna*

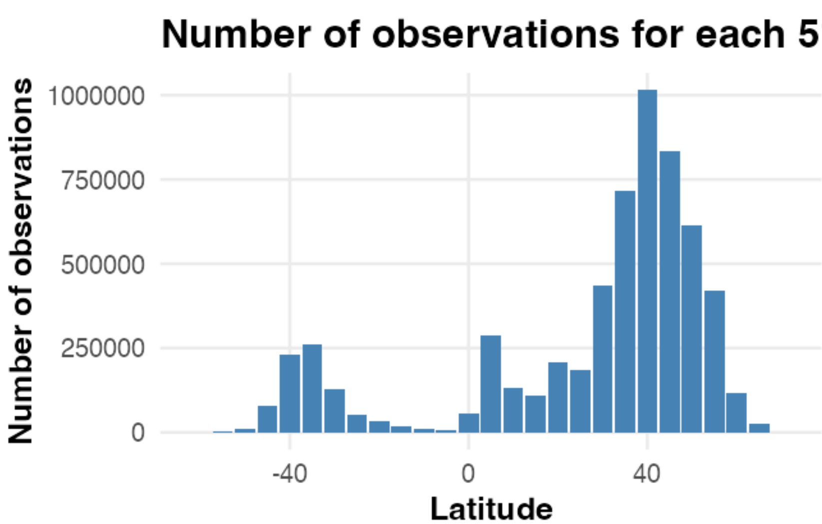
How are the trees distributed depending on the latitude?





The histogram preserves absolute observation counts in each latitude interval, while the density curve emphasises the overall distribution shape. Both show substantially more recorded observations in the Northern Hemisphere, which may reflect differences in land area and sampling effort as well as tree distribution.

Larger representations are also often used in ecological studies, because 5° latitudinal bands contain a satisfying amount of information without showing too much noise.



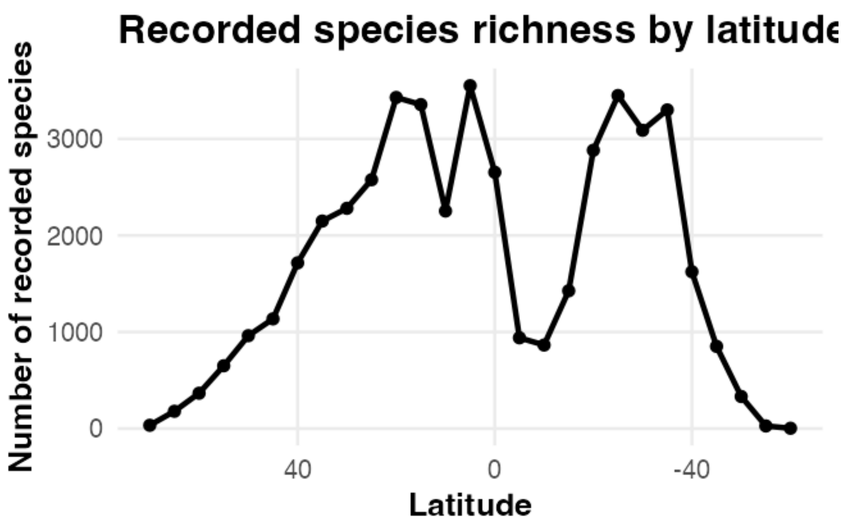
The same conclusions can be drawn as before.

We usually take 5° slices because it is small enough to find local tendencies but also big enough to avoid noise and too much non useful information and generally fine enough to reveal geographical patterns (climatic gradients, biodiversity, species distribution, etc.) without creating too much noise.

6.3 Geographical observations of species

> — *Anna*

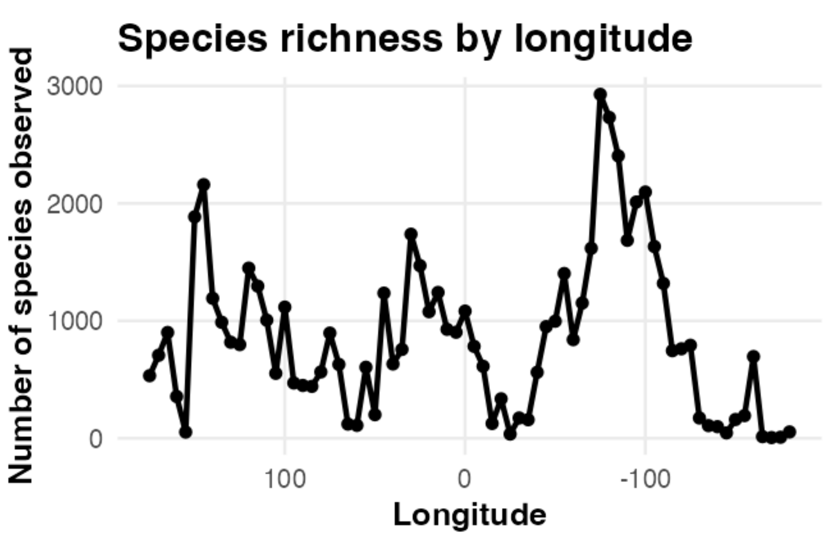
How does latitude influence the different species observed (latitude and species richness)?



Species richness is the number of distinct species recorded in a particular area. Here, recorded richness is higher across many Northern Hemisphere latitude bands and lower near the equator and poles; because observation effort is uneven, this is not a direct estimate of biological richness.

We see here that the maximum species richness are very close in the two hemisphere, but that the distribution is broader in the northern hemisphere than in the southern hemisphere (it is also because trees grow at higher latitudes in the northern hemisphere than in the southern hemisphere). Similar environmental conditions can be found in the two hemispheres but they are not spread following the same pattern.

How does longitude influence the different species observed (longitude and species richness)?



Areas with high recorded species richness correspond to continental areas, but little more can be concluded from longitude alone.

What is the local recorded species richness across the world?

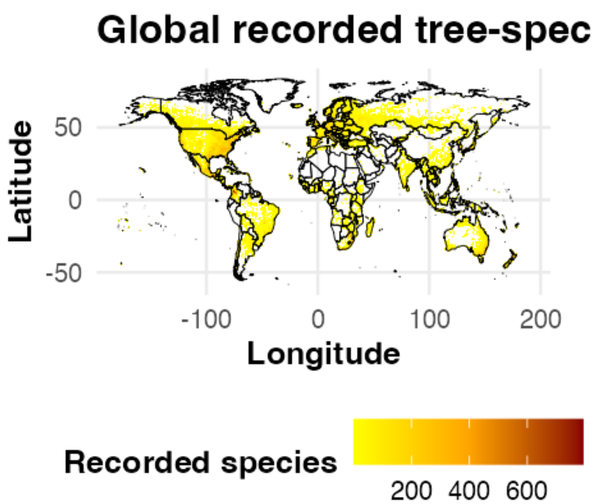
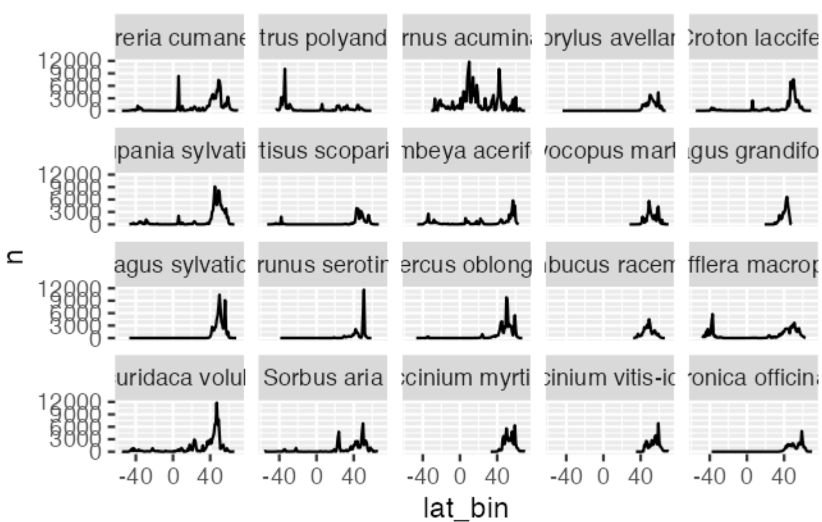


Figure 3: Recorded species richness per 1° grid cell. Counts use all observations; the map shows recorded richness rather than estimated biological richness.

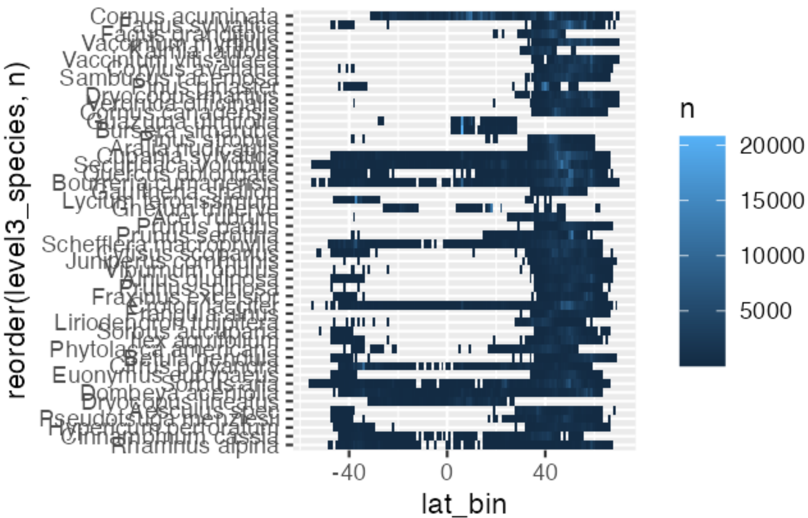
Within sampled areas, many grid cells show similar recorded richness, while the species involved vary geographically. Observation effort also varies strongly between cells, so the map should not be read as showing homogeneous underlying biodiversity.

How are the different tree species distributed depending on the latitude? Here, we only check for the most represented species.



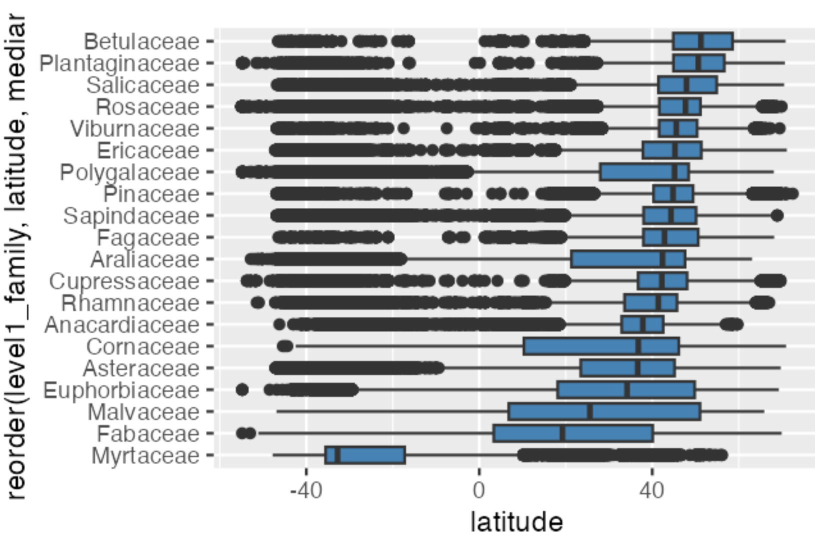
With this figure, we see that among the most represented species in this dataset, they present different world distributions. Most of them are mostly present in the northern hemisphere, but some are also present in the southern hemisphere (such as Dombeya acrifolia), and some are almost evenly present in the two hemispheres, such as Cornus acuminata.

Another representation could be a heatmap representing the species depenting on the latitude:

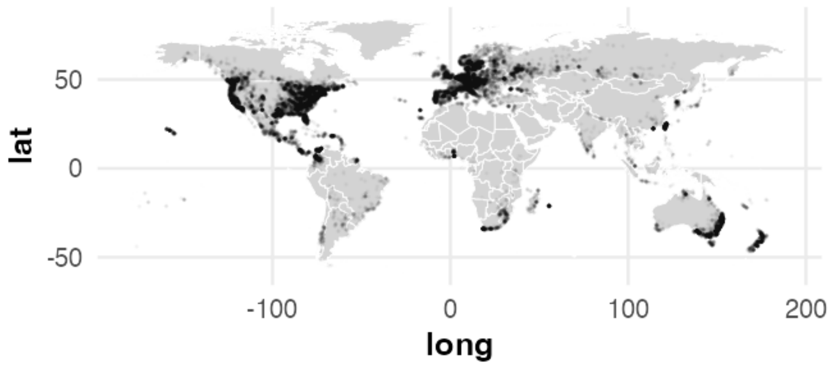


Again, we see here that most species are present mostly in the northern hemisphere, but some can also be present in both hemisphere or only in the southern hemisphere or near the equator (Guazuma umifolia, a tree largely present in Latin America and really rare in other areas).

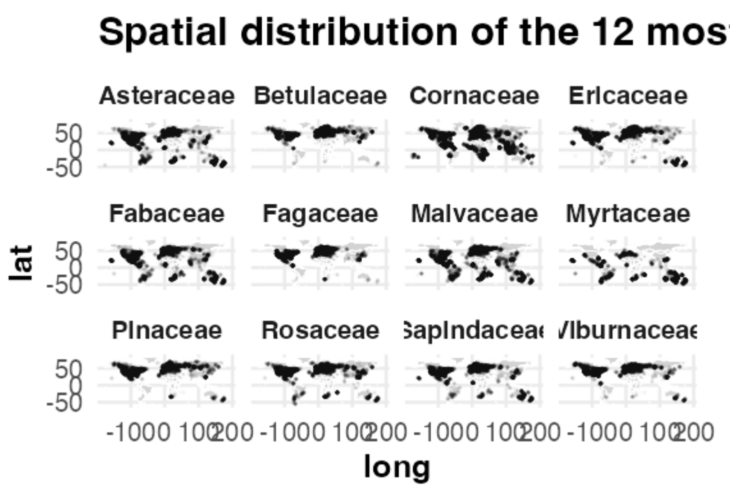
boxplot family x latitude



Same but on a map. Both maps below use the shared worldwide point-map subsample (see Methodology for subsampling details).



Displayed points: 40,000 sampled observations.



Displayed points: up to 40,000 per family.

On this visualization, we can see the distribution of the different taxonomical families across the world: - families are very widely spread across the world such as the Asteraceae: this family contains a very large amount of different species, that live in temperate areas. They can survive various non-extreme conditions. - families are observed in more restrictive areas, such as the families Ericaceae and Pinaceae: these families are very adapted to cold and unwelcoming places such as the tundra and boreal forests or places that are at a high altitude.

7 Worldwide Distribution of the Data

— *Camille*

How are observations distributed globally, and what does this reveal about sampling bias and taxonomic diversity? This section addresses four guiding questions:

1. Where are observations geographically concentrated, and does the coverage reflect ecological reality or observer activity?

sample_id	country_code	level0	level1_family	level2_genus	level3_species	source	year	continent
1	MW	Evergreen Broadleaf	Acanthaceae	Anisotes	Anisotes formosissimus	iNaturalist Research-grade Observations	2024	Africa
2	MZ	Evergreen Broadleaf	Acanthaceae	Anisotes	Anisotes formosissimus	iNaturalist Research-grade Observations	2018	Africa
3	MZ	Evergreen Broadleaf	Acanthaceae	Anisotes	Anisotes sessiliflorus	iNaturalist Research-grade Observations	2022	Africa
4	MZ	Evergreen Broadleaf	Acanthaceae	Anisotes	Anisotes sessiliflorus	iNaturalist Research-grade Observations	2015	Africa
6	KE	Evergreen Broadleaf	Acanthaceae	Avicennia	Avicennia spec	naturgucker	2023	Africa
7	NG	Evergreen Broadleaf	Acanthaceae	Avicennia	Avicennia spec	Biodiversity of Some Coastal Communities in Akwa-Ibom State, Nigeria	2021	Africa

Table 1: An overview of the first few rows of the dataset

2. How does the distribution of foliage types vary across latitudes and continents?
3. How does taxonomic richness compare across continents once differences in sampling effort are accounted for?
4. Has the spatial coverage of the dataset expanded or merely intensified over the years?

In this section, the aim is to understand how observations are distributed globally, both spatially and temporally. This distribution matters because it directly shapes how the rest of the dataset should be interpreted: an over-representation of certain regions may reflect sampling bias (the activity of contributors in the field) rather than the true state of global tree biodiversity. It is therefore essential to characterize where and when the data was collected, and to identify areas of high and low sampling density. We start by examining the overall geographical spread of observations across continents, before looking more closely at how taxonomic diversity (families, genus, species) is distributed in space. Finally, we analyze how sampling effort has evolved over time, both globally and by continent, to assess whether spatial biases have shifted across the years.

7.1 Pre-Processing

For this analysis, observations are grouped by continent rather than by country. With data spanning more than 210 countries, a country-level analysis would fragment the dataset into categories too small and numerous to reveal meaningful patterns, while a continent-level grouping provides a small number of interpretable regions, each large enough to support robust comparisons of taxonomic diversity and temporal trends. The continent for each observation was derived by spatially joining the point coordinates (longitude, latitude) with a world countries shapefile (rnaturalearth). This approach, however, relies on administrative continent boundaries, which do not always align with the geographical logic relevant to this study. Two adjustments were therefore necessary:

- Russia is administratively classified as a single entity, but spans two geographical continents. Observations were reassigned to Europe or Asia based on the conventional Ural Mountains boundary (60°E longitude).
- Réunion Island, despite being a French overseas territory, is geographically located in the Indian Ocean off the African coast, and was therefore reassigned to Africa rather than being grouped with other European territories.

Observations for which no continent could be determined, as well as those located over open ocean (e.g., isolated island territories not covered by the country shapefile), were excluded from the dataset.

The resulting dataset, with the added continent column, is previewed in Table 1.

To ensure visual consistency throughout this section, each continent is assigned a fixed color, shown in Figure 4. This color scheme is derived from the Okabe-Ito palette, chosen for its colorblind-friendly properties, and is applied consistently across all subsequent maps and charts to facilitate comparison between figures.

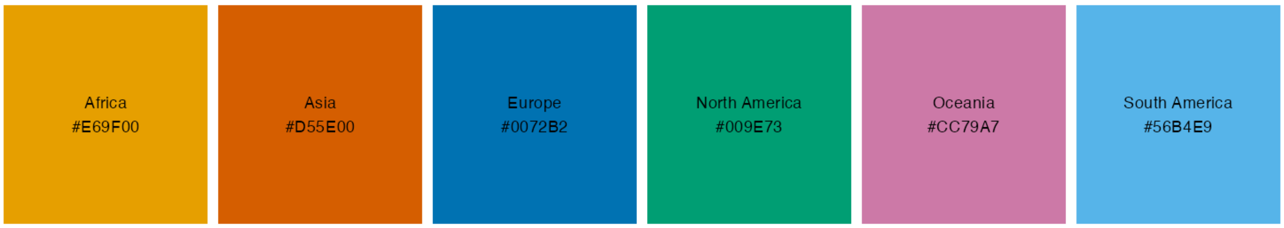


Figure 4: Colour scheme used to represent the continents in the following figures

7.2 Global overview

This first subsection provides a global overview of where observations are located. The goal is twofold: first, to visualize the overall spatial coverage of the dataset and assess which regions of the world are represented; second, to quantify this coverage at the continent level in order to detect potential imbalances in sampling effort. Figure 5 projects a reproducible subsample of observations on a world map, colored by continent, offering a first visual impression of the dataset's geographical spread without plotting millions of individual points. Figure 6 then complements this map with a quantitative summary, showing the total number of observations recorded per continent. Finally, Figure 7 goes one step further by highlighting areas of high and low observation density within continents, revealing local concentrations that a simple point map or continent-level count cannot capture on their own. For instance, distinguishing a continent with many observations spread evenly across its territory from one where the same volume of observations is concentrated in a few localized hotspots.

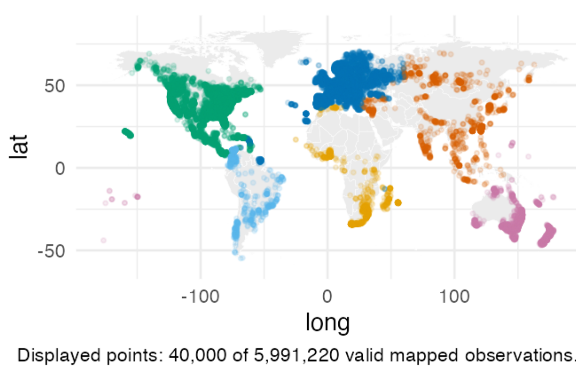


Figure 5: Geographical projection of observations by continent

Figure 5 shows a strongly uneven geographical coverage of observations. North America (green) and Europe (blue) display a dense, near-continuous coverage, closely following major population centers and road networks (a pattern typical of citizen science platforms like iNaturalist, where sampling effort is driven by observer density rather than tree distribution alone). Asia (orange) covers a considerable geographical extent, but coverage is markedly uneven: it is dense in East Asia, India, and along the Trans-Siberian corridor, yet visibly sparse across Central Asia and Siberia's interior. Africa (yellow) shows a similar contrast, with observations concentrated in South Africa, East Africa, and coastal West Africa, while the Congo Basin and the Sahara remain largely unsampled. South America (light blue) is dominated by coastal Brazil and the Andean corridor, with the Amazon interior almost entirely absent from the dataset despite hosting one of the highest levels of tree biodiversity on Earth. Oceania (pink) is essentially limited to Australia's populated coastline and a scattering of Pacific islands. Overall, this map already suggests that observation density mirrors human accessibility and citizen-science activity more closely than it reflects actual tree biodiversity (a hypothesis further examined in the following density map Figure 7).

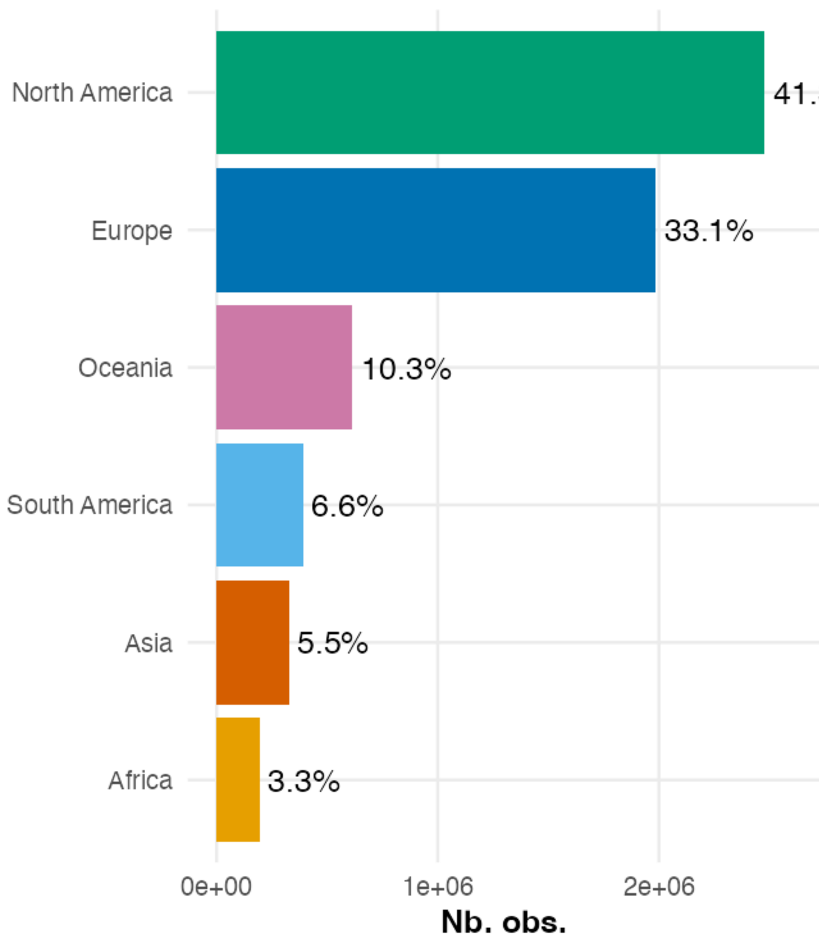


Figure 6: Total number of observations by continent

Figure 6 quantifies the imbalance already suggested by Figure 5. North America and Europe together account for 74.4% of all observations (41.3% and 33.1% respectively), despite representing only a fraction of the world's land area and tree biodiversity. Oceania follows at 10.3%, largely driven by Australia, before a sharp drop to South America (6.6%), Asia (5.5%) and Africa (3.3%). This ranking reveals an important nuance compared to the previous map. Although Asia appeared visually extensive in Figure 5 (spanning a wide longitudinal range with a dense cluster of points), it accounts for only 5.5% of total observations, far below Europe or North America. This apparent contradiction is simply a matter of scale: individual points can appear numerous and spread out on a map while still representing a small share of a dataset containing several million observations overall. This illustrates why the two figures are complementary. The map shows where observations are located, while the bar chart shows how much each region actually contributes to the dataset.

The density map below is computed on a reproducible random subsample (see Methodology for the subsampling rationale).

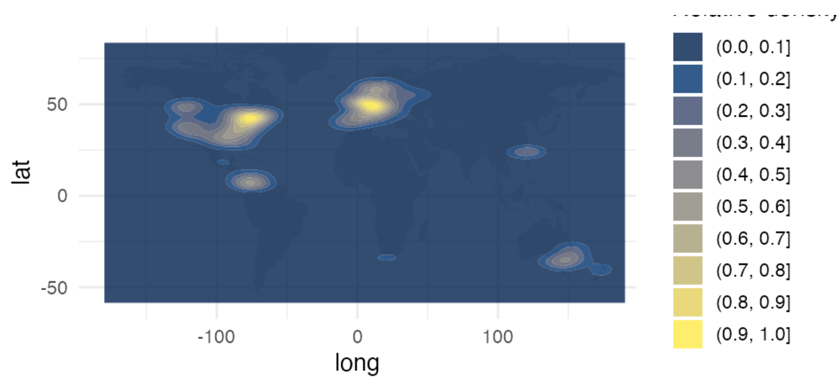


Figure 7: Geographical density of observations

Note: the density estimate is based on a reproducible random subsample of 50,000 observations (drawn with the report-level map seed from the full dataset), rather than all 6.26 million records. Passing the full dataset to `geom_density_2d_filled()` *caused a crash in ggplot2 3.5.2 due to memory limitations in the internal stat pipeline. At this sample size, kernel density estimation produces a visually identical result, as the smoothing operation is stable well below this threshold.*

Figure 7 confirms and refines the pattern already observed in the previous two figures by revealing that observations are not only concentrated on certain continents, but highly clustered within a small number of localized hotspots. The most intense concentration is found in Western Europe (particularly around France, Germany and the United Kingdom), followed closely by the northeastern United States. Secondary hotspots appear along the U.S. West Coast, in Central America, in southeastern Australia, and around Taiwan / southern Japan, while the vast majority of the remaining land surface, including most of Africa, Asia and South America, shows minimal to no density. This concentration pattern reinforces the hypothesis raised earlier: sampling effort appears strongly correlated with population density and citizen-science activity rather than with the underlying distribution of tree biodiversity. Notably, even within high-observation continents such as North America (Figure 6), density is far from uniform. Most observations are concentrated along the East and West coasts, while large inland areas remain comparatively under-sampled.

Together, these three figures paint a consistent picture. The dataset's geographical coverage reflects observer accessibility and activity far more than actual global tree distribution, a limitation that should be kept in mind, particularly when interpreting taxonomic diversity by continent in the following sections.

7.3 Spatial distribution of foliage types

Having established the overall geographical coverage of the dataset, this subsection shifts focus from where observations occur to what is being observed where. Each tree in the dataset is classified into one of four foliage types (Evergreen Broadleaf, Evergreen Needleleaf, Deciduous Broadleaf, or Deciduous Needleleaf). This classification closely tied to climate, since foliage strategy reflects how trees adapt to seasonal variation in temperature and water availability. The goal is therefore to examine whether these foliage types are distributed uniformly across the sampled regions, or whether they follow the climatic gradients expected from ecological theory, and to what extent this pattern can still be observed despite the sampling bias identified previously.

Three complementary perspectives are used to explore this question. Figure 8 first projects a reproducible subsample of observations for each foliage type, allowing their geographical range to be visualized directly on a world map. Figure 9 then aggregates this information into a quantitative comparison, showing how foliage composition varies from one continent to another. Finally, Figure 10 moves away from continental boundaries altogether, examining how the density of each foliage type is distributed along a continuous latitudinal gradient. This perspective is particularly relevant here, since climate (and therefore foliage strategy) is primarily structured by latitude rather than by administrative continental divisions.

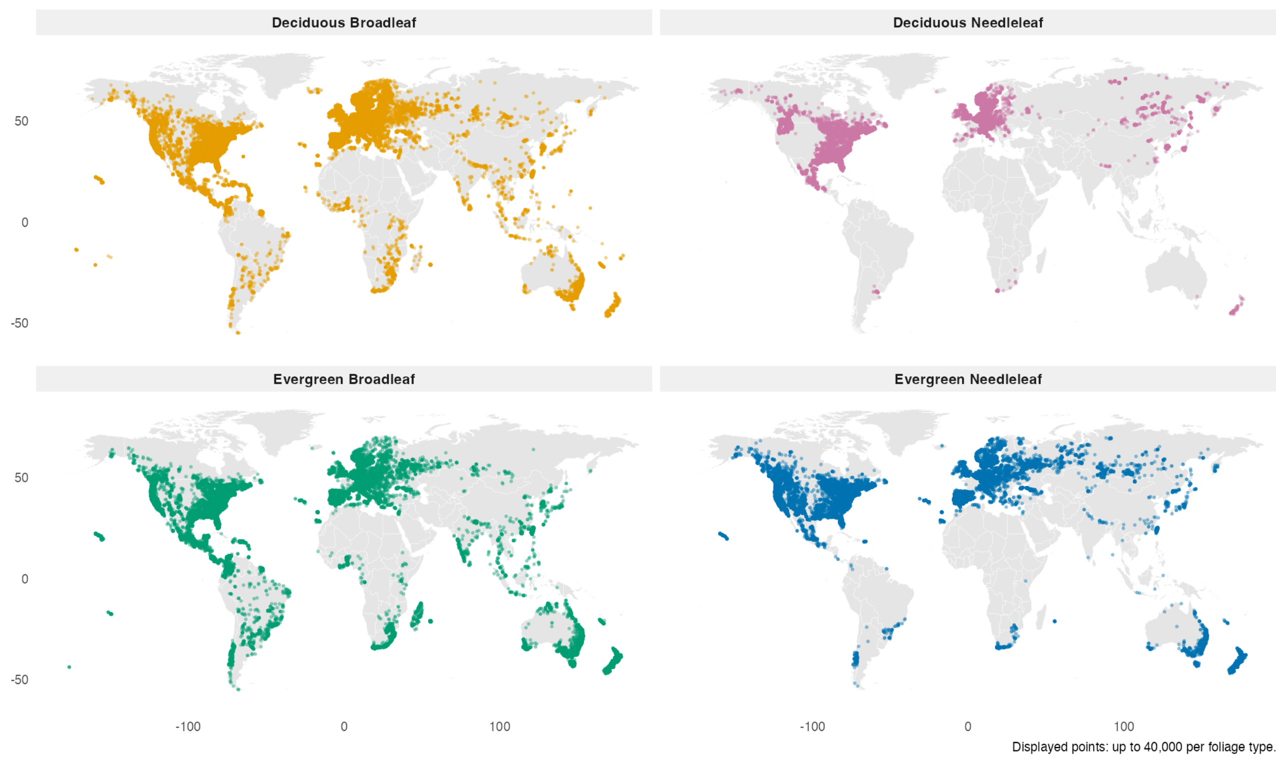


Figure 8: Geographical distribution by type of foliage

Figure 8 reveals that Deciduous Broadleaf (orange) and Evergreen Broadleaf (green) share a broadly similar geographical footprint, both present across nearly all sampled continents and spanning a wide range of latitudes, from temperate to tropical regions. This similarity suggests that, at the continental scale visualized here, geographical presence alone does not clearly separate these two categories. A finer, latitude-based view is needed to detect any underlying climatic gradient between them, which is examined in Figure 10. Evergreen Needleleaf (blue) is most heavily concentrated across the boreal and temperate zones of the Northern Hemisphere (North America, Europe, and northern Asia) but is also present, at lower density, in the southern parts of South America, southern

Africa, and Oceania, suggesting an association with temperate conditions more broadly rather than strictly cold northern climates. Deciduous Needleleaf (pink) remains overwhelmingly concentrated at high latitudes, primarily across Siberia and parts of Canada, with only a lower-density secondary presence in the United States and Europe, confirming that, unlike Evergreen Needleleaf, this foliage type stays largely confined to high-latitude regions overall.

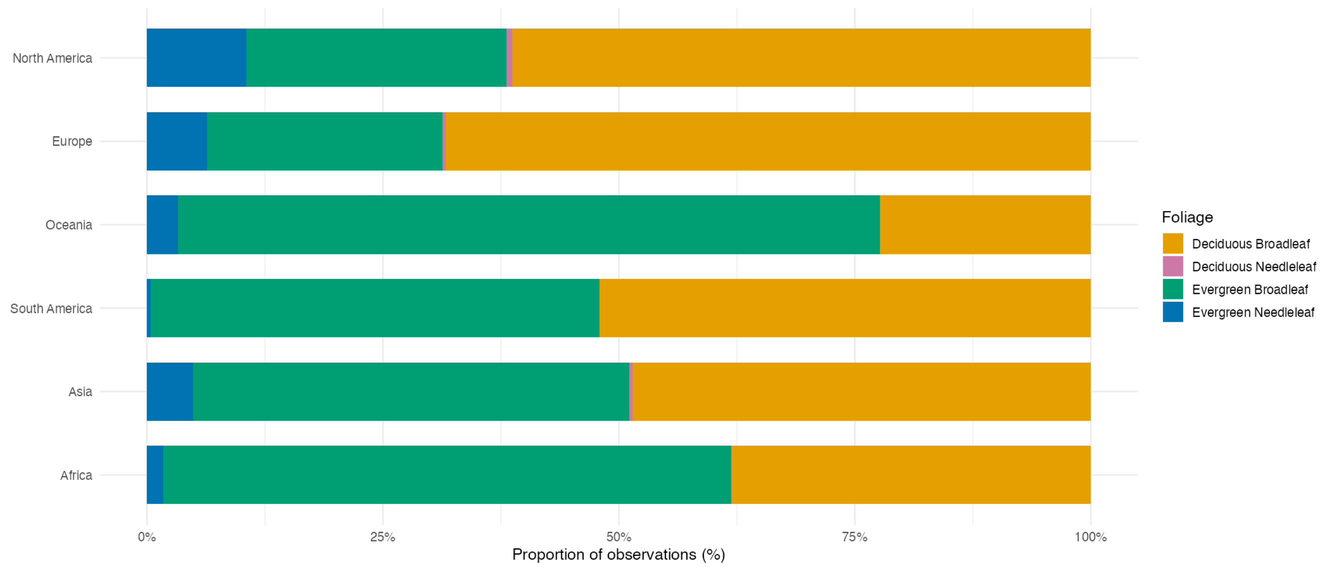


Figure 9: Geographical distribution by type of foliage

Figure 9 complements the previous maps by quantifying, for each continent, the relative composition of foliage types, independent of each continent's overall sampling volume. A clear contrast emerges between temperate and tropical/subtropical continents. North America and Europe share a similar profile, dominated by Deciduous Broadleaf (around 60–70% of observations), followed by Evergreen Broadleaf as the second-largest category (roughly 25–30%), with Evergreen Needleleaf making up a smaller but still notable share (around 10%), a share not found at comparable levels on any other continent, consistent with the boreal/temperate conifer forests characteristic of these regions. Oceania and Africa, by contrast, are clearly dominated by Evergreen Broadleaf, which accounts for the majority of observations in both continents (over 75% in Oceania, around 60% in Africa), with Deciduous Broadleaf as a smaller secondary category. South America and Asia present an intermediate profile, with Evergreen Broadleaf and Deciduous Broadleaf nearly evenly split at close to 50% each, suggesting that these two continents span a broader mix of climatic zones, from tropical to temperate, rather than being dominated by a single climate regime. Across all four of these continents, Evergreen Needleleaf drops to a marginal share compared to North America and Europe, and Deciduous Needleleaf is nearly absent everywhere except for a very thin presence in North America, Europe, and Asia, consistent with the highly restricted geographical range already observed in Figure 8. This continent-level breakdown reinforces the latitudinal pattern suggested by the maps: continents located predominantly at temperate latitudes (North America, Europe) show a foliage composition weighted toward deciduous and needleleaf strategies, while continents dominated by tropical or subtropical land area (Oceania, Africa) are dominated by evergreen broadleaf species, with South America and Asia sitting in between due to their wider climatic range. This distinction by continent, however, remains an approximation, since large countries such as the United States or China span multiple climatic zones. This limitation is directly addressed by

Figure 10, which examines this relationship along a continuous latitudinal axis rather than through continental boundaries.

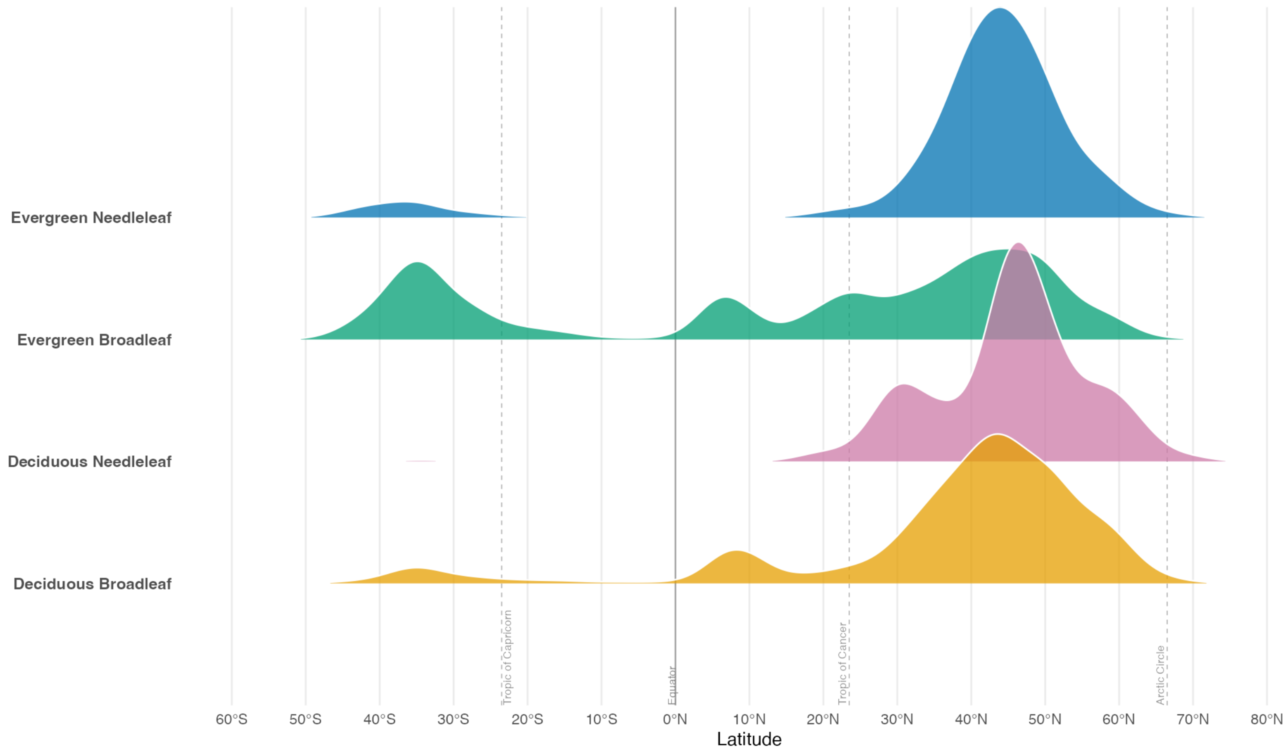


Figure 10: Distribution of foliage types by latitude

Figure 10 refines the patterns suggested by the previous map and bar chart. Evergreen Needleleaf (blue) shows a sharp peak around 45–50°N, with only a small secondary bump around 30–40°S. Deciduous Needleleaf (pink) is similarly concentrated at high northern latitudes, with a pronounced peak around 45–55°N and little recorded presence in the Southern Hemisphere. Deciduous Broadleaf (orange) has a large peak around 45–50°N, while Evergreen Broadleaf (green) is more widely distributed and has a more substantial tropical presence. Unlike the continent-level chart, the ridge plot exposes the continuous latitudinal shape of each category. These patterns are consistent with an association between foliage strategy and climate, but the plot alone cannot establish latitude as the primary driver because observation effort and other environmental variables are not controlled.

7.4 Taxonomic richness by continent

This subsection turns from the spatial and climatic distribution of foliage types to a different question: "how much taxonomic diversity does each continent actually contain ?" This question, however, cannot be answered directly by simply counting the number of distinct species or genera observed per continent, precisely because of the sampling imbalance identified in the first subsection of this report. A continent with more observations is mechanically more likely to have recorded more distinct taxa, even if its underlying biodiversity is not necessarily higher. Comparing raw taxonomic counts between continents therefore risks conflating genuine biodiversity differences with differences in sampling effort alone.

Three complementary views are used to address this issue. Table 2 first reports the raw number of distinct taxa recorded per continent, providing a simple numerical summary of the dataset's taxonomic content. Figure 11 then visualizes these same raw counts as a bar chart, making it easier to compare continents at a glance, while keeping in mind that, at this stage, these values remain directly influenced by each continent's sampling volume, as shown in Figure 6. Finally, Figure 12 addresses this limitation directly through rarefaction curves, which estimate the expected number of unique species as a function of sample size, allowing continents to be compared on an equal footing regardless of how many observations were actually collected for each.

continent	Foliage	Families	Genus	Species	Complete_taxa
Africa	4	168	987	3831	3831
Asia	4	180	958	3809	3809
Europe	4	173	737	2437	2437
North America	4	203	1219	6528	6528
Oceania	4	175	886	4900	4900
South America	4	183	1099	4968	4968

Table 2: Table of the taxonomic richness by continent

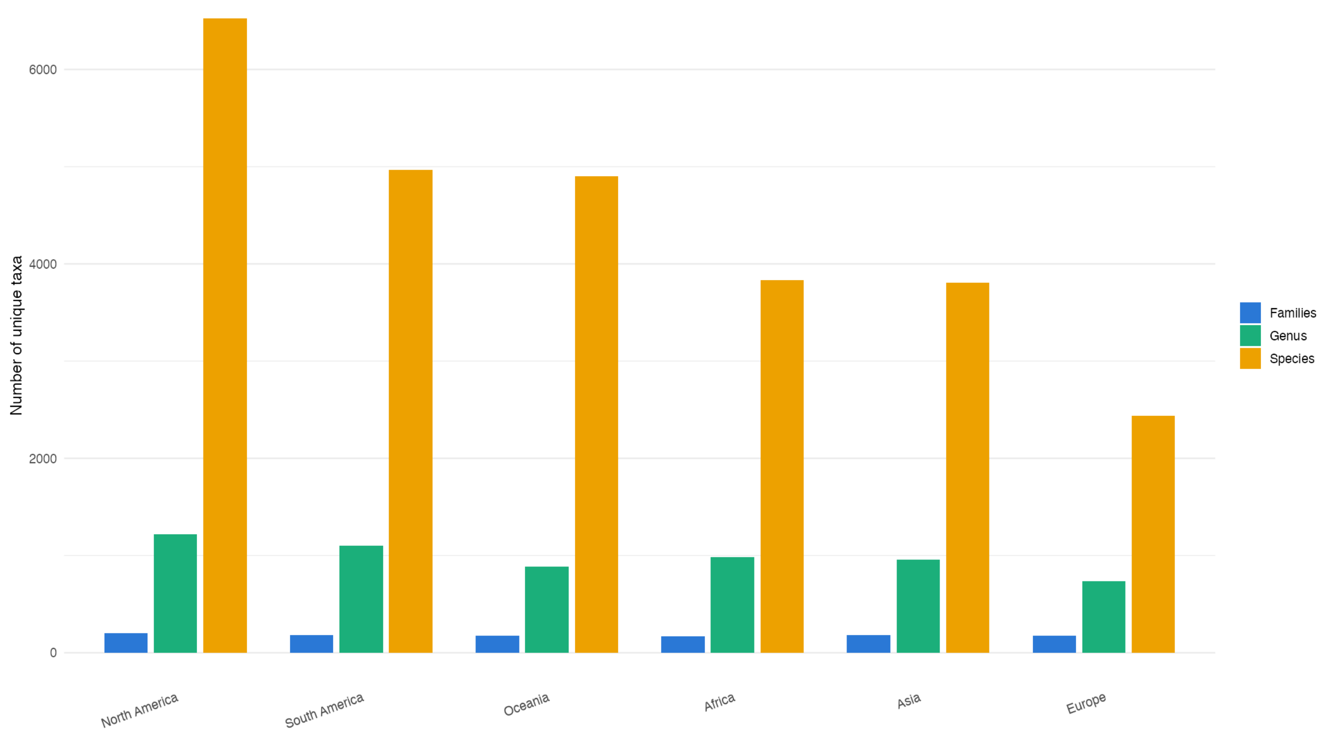


Figure 11: Taxonomic richness by continent (raw values)

Table 2 and Figure 11 report the raw number of distinct taxa recorded per continent, at three taxonomic levels: family, genus, and species. All six continents show a comparable number of families (168–203), suggesting that the dataset captures a broadly similar range of major taxonomic groups everywhere. This similarity breaks down markedly at finer taxonomic resolution : genus counts range from 737 (Europe) to 1219 (North America), and species counts range even more widely, from 2437 (Europe) to 6528 (North America). North America stands out as the richest continent at every taxonomic level, while Europe consistently ranks lowest.

Comparing this ranking to the observation volume documented in Figure 6 reveals an interesting discrepancy. Europe, despite being the second most heavily sampled continent overall (33.1% of observations), records the lowest species richness of all six continents (even lower than Africa and Asia, which together account for less than 9% of total observations). Conversely, South America and Oceania, which together represent under 17% of observations, show species counts (4968 and 4900 respectively) roughly double that of Europe, despite Europe having been sampled nearly four times more intensively.

This pattern suggests that raw taxonomic richness, as shown here, is not simply proportional to sampling volume, but it does not yet establish whether these differences reflect genuine biodiversity patterns or are still partly shaped by uneven sampling effort within each continent. A question addressed directly by the rarefaction analysis in Figure 12.

Methodological note: The `vegan::rarefy()` function calculates the expected number of recorded species in a random sample of n observations, drawn without replacement from each continent's species-abundance counts. It evaluates this expectation analytically rather than performing repeated random simulations. Calculating the expectation across increasing sample sizes produces one rarefaction curve per continent. This standardises the number of observations being compared, but it does not correct for geographic coverage, source composition, detectability, or other differences in sampling design.

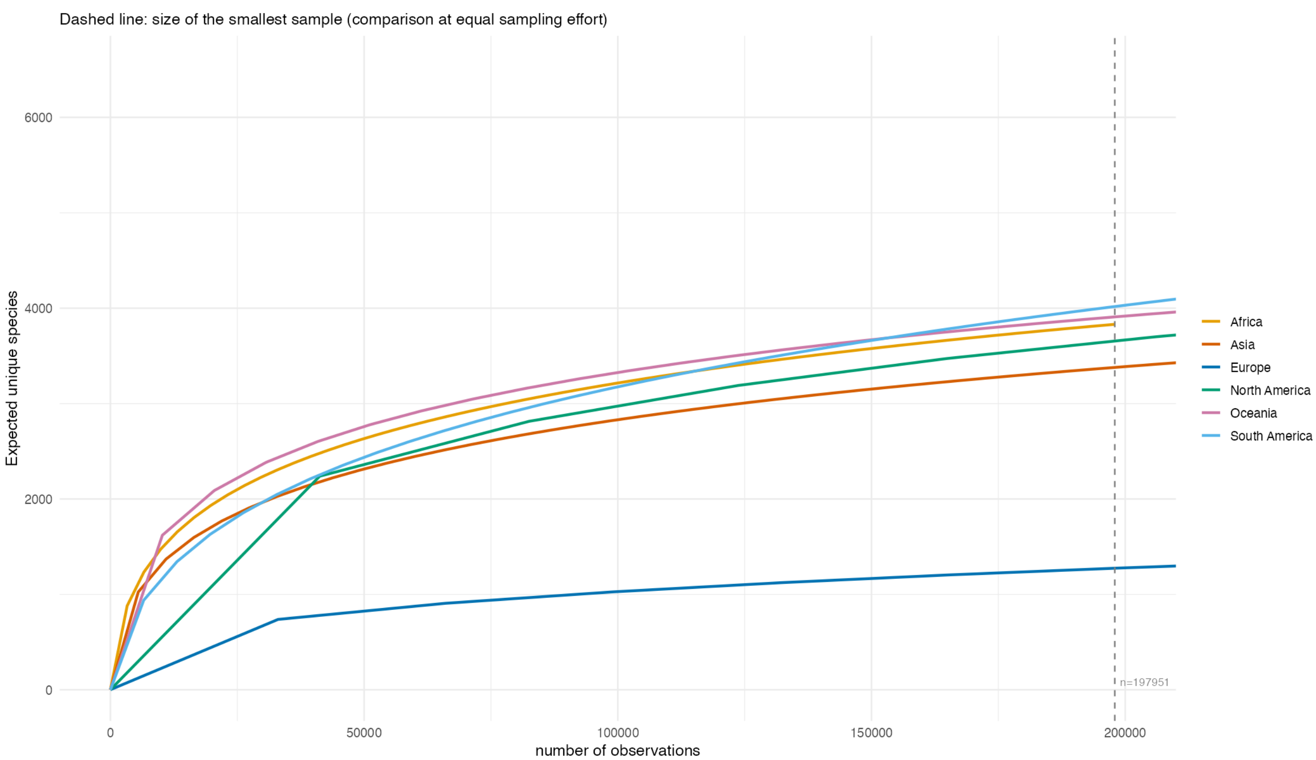


Figure 12: Rarefaction curves by continent

Figure 12 provides an equal-observation comparison that substantially revises the picture suggested by the raw counts in Figure 11. At the shared sampling effort marked by the dashed line (n=198,000, the size of Africa's full sample), the ranking of recorded richness changes considerably. Oceania and South America reach close to 4,000 expected recorded species, followed by Africa (around 3,900). North America and Asia form an intermediate group (roughly 3,700 and 3,400 respectively), while Europe has around 1,300 expected recorded species at this sample size.

The most striking change concerns North America. While it has the highest raw species count in Figure 11 (6,528 species), this advantage largely disappears once observation count is equalised. Its rarefaction curve is overtaken by Oceania, South America, and Africa before reaching the comparison point. This suggests that North America's raw lead is strongly influenced by its larger sample size (Figure 6), without establishing how the continents compare in underlying biological diversity.

Europe's rarefaction curve remains below the other continents across the range shown. This indicates lower recorded richness even at equal observation counts. The difference may be consistent with ecological patterns, but it may also reflect source composition, geographic concentration, detectability, or other sampling differences that rarefaction does not remove.

Finally, Africa, Oceania, and South America remain fairly closely clustered across the observed range, suggesting comparable recorded richness after standardising observation count. This adjustment addresses sample size rather than all forms of sampling bias.

7.5 Changes in the distribution over time

The two previous subsections examined the spatial and taxonomic distribution of observations as a static snapshot, without considering their recorded year. Given the strong reliance on citizen-science platforms such as iNaturalist, the number and coverage of dated observations are unlikely to be constant across the period. This final subsection therefore examines how observation volume and represented geography differ by observation year. The figures do not show when records were ingested into GlobalGeoTree.

Figure 13 first tracks the total number of observations dated to each year, alongside the number of distinct countries represented, allowing changes in record volume to be compared with changes in geographic representation.

Figure 14 then breaks this trend down by continent, revealing whether this growth has been shared evenly across regions or driven primarily by a subset of continents, which would in turn affect the balance of the geographical biases identified earlier.

Finally, Figure 15 brings the temporal and spatial dimensions together, mapping sampled observations over successive two-year periods to directly visualize how the geographical coverage of the dataset has expanded, or remained concentrated, over time.

Taken together, these figures assess whether the spatial imbalances observed in the previous subsections are a stable, structural feature of the dataset, or whether they are progressively narrowing as data collection continues to grow.

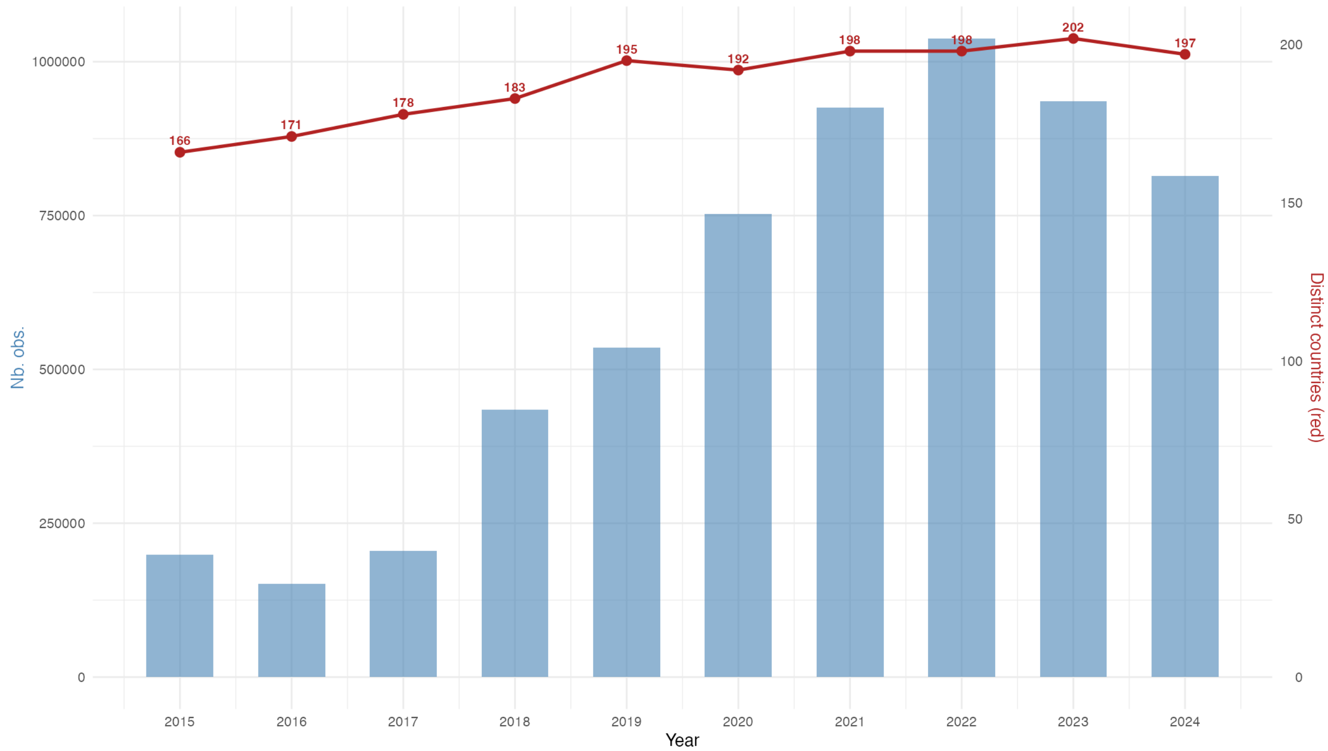


Figure 13: Observation volume and represented countries by observation year

Figure 13 reveals a marked increase in observations dated from 2018 onward. After remaining between roughly 150,000 and 200,000 observations per year from 2015 to 2018, the count rises sharply and peaks in 2022 at just over 1,000,000. This pattern is consistent with changes in citizen-science activity and source coverage, but observation year alone cannot establish when or why records were added to the combined dataset.

The number of distinct countries represented by dated observations (red line) also increases, from 166 in 2015 to over 198 by 2021, before stabilising around 197–202 through 2024. This broad representation should be interpreted carefully alongside the sharp difference in observation volume: one observation and hundreds of thousands of observations both count as country presence. The chart therefore shows breadth of representation by observation year, not equal sampling within or between countries.

Whether this recent growth in volume has been accompanied by a more even distribution across continents, or remains concentrated in already-dominant regions, is examined more directly in the next figure (Figure 14).

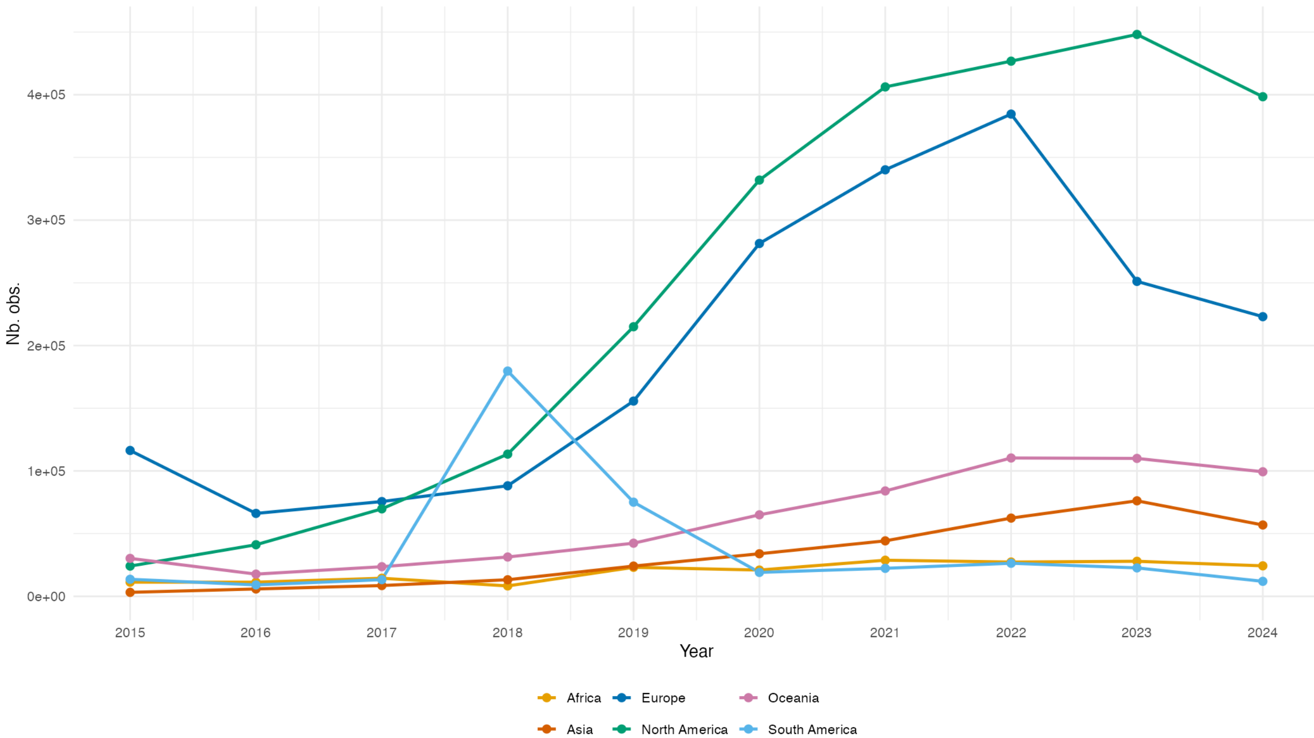


Figure 14: Observation volume by continent and observation year

Figure 14 refines the pattern in the previous figure. The increase in observations dated since 2018 is concentrated mainly in North America and Europe, while the other four continents remain comparatively flat. North America rises from roughly 20,000 observations dated 2015 to over 450,000 dated 2023, followed by a slight decline in 2024. Europe follows a similar trajectory until 2022 and then declines. Asia, Africa, and Oceania remain at lower levels, while South America has a distinct spike in 2018. These differences show that the imbalance between continents persists across observation years. They may reflect changes in platform activity, source coverage, or the temporal completeness of contributed records rather than changes in tree populations.

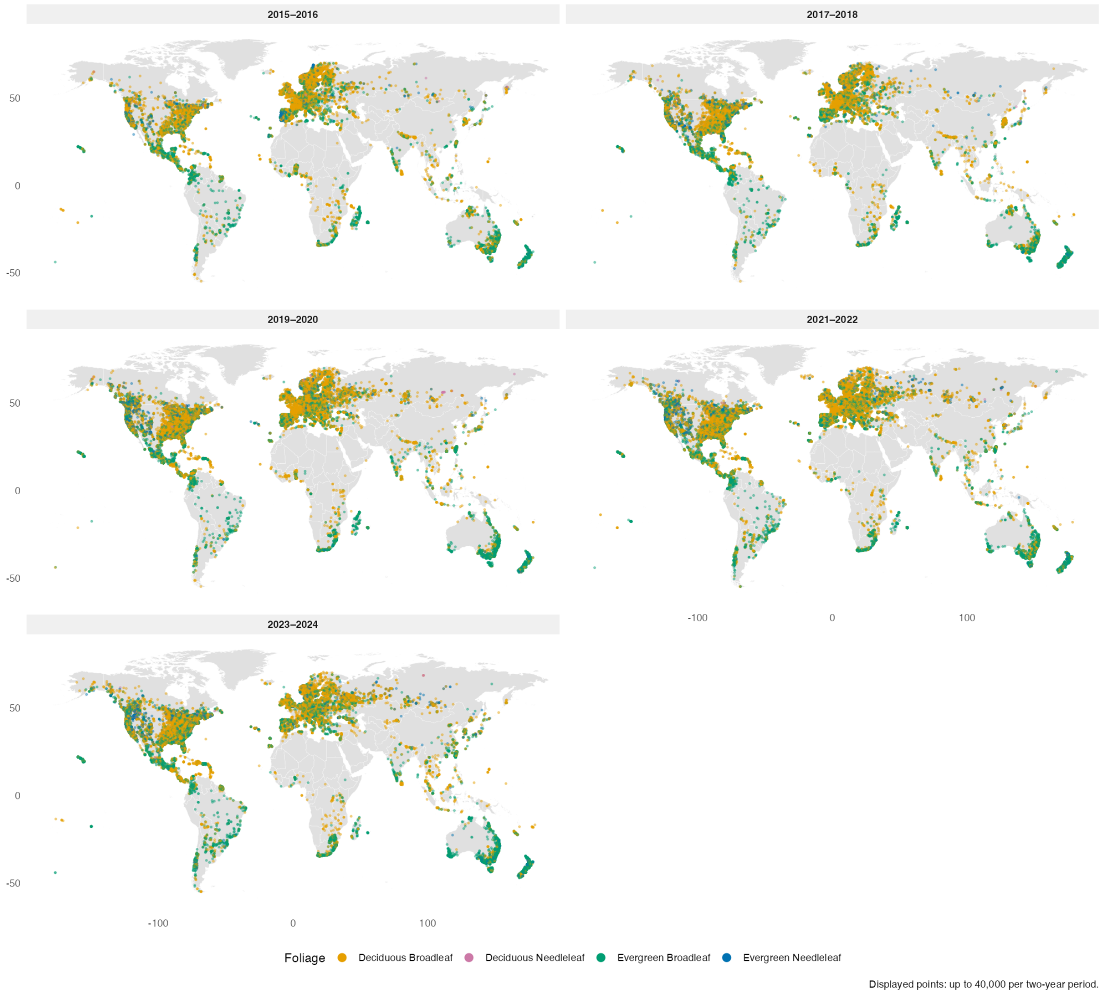


Figure 15: Geographical scope of data collection by period (5 two-year periods)

Figure 15 provides a direct spatial confirmation of the trends identified in the two previous figures, while also introducing an additional dimension: foliage type (Evergreen Broadleaf, Evergreen Needleleaf, Deciduous Broadleaf). Across all five sampled period panels, the overall geographical footprint of the dataset remains remarkably stable. The same broad regions (North America, Europe, coastal South America, parts of Africa and Southeast Asia/Oceania) are already represented as early as 2015–2016, with no major new region appearing in later periods. This is consistent with the near-total country coverage observed in Figure 13: geographical breadth was established early, and subsequent growth in volume (Figure 14) has taken the form of densification within already-sampled areas rather than expansion into new ones. This densification is visually apparent when comparing early and recent periods: point coverage becomes progressively denser and more continuous over North America and Europe, particularly for deciduous broadleaf (orange) and evergreen broadleaf (green) foliage, while regions such as central Africa, the Amazon interior, and Central Asia remain visibly sparse throughout all five periods, showing little to no improvement over the decade.

Overall, this final figure shows that the geographic imbalance is visible throughout the full range of observation years. This is consistent with a persistent structural bias in the dated records, while not describing when the contributing records were actually ingested into GlobalGeoTree.

Part II: Germany and Berlin Analysis

> — *Nataliia*

Part II examines tree observations within Germany. Germany was selected because it has relatively high observation density and contributions from multiple data sources, making it a good case for studying how global biases play out at the national and sub-national level. Berlin is examined in detail because this is where our team is based. Baden-Württemberg and NRW are included only as short map-based comparisons.

Table 3: Germany data check

item	value
Total observations (all countries)	5,991,220
Germany observations used in Part II	204,231

8 Germany: Species and Common-Name Lookup

To make charts easier to read, English common names are shown next to Latin names wherever possible. As described in the Methodology section, the GlobalGeoTree dataset contains `species_key`, a unique GBIF identifier for each species. This key is used to query the GBIF vernacular-names API once and cache the result in `data/external/species_common_names.csv`. On every subsequent render, the report reads from that file.

Table 4: German species overview

metric	value
Unique species in German observations	464
Species with a valid GBIF species key	464

Table 5: GBIF common names lookup result

metric	value
Species queried from GBIF	464
Species with an English common name	420

Species without an English name in GBIF are shown with their Latin name only in charts. The cached file can be deleted and re-generated at any time by deleting `data/external/species_common_names.csv` and re-rendering the report.

9 Germany: National Patterns

This section examines how tree observations are distributed across Germany as a whole: spatial concentration, temporal trends, source structure, and Bundesland-level variation.

9.1 Load Bundesländer boundaries

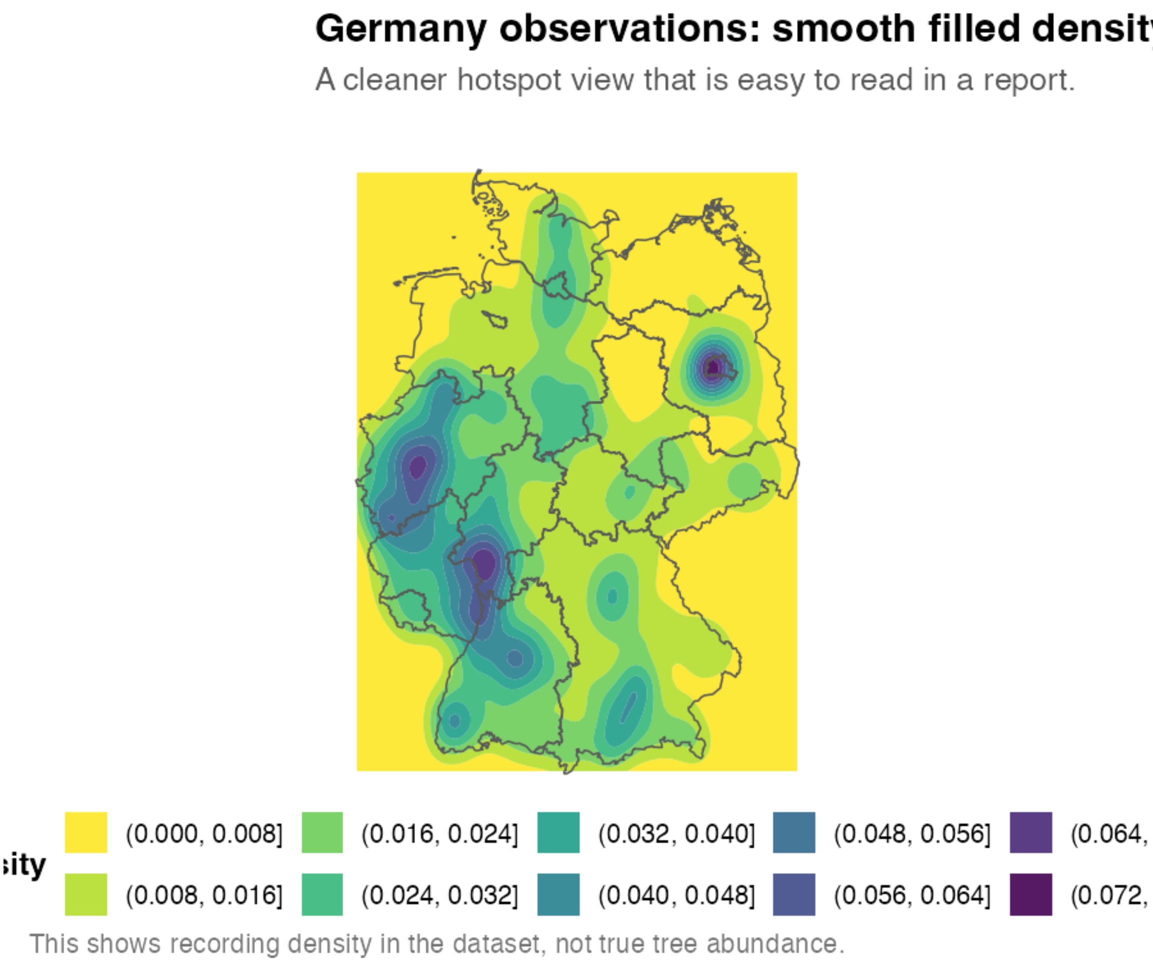
Bundesländer polygons are needed to draw state boundaries on Germany maps and to assign observations to federal states. The code first looks for a local GeoJSON file; if it is not found, the public GISCO NUTS1 file is downloaded.

Table 6: Bundesländer boundary check

item	value
Boundary status	Loaded Bundesländer / NUTS1 boundaries from data/external/germany_nuts1_2021.geojson
Number of Bundesländer polygons	16

9.2 Distribution of observations in Germany

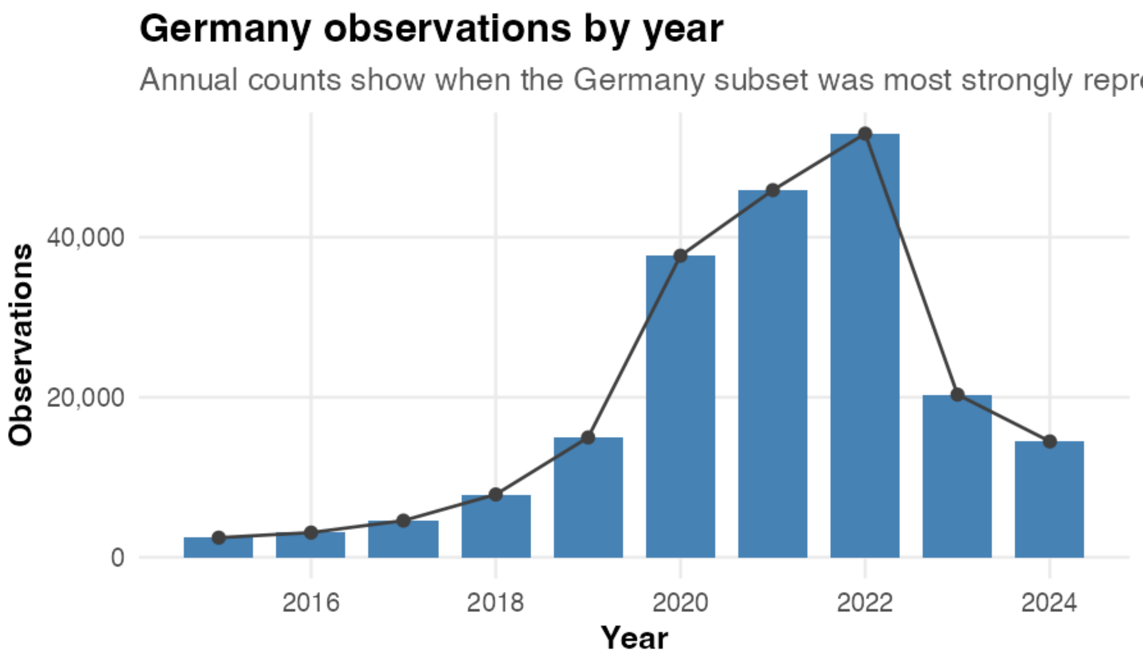
Where are observations spatially concentrated across Germany? The smooth filled density map below provides a first overview of recording hotspots across the country.



Darker areas show where more observations were recorded. These hotspots should be read as observation intensity rather than tree abundance: a location can appear dense because observers are active there, because a platform has strong coverage, or simply because access is easier.

9.3 Observations by year

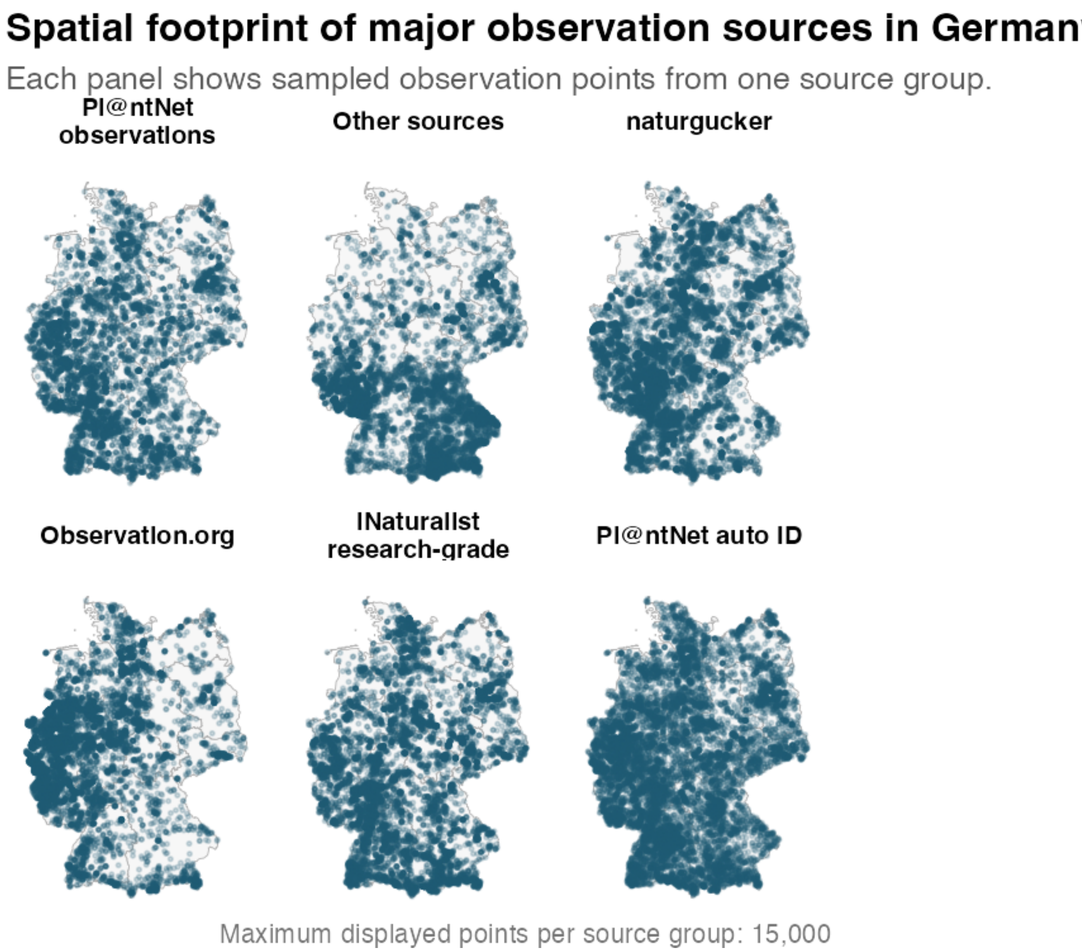
How has recording activity changed over time in Germany? A dataset dominated by a few recent years may show spatial patterns that reflect current platform growth rather than long-term tree distribution.



Peaks in this chart reflect peaks in recording activity, not necessarily more trees. A strong upward trend in recent years is common in citizen-science datasets as more users join the platforms.

9.4 Observations by source on the country map

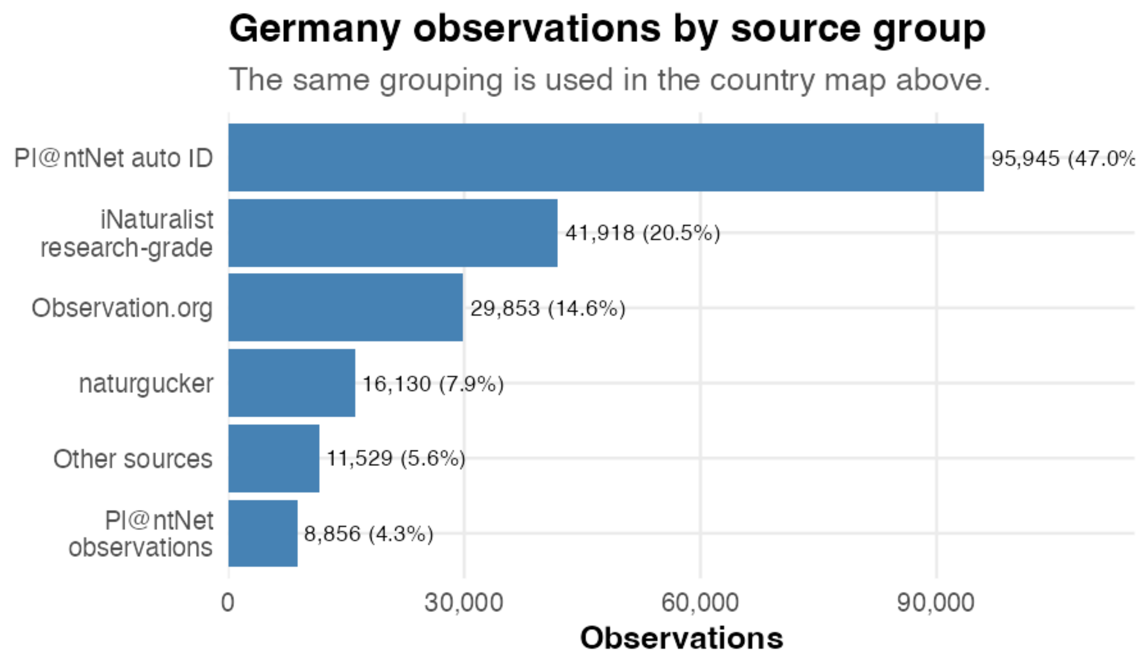
The year distribution above shows *when* Germany observations were recorded. The next question is *who* recorded them — which platforms contribute the most records, and do they cover the country differently? Each panel shows where one source group collected its records.



If source panels look visually different, this is evidence of source bias: the national density map then reflects a mix of ecological signal and platform coverage.

9.5 Observations by source (bar chart)

The map above shows *where* each source records; this bar chart shows *how many* records each source contributes.



The largest bar identifies the dominant source. A single dominant source can strongly shape the density patterns seen on all subsequent maps.

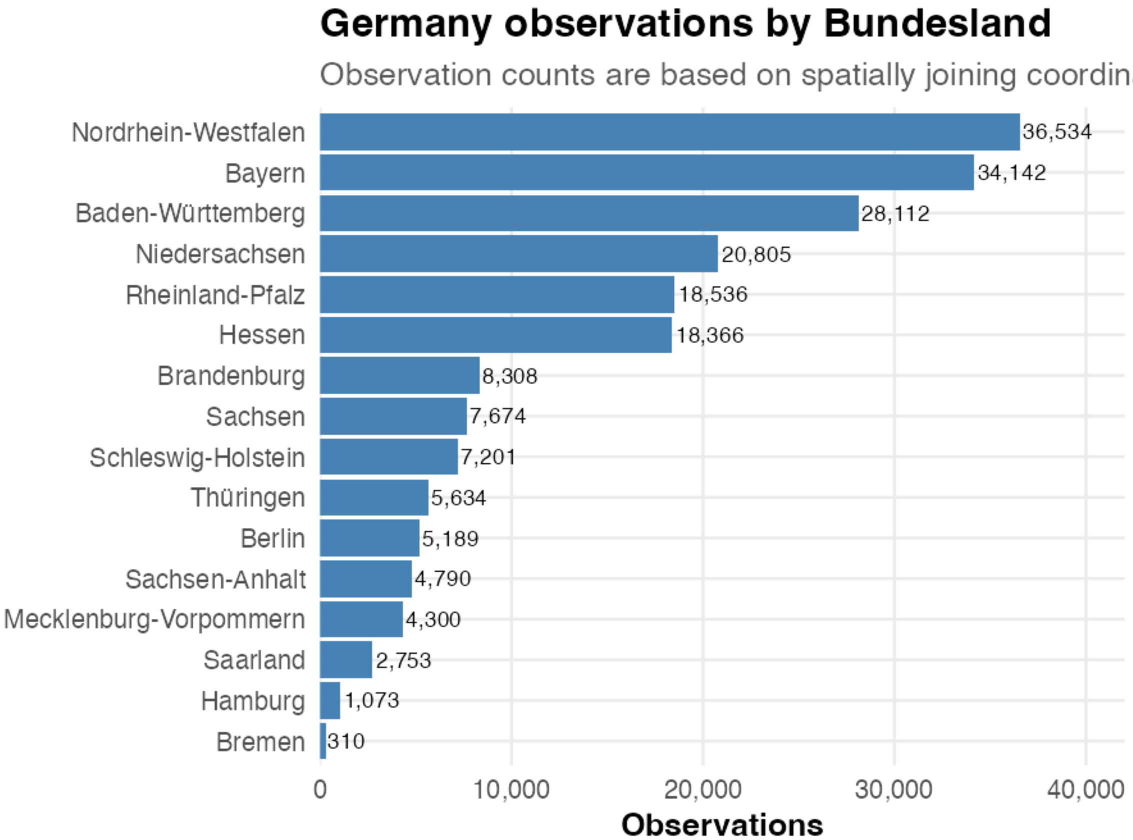
> **Note**
>
> **Two Pl@ntNet source names.** `Pl@ntNet observations` and `Pl@ntNet auto ID` are both Pl@ntNet-derived but represent different validation routes: one is the regular shared-observation dataset, and the other contains automatically identified occurrences. They are therefore kept separate throughout this report.

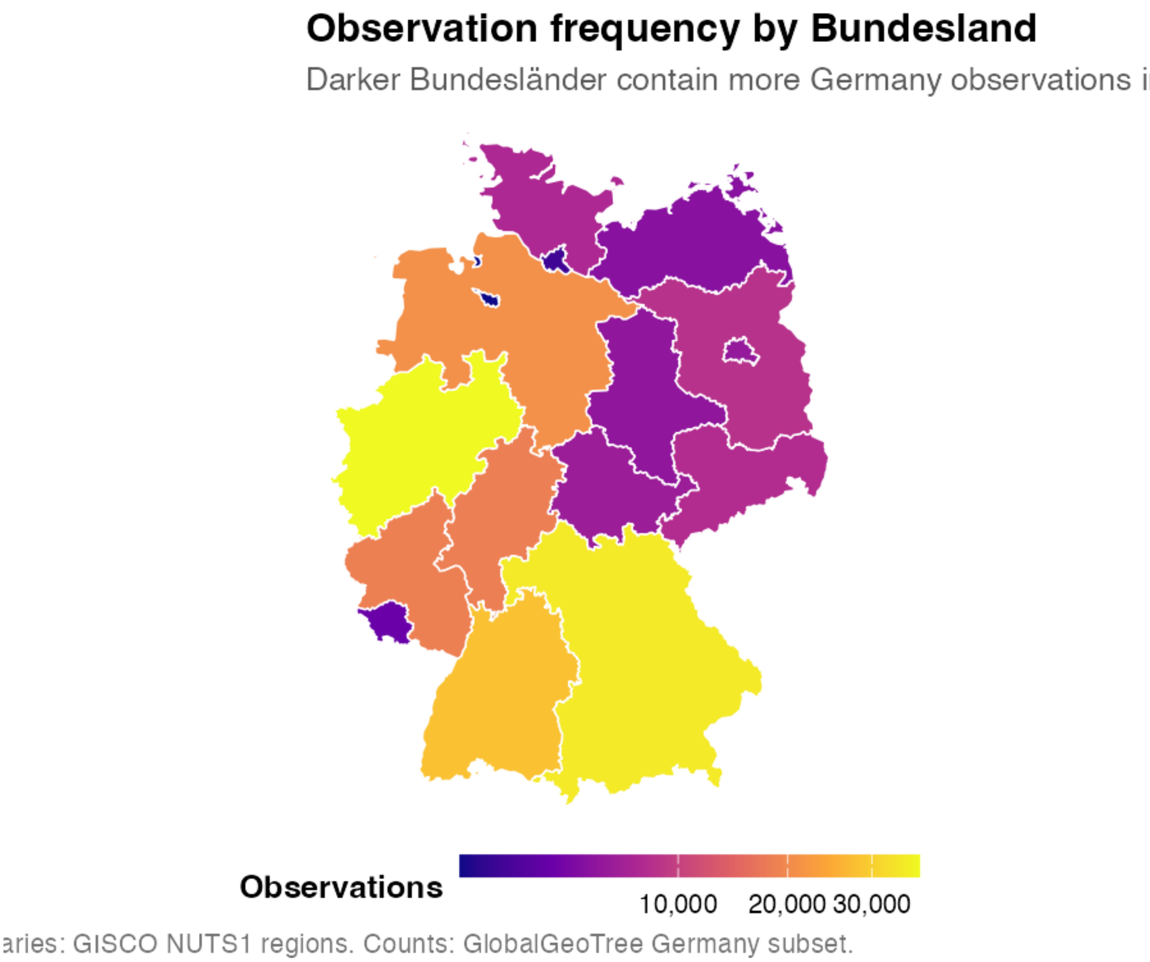
9.6 Observations mapped to Bundesländer

Having seen the sources, we now look at where their records land across Germany's sixteen federal states. Observations are assigned to Bundesländer using a spatial join — each observation's coordinates are matched to the boundary polygon it falls within.

Table 7: Bundesland spatial join check

metric	value
Germany observations before join	204,231
Matched to a Bundesland	203,727
Bundesländer with observations	16


Germany observations by Bundesland
Observation counts are based on spatially joining coordina
Nordrhein-Westfalen
36,534
Bayern
34,142
Baden-Württemberg
28,112
Niedersachsen
20,805
Rheinland-Pfalz
18,536
Hessen
18,366
Brandenburg
8,308
Sachsen
7,674
Schleswig-Holstein
7,201
Thüringen
5,634
Berlin
5,189
Sachsen-Anhalt
4,790
Mecklenburg-Vorpommern
4,300
Saarland
2,753
Hamburg
1,073
Bremen
310
0
10,000
20,000
30,000
40,000
Observations



Together, the bar chart provides exact rankings while the map shows whether high-record states cluster in one part of Germany. Differences between Bundesländer reflect population density, platform coverage, and observer activity rather than differences in actual tree abundance. Nordrhein-Westfalen and Baden-Württemberg — the first and third largest Bundesländer by observation count — are revisited later as spatial comparison points for Berlin.

10 Berlin

Berlin is the city of our university, which is why it was selected as the main target for regional analysis in Germany. As a city-state, Berlin's entire area is urban, which may produce recording patterns quite different from larger, more rural federal states.

10.1 Load Berlin district boundaries

Berlin is handled separately because a finer boundary layer than Bundesländer is needed for city-level analysis. The district boundaries come from an open-data project and provide polygons for Berlin's local areas (Ortsteile).

Table 8: Berlin boundary check

item	value
Berlin boundary status	Loaded Berlin district/local-area boundaries from data/external/berlin_districts.geojson
Number of Berlin polygons	97

10.2 Berlin observations

Table 9: Berlin observation summary

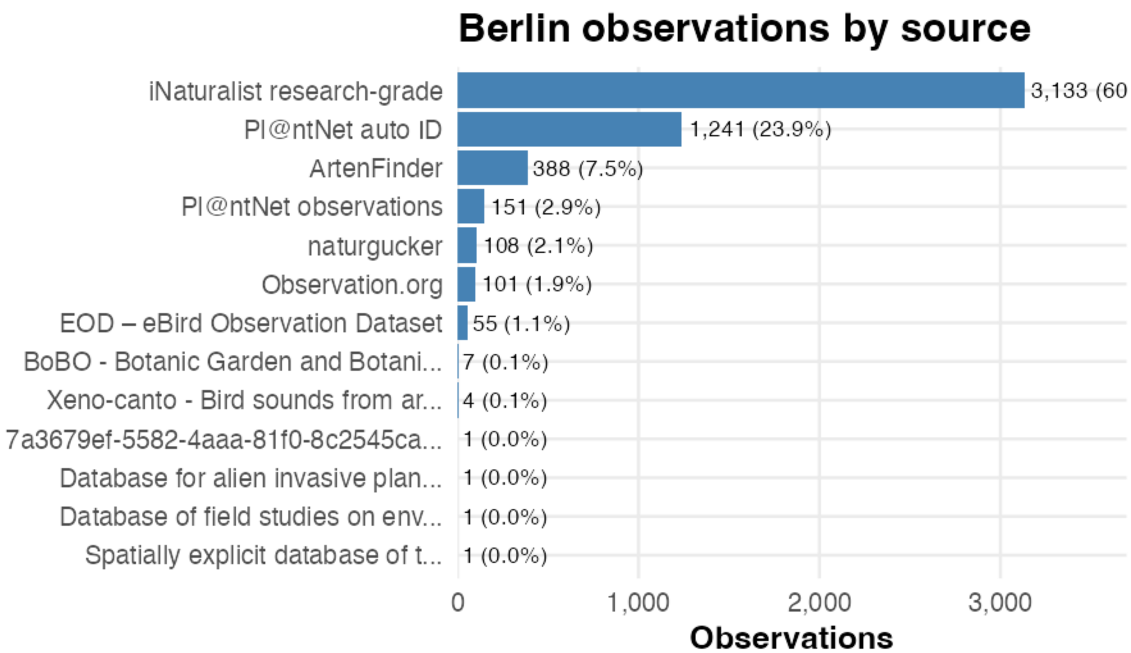
metric	value
Berlin filtering method	Spatial join to Berlin district/local-area polygons
Berlin observations	5,192
Berlin recorded species	158
Berlin recorded genera	85
Berlin recorded families	43
Available source values	13
Available years	2015-2024

10.3 Berlin observation sources

Source composition is important context for every Berlin map. If one platform dominates, Berlin hotspots may partly reflect that platform's recording behaviour rather than tree distribution.

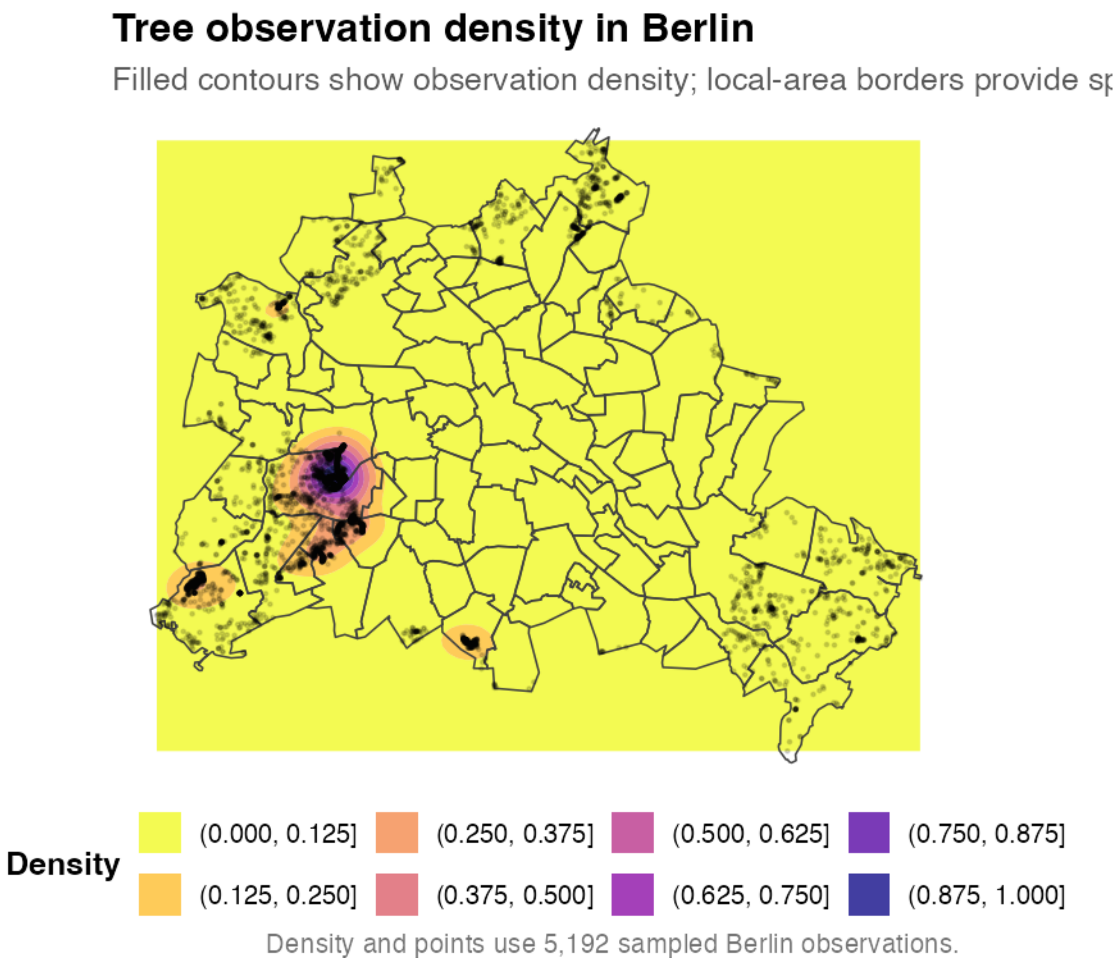
Table 10: Berlin observations by source

source_short	observations	share	label
iNaturalist research-grade	3,133	60.3%	3,133 (60.3%)
Pl@ntNet auto ID	1,241	23.9%	1,241 (23.9%)
ArtenFinder	388	7.5%	388 (7.5%)
Pl@ntNet observations	151	2.9%	151 (2.9%)
naturgucker	108	2.1%	108 (2.1%)
Observation.org	101	1.9%	101 (1.9%)
EOD – eBird Observation Dataset	55	1.1%	55 (1.1%)
BoBO - Botanic Garden and Botani…	7	0.1%	7 (0.1%)
Xeno-canto - Bird sounds from ar…	4	0.1%	4 (0.1%)
7a3679ef-5582-4aaa-81f0-8c2545ca…	1	0.0%	1 (0.0%)
Database for alien invasive plan…	1	0.0%	1 (0.0%)
Database of field studies on env…	1	0.0%	1 (0.0%)
Spatially explicit database of t…	1	0.0%	1 (0.0%)



10.4 Spatial distribution of Berlin observations

Which parts of Berlin show the highest recording density? Because Berlin is geographically compact, a single density map with overlaid district boundaries is sufficient to show spatial patterns.



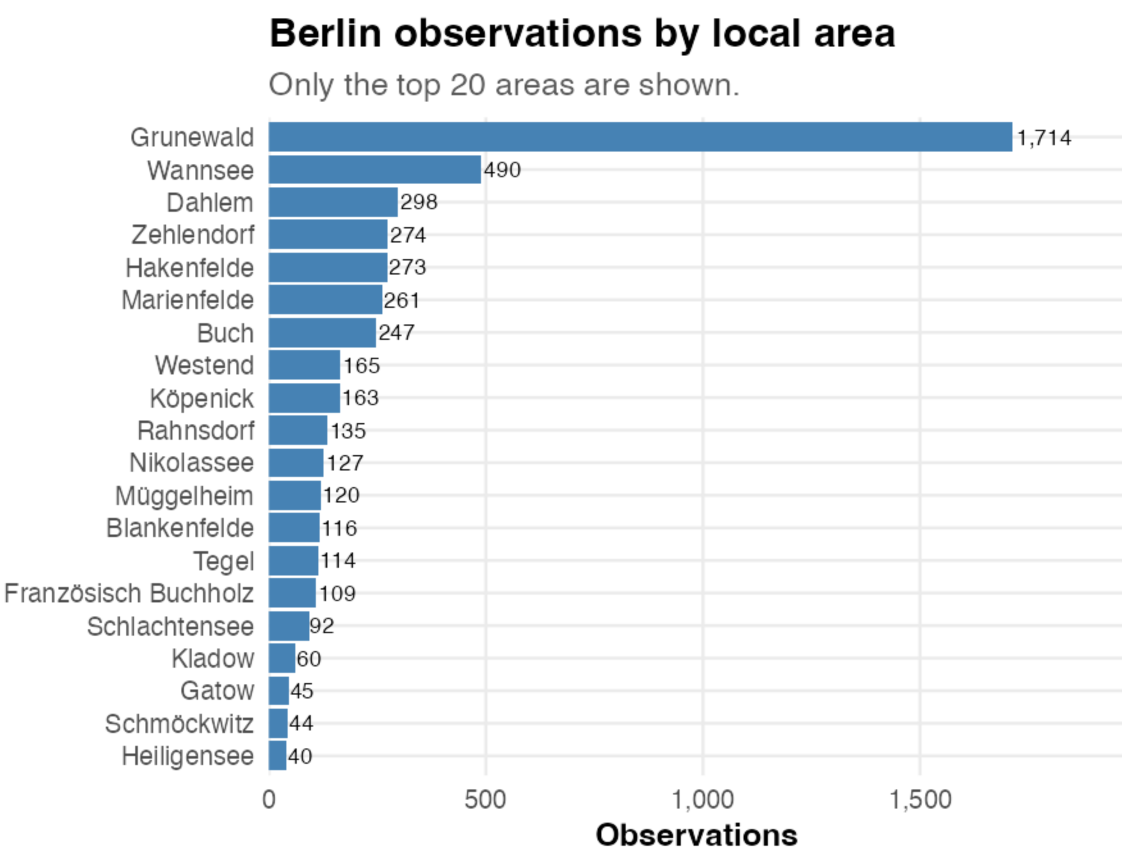
The density map highlights which parts of Berlin have the most recorded tree observations. High-density areas are likely to correspond to parks, green corridors, and neighbourhoods with active observers rather than necessarily having more trees than other areas.

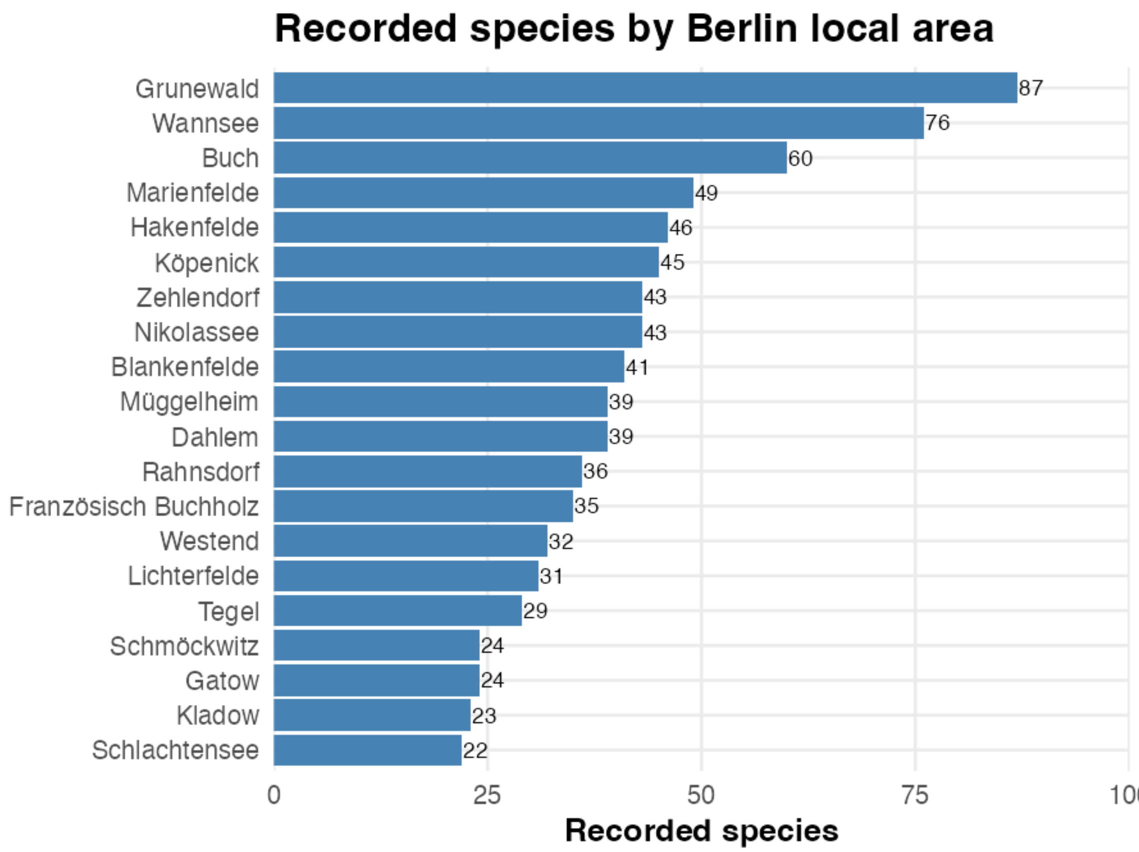
10.5 Berlin local areas: observations and species

Table 11: Top Berlin local areas by observation count

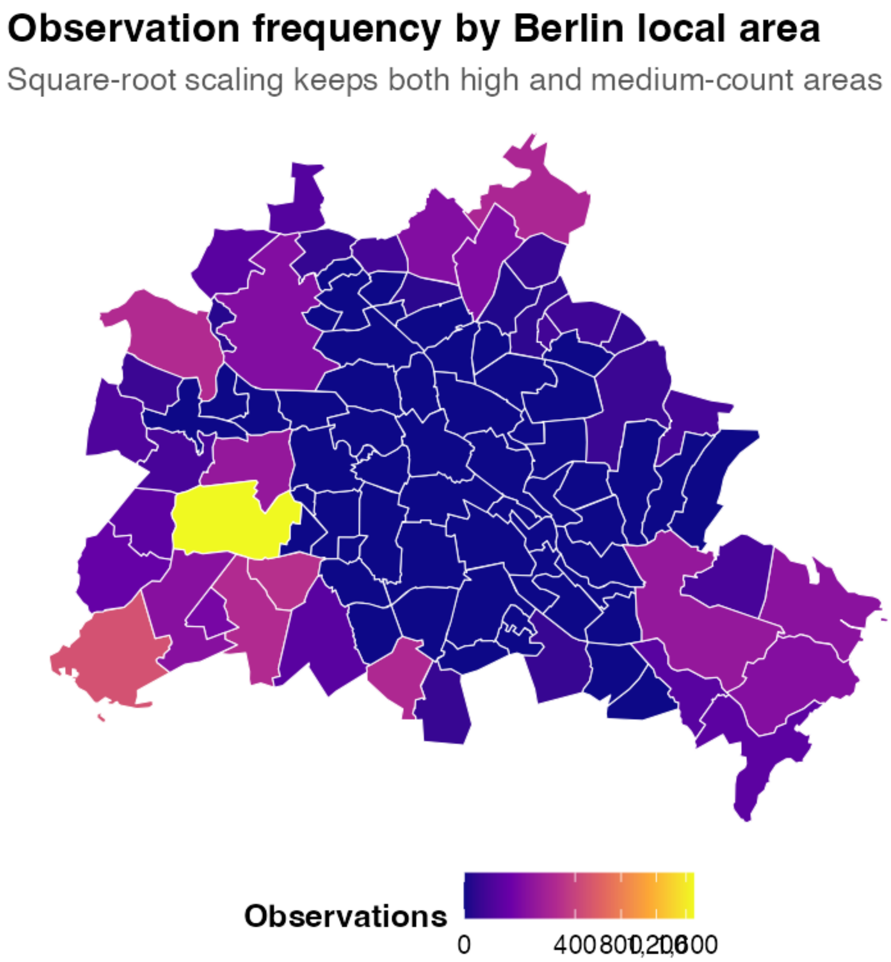
district	observations	species	sources
Grunewald	1,714	87	7
Wannsee	490	76	8
Dahlem	298	39	7
Zehlendorf	274	43	6
Hakenfelde	273	46	8
Marienfelde	261	49	6
Buch	247	60	7
Westend	165	32	5
Köpenick	163	45	7
Rahnsdorf	135	36	5

district	observations	species	sources
Nikolassee	127	43	7
Müggelheim	120	39	8
Blankenfelde	116	41	7
Tegel	114	29	8
Französisch Buchholz	109	35	5
Schlachtensee	92	22	4
Kladow	60	23	5
Gatow	45	24	5
Schmöckwitz	44	24	7
Heiligensee	40	15	5





Recorded species by Berlin local area
Grunewald
Wannsee
Buch
Marienfelde
Hakenfelde
Köpenick
Zehlendorf
Nikolassee
Blankenfelde
Müggelheim
Dahlem
Rahnsdorf
Französisch Buchholz
Westend
Lichterfelde
Tegel
Schmöckwitz
Gatow
Kladow
Schlachtensee
87
76
60
49
46
45
43
43
41
39
39
36
35
32
31
29
24
24
23
22
0
25
50
75
100
Recorded species



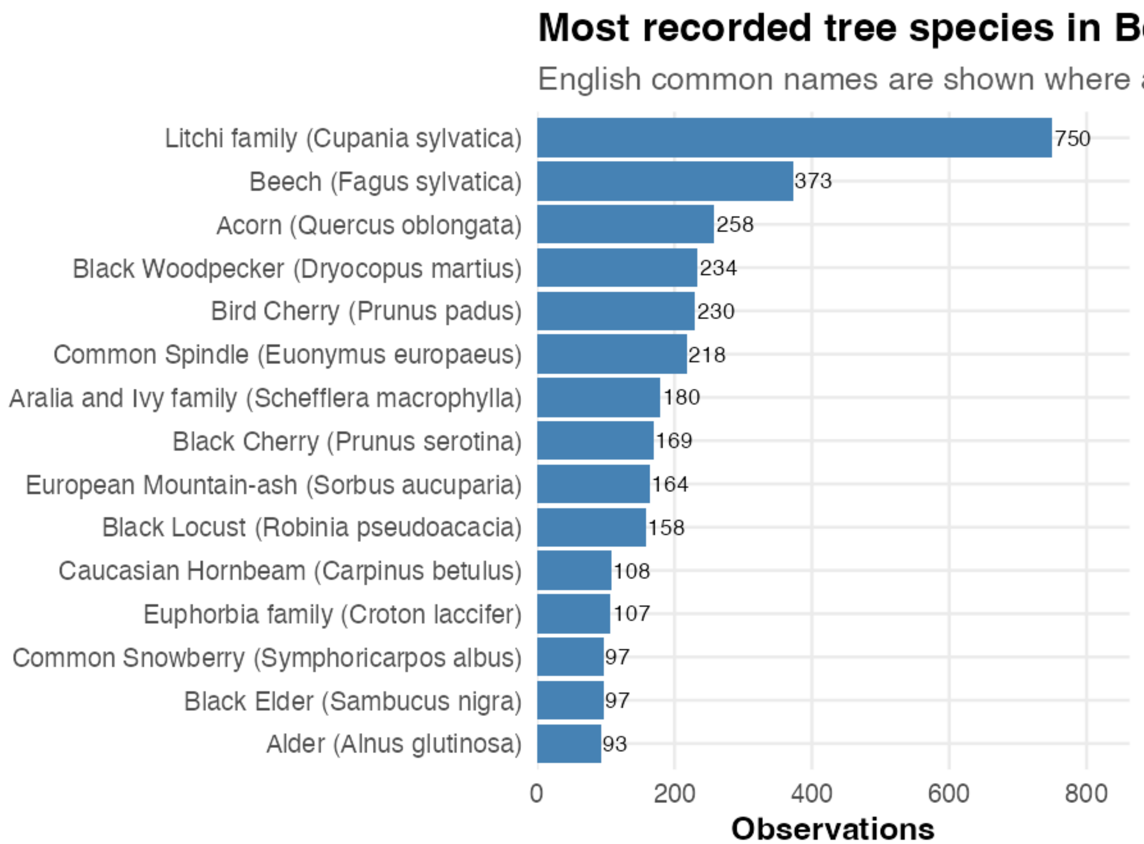
The bar charts rank local areas by observation count and species count separately, while the choropleth map shows the spatial pattern. Local areas with both high observation counts and high species richness may genuinely be more biodiverse or may simply have more active recording communities.

10.6 Top recorded species in Berlin

Table 12: Top recorded species in Berlin

scientific_name	common_name_en	observations	share
Cupania sylvatica	Litchi family	750	14.4%
Fagus sylvatica	Beech	373	7.2%
Quercus oblongata	Acorn	258	5.0%
Dryocopus martius	Black Woodpecker	234	4.5%
Prunus padus	Bird Cherry	230	4.4%
Euonymus europaeus	Common Spindle	218	4.2%
Schefflera macrophylla	Aralia and Ivy family	180	3.5%
Prunus serotina	Black Cherry	169	3.3%
Sorbus aucuparia	European Mountain-ash	164	3.2%

scientific_name	common_name_en	observations	share
Robinia pseudoacacia	Black Locust	158	3.0%
Carpinus betulus	Caucasian Hornbeam	108	2.1%
Croton laccifer	Euphorbia family	107	2.1%
Sambucus nigra	Black Elder	97	1.9%
Symphoricarpos albus	Common Snowberry	97	1.9%
Alnus glutinosa	Alder	93	1.8%



The most frequently recorded species in Berlin reflect the city's urban tree stock, which includes both native species and commonly planted street and park trees. High observation counts for a species can indicate that it is genuinely common, that it is easy to identify and therefore often recorded by citizen-science users, or both.

10.7 Berlin oak tree groups

This section demonstrates a possible practical application of the dataset. The oak processionary moth (*Thaumetopoea processionea*) can cause allergic reactions and skin irritation near affected oak trees. Mapping recorded oak locations provides a starting point that could be compared with official tree inventories and pest-surveillance data; GlobalGeoTree observations alone do not show whether the moth is present or whether a public warning is needed.

Table 13: Berlin oak observation summary

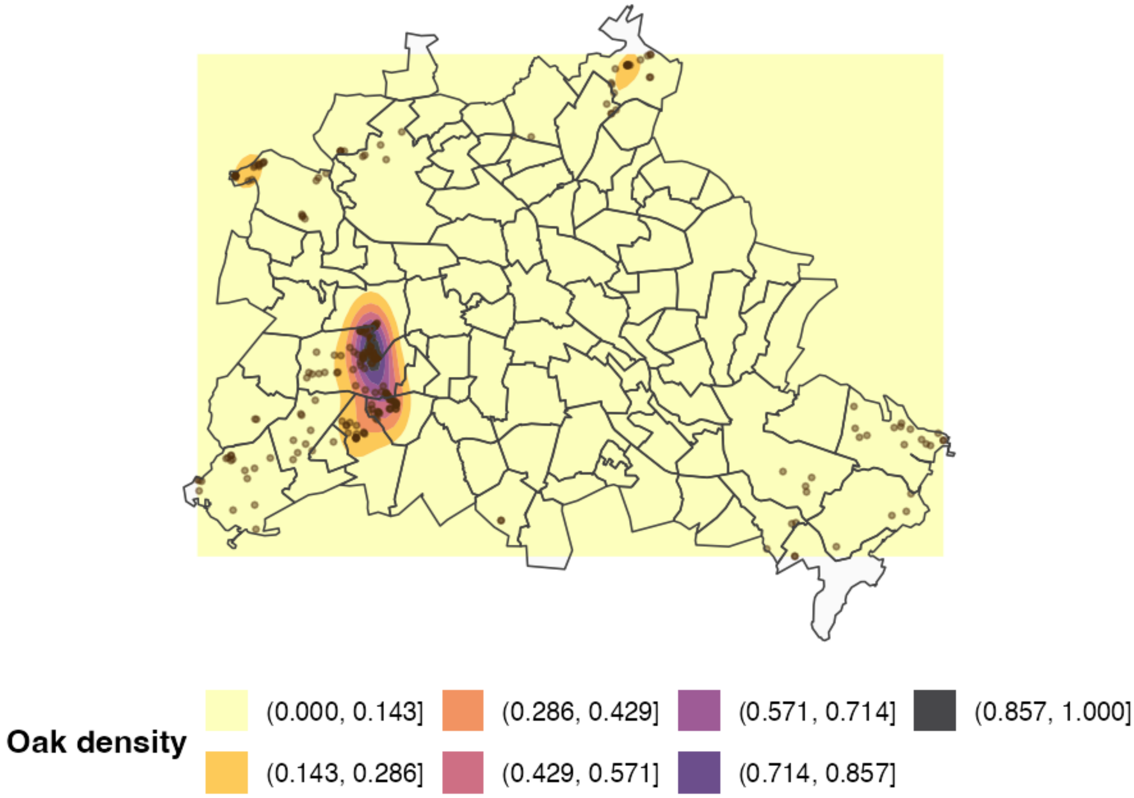
metric	value
Berlin oak observations	258
Berlin recorded oak species	1
Share of Berlin observations that are oak	5.0%

Table 14: Top Berlin local areas by recorded oak observations

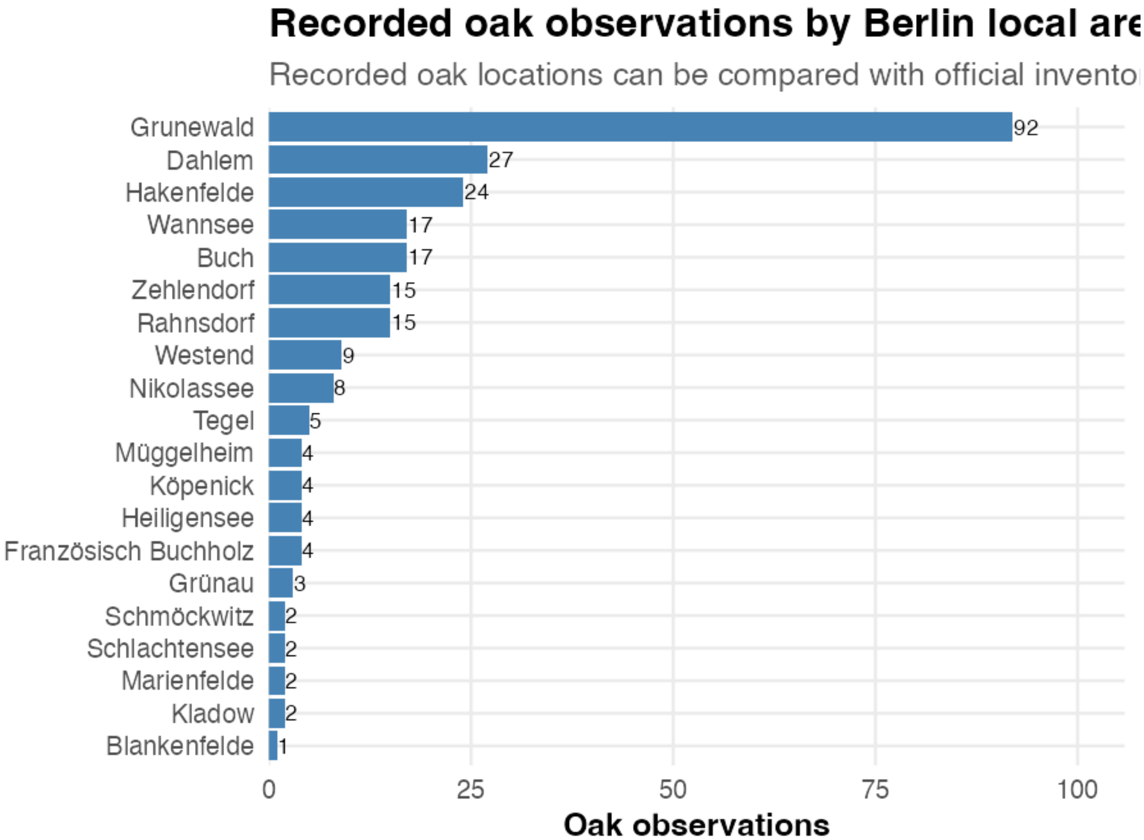
district	oak_observations
Grunewald	92
Dahlem	27
Hakenfelde	24
Buch	17
Wannsee	17
Rahnsdorf	15
Zehlendorf	15
Westend	9
Nikolassee	8
Tegel	5
Französisch Buchholz	4
Heiligensee	4
Köpenick	4
Müggelheim	4
Grünau	3
Kladow	2
Marienfelde	2
Schlachtensee	2
Schmöckwitz	2
Blankenfelde	1

Recorded oak tree groups in Berlin

Oak observations are filtered from Quercus records and mapped with l



Displayed oak points: 258 of 258 Berlin oak observations.



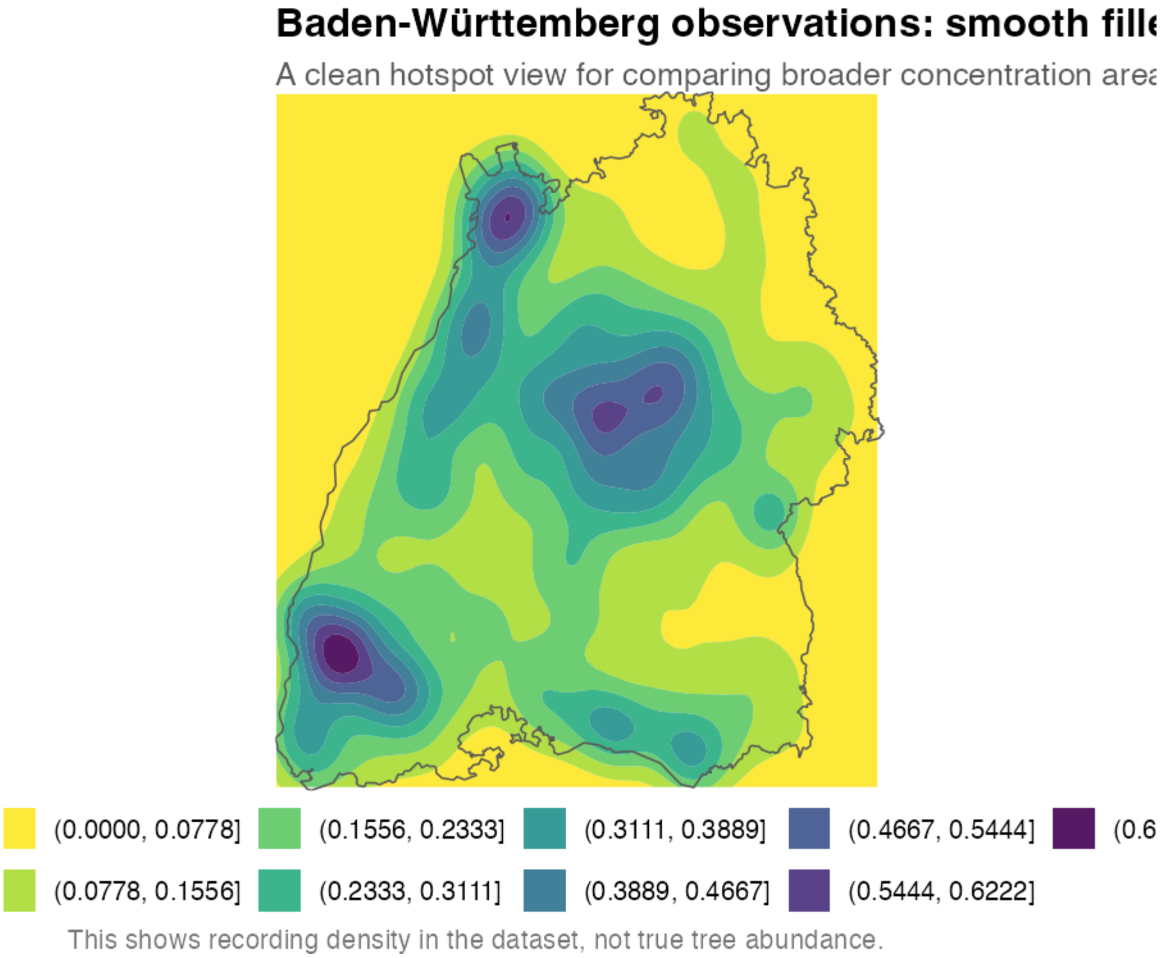
The oak map and local-area ranking show where recorded oak groups are concentrated across Berlin. Because these records reflect observer activity and are not a complete oak census, they cannot determine pest presence or risk by themselves. The visualization instead demonstrates how GlobalGeoTree locations could be combined with official inventories and surveillance records in a practical follow-up analysis.

11 Baden-Württemberg and Nordrhein-Westfalen

The Berlin analysis above revealed that observations are heavily concentrated in a single location, raising questions about whether this pattern reflects a Berlin-specific data quality issue or is typical for German regions. To investigate, we compare Berlin with Nordrhein-Westfalen (NRW) and Baden-Württemberg — the first and third largest Bundesländer by observation count. For each state, a density map and a NUTS3 district choropleth show where observations are concentrated. Both should be read as recording intensity, not true tree abundance.

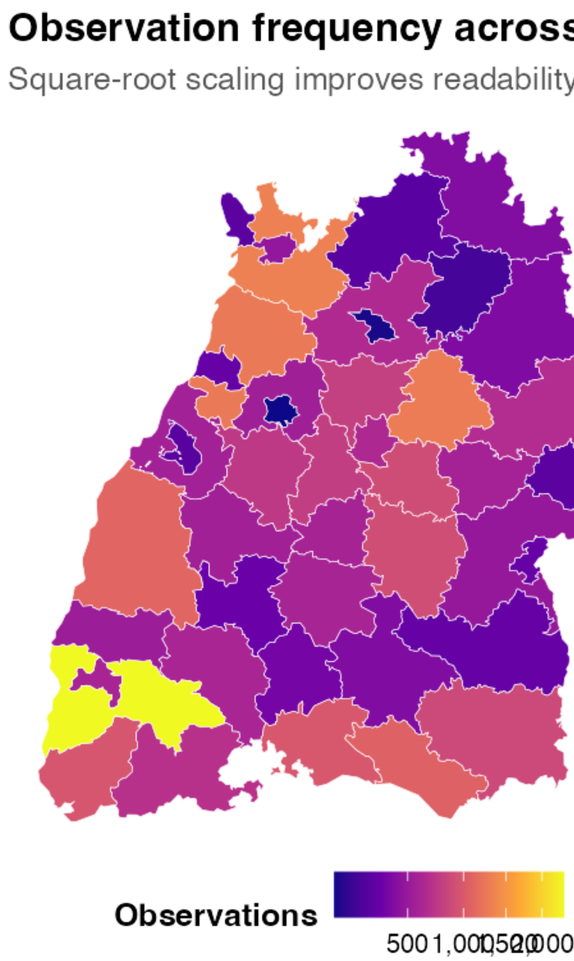
11.1 Baden-Württemberg

Baden-Württemberg is kept as a compact visual reference for Berlin. The two maps below show where observations are densest across the state and how observation frequency varies across NUTS3 districts.



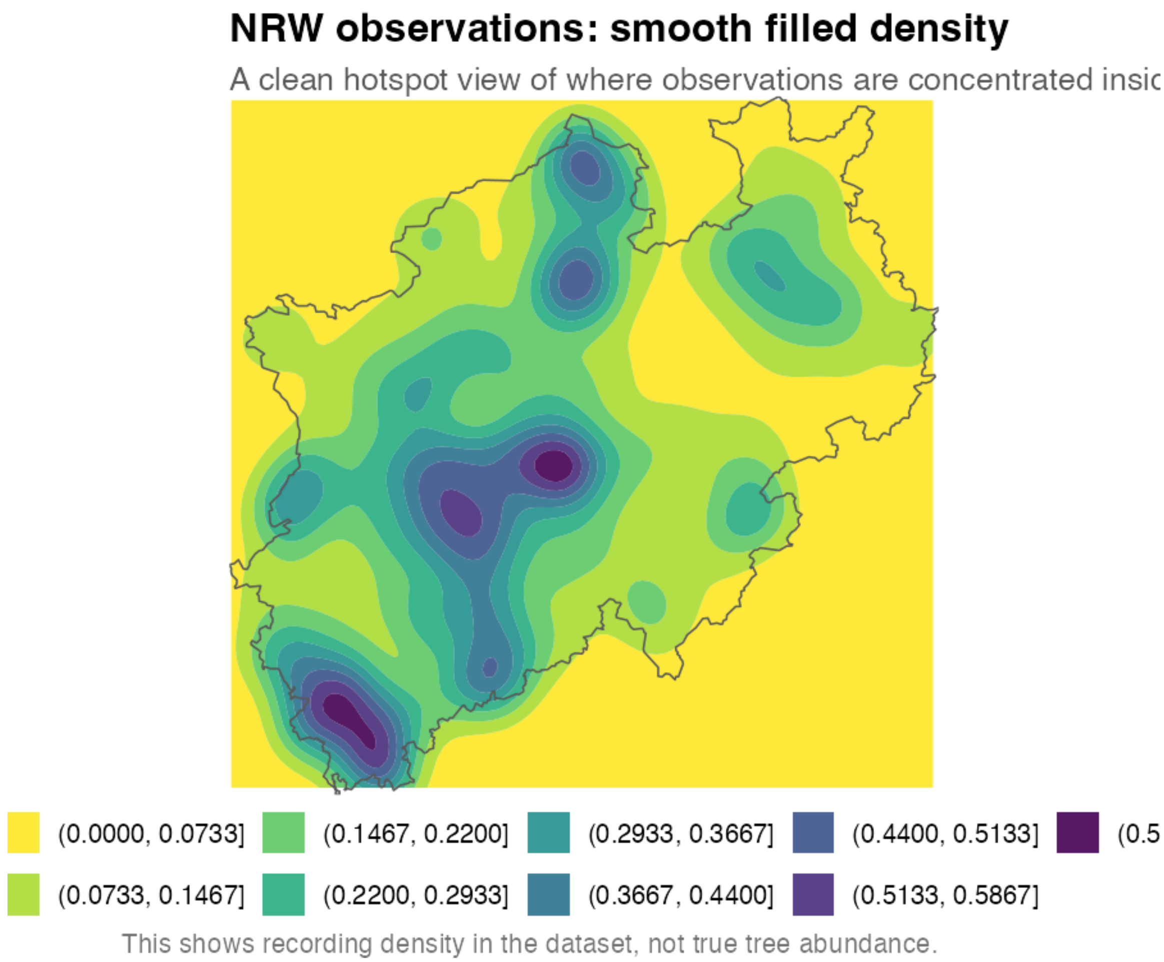
The filled density map is included only as a broad visual comparison with Berlin's observation-density pattern. It should be read as recording intensity, not true tree abundance.

11.1.1 Baden-Württemberg observations by NUTS3 district



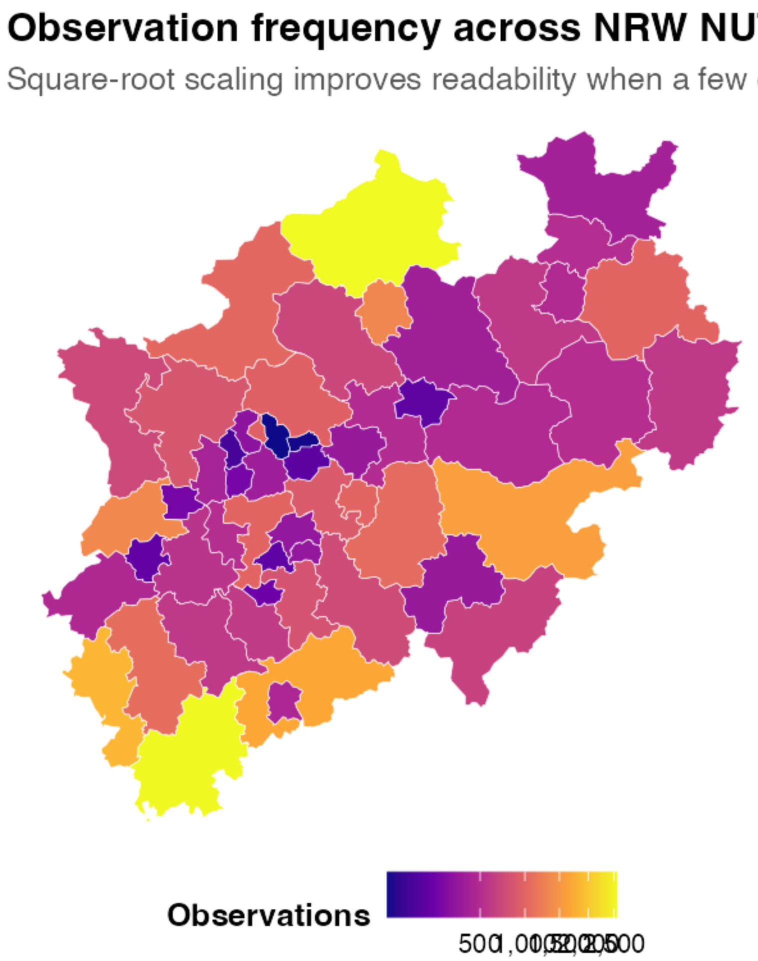
11.2 Nordrhein-Westfalen (NRW)

NRW serves the same role as Baden-Württemberg: a state-level comparison point for Berlin. The density map and NUTS3 district choropleth below follow the same layout.



As with Baden-Württemberg, this density map should be read as recording intensity rather than true tree abundance.

11.2.1 NRW observations by NUTS3 district



Comparing the density maps of Baden-Württemberg and NRW with that of Berlin highlights a notable difference in spatial distribution. While observations in BW and NRW are spread much more evenly across their respective territories — with multiple distinct hotspots corresponding to major cities and populated areas — Berlin's observations are heavily concentrated in a single location. This contrast reinforces the data quality concern raised earlier: the Berlin subset appears to reflect localised recording activity rather than a representative spatial sample of the city's tree cover.

Part III: Repartition of the data in the United States

> — *Santiago*

This part focuses on the U.S. records and asks whether their distribution reflects ecological patterns, the way the data was collected, or a combination of both. Knowing which places and years contribute most of the observations is necessary before drawing conclusions about tree diversity from the dataset.

The analysis first describes national coverage and its regional imbalance. It then compares foliage categories, latitude profiles, and taxonomic richness. The final figures add the time dimension to show whether the geographical pattern changed as the number of records increased.

11.1 Pre-Processing

Rows labelled with the `US` country code were retained. Their coordinates were matched to state polygons from the `maps` package; points missed by the simplified coastal boundaries were linked to the closest state centre. The state names were then converted into the four Census regions—Northeast, South, Midwest, and West—so that comparisons would not be fragmented across fifty individual states.

Table 15 displays the working variables after this preparation.

sample_id	level0	level1_family	level2_genus	level3_species	year	longitude	latitude	state	region
2638136	Deciduous Broadleaf	Amaranthaceae	Chenopodium	Chenopodium album	2021	-73.93528	44.60028	new york	Northeast
2638137	Deciduous Broadleaf	Amaranthaceae	Chenopodium	Chenopodium album	2020	-69.90992	44.22425	maine	Northeast
2638204	Deciduous Broadleaf	Amaranthaceae	Chenopodium	Chenopodium album	2024	-73.08378	44.52931	vermont	Northeast
2638205	Deciduous Broadleaf	Amaranthaceae	Chenopodium	Chenopodium album	2024	-73.01671	44.49624	vermont	Northeast
2638206	Deciduous Broadleaf	Amaranthaceae	Chenopodium	Chenopodium album	2024	-72.91761	44.99943	vermont	Northeast
2638207	Deciduous Broadleaf	Amaranthaceae	Chenopodium	Chenopodium album	2024	-68.50912	44.68911	maine	Northeast

Table 15: First observations after preparing the U.S. subset

The regional colors in Figure 16 remain unchanged from one graph to another. They come from the colorblind-friendly Okabe-Ito palette.

Figure 16: Regional colors used throughout the U.S. analysis

11.2 United States overview

The initial overview separates national reach from sampling intensity. A dataset may include all major parts of the country and still be dominated by a few places.

For that reason, Figure 17 shows the individual coordinates, Figure 18 compares the regional totals, and Figure 19 identifies the cells where records accumulate most strongly.

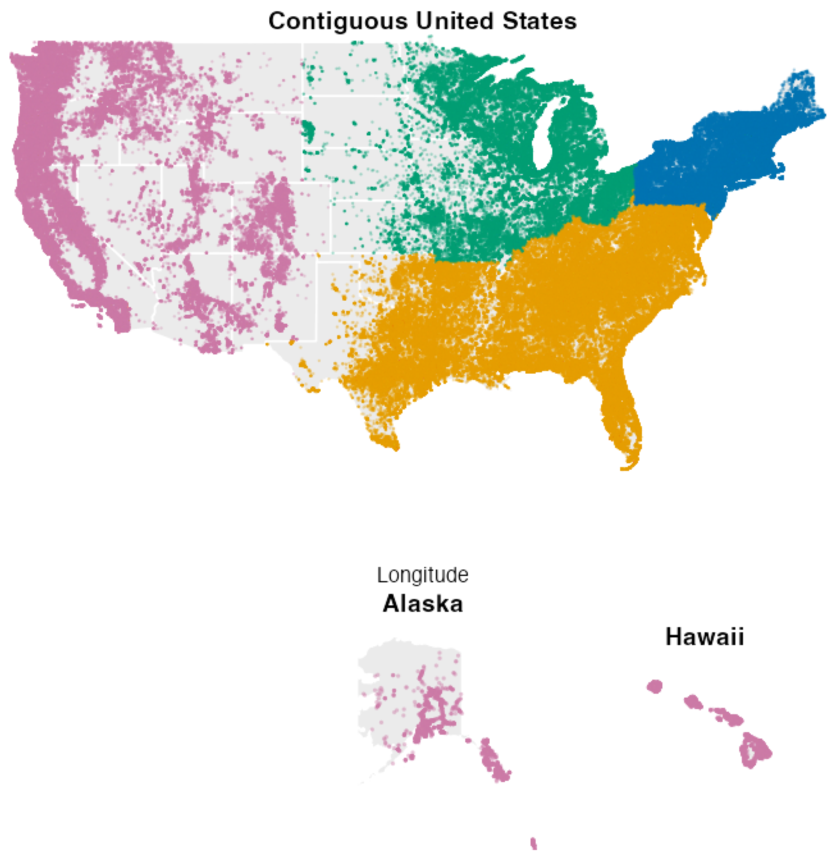


Figure 17: Location of U.S. observations, colored by Census region

The points in Figure 17 cover most of the country, but not with the same intensity. Texas, California, Florida, and the Northeast form visible clusters, whereas the Great Plains and much of the interior West are sparsely represented. Accessibility and the location of contributors probably explain part of this contrast, so the map cannot be read as a map of tree abundance.

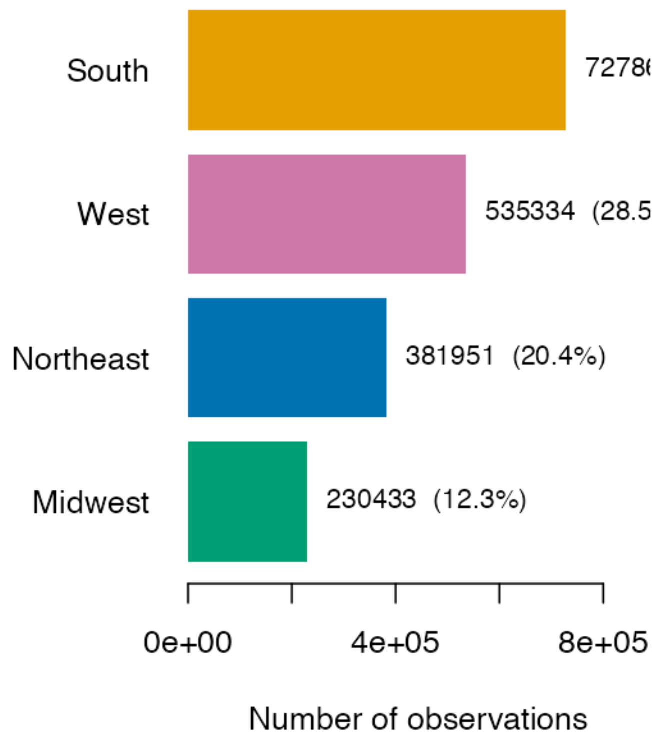


Figure 18: Contribution of each U.S. region to the complete U.S. subset

The regional counts in Figure 18 make the imbalance more explicit. The South supplies the largest share of records, followed by the West. The Northeast and Midwest have considerably fewer rows. These percentages refer to dataset participation and should not be interpreted as regional forest shares.

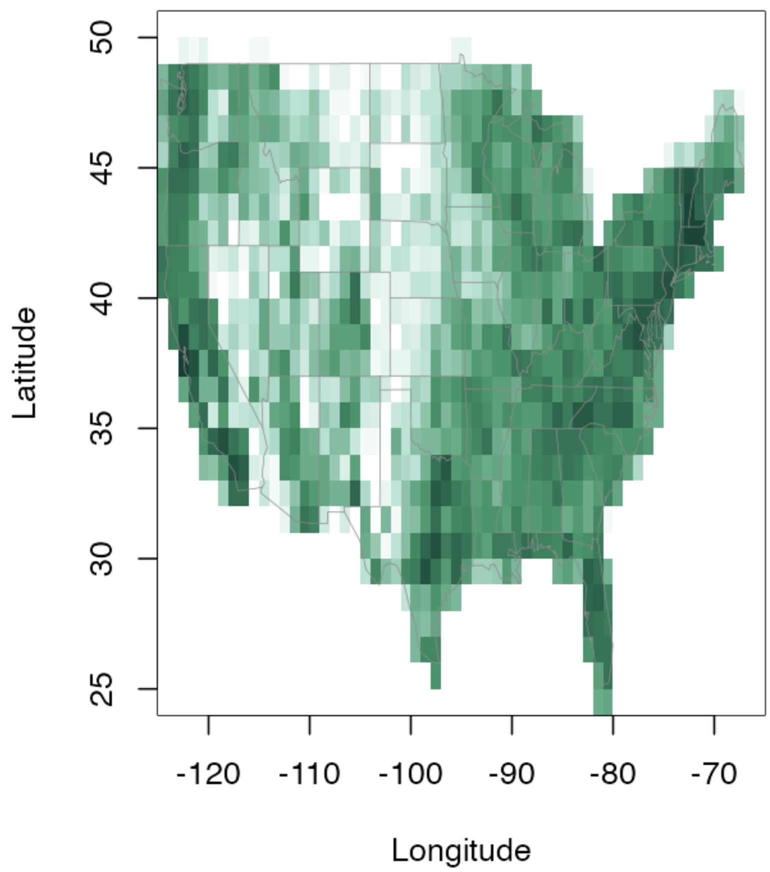


Figure 19: Local concentration of records across the contiguous United States

The grid representation in Figure 19 isolates the main hotspots around central Texas, southern California, Florida, and the northeastern corridor. Pale inland cells indicate limited sampling rather than an absence of trees.

Overall, the U.S. subset has a wide footprint but an uneven internal balance. The following comparisons are therefore interpreted as patterns within the available records, not as a national census.

11.3 Spatial distribution of foliage types

The next step is to compare the kinds of trees recorded in different places. Foliage categories summarize important ecological strategies: broad or needle-shaped leaves and evergreen or deciduous behavior respond differently to cold, drought, and seasonality.

The small maps in Figure 20 preserve the spatial detail. Figure 21 converts those records into regional percentages, while Figure 22 removes the administrative boundaries and follows the north–south gradient directly.

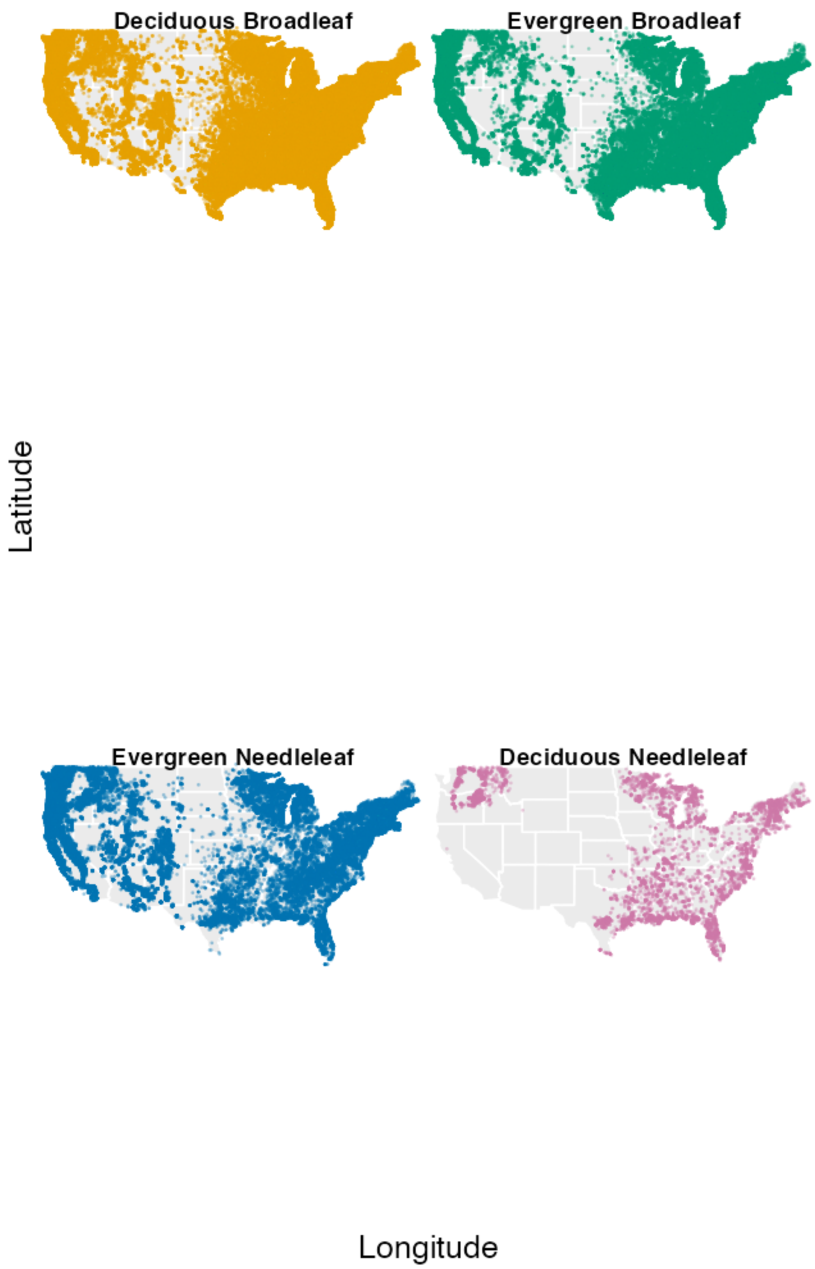


Figure 20: U.S. locations recorded for each foliage category

The categories in Figure 20 do not occupy identical spaces. Deciduous Broadleaf is found throughout the contiguous states and is particularly visible in the East and South. Evergreen Broadleaf is concentrated farther south and on the Pacific coast, while Evergreen Needleleaf forms strong western and northern clusters. Deciduous Needleleaf is present in pink, but its 10,951 records make it much rarer than the other three categories.

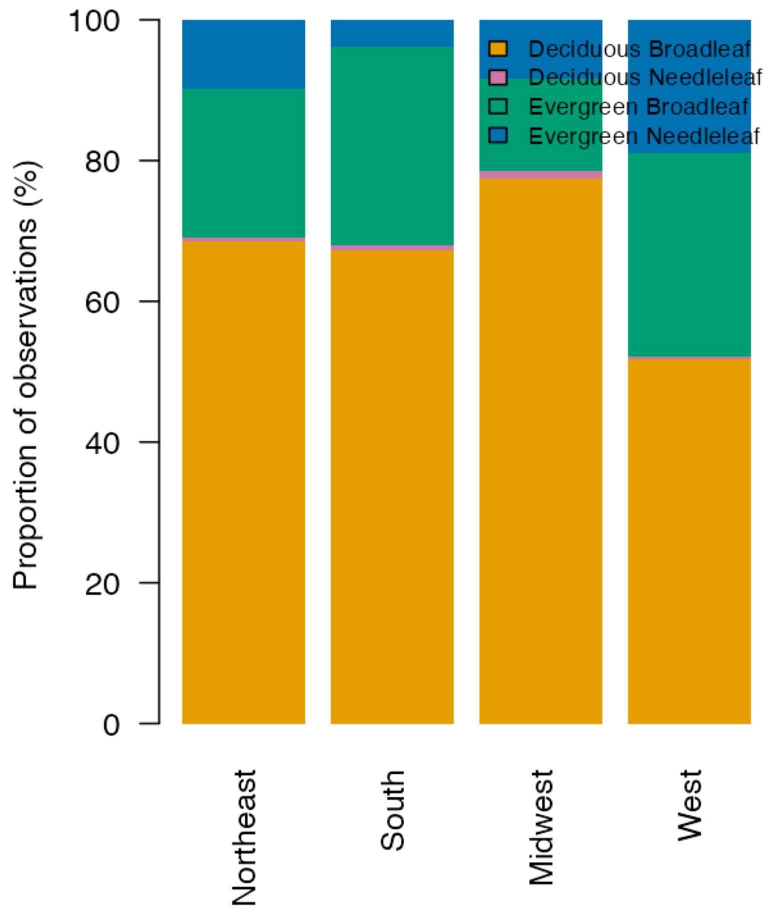


Figure 21: Foliage composition within each U.S. region

Using percentages in Figure 21 prevents the South's larger sample from dominating the comparison. Deciduous Broadleaf remains the largest group everywhere, although its share varies greatly. Evergreen Broadleaf reaches its largest regional shares in the South and West, while Evergreen Needleleaf is especially prominent in the West. Deciduous Needleleaf occurs in every region but never exceeds a little over 1%. Regional climate is one plausible contributor to these differences.

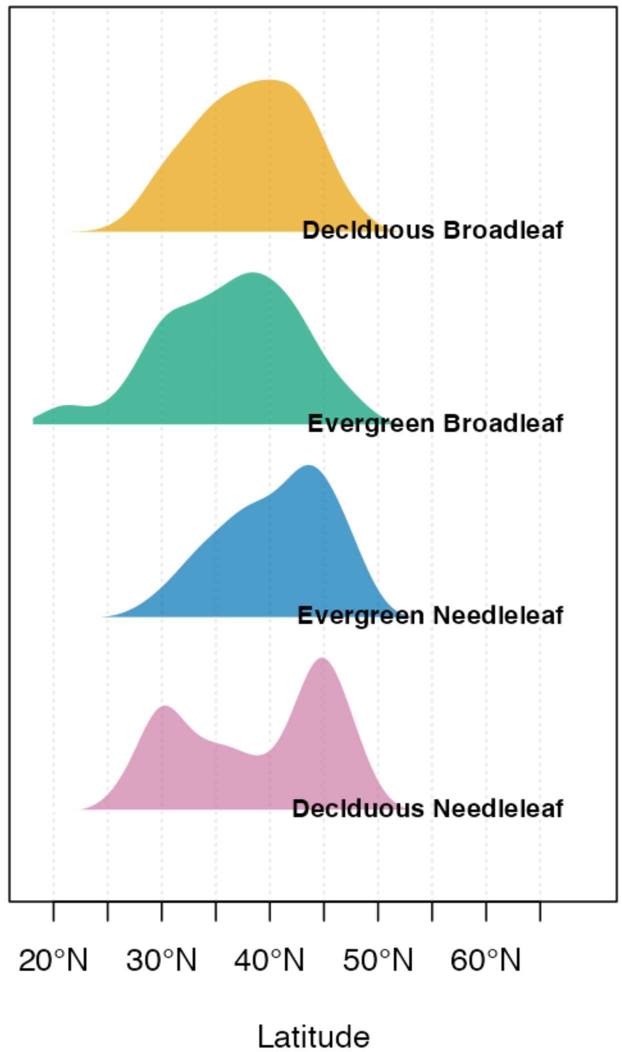


Figure 22: Latitude profiles of the foliage categories

The curves in Figure 22 overlap, but their peaks occur in different places. Evergreen Broadleaf has a clear southern component, including Hawaii, whereas Evergreen Needleleaf is weighted toward the northern half of the country. Deciduous Broadleaf covers the broadest span of the contiguous states. Deciduous Needleleaf is much less frequent but is not absent: its pink curve has concentrations around 30°N and 45°N.

These shifts support the idea that climate helps structure foliage type. Moving north usually changes temperature, season length, and frost exposure. Still, latitude does not measure climate directly, and other factors—precipitation, elevation, soil, land use, historical range, and sampling density—change at the same time. The figure therefore provides evidence of association, not a causal estimate.

11.4 Taxonomic richness by region

Taxonomic richness is measured here at three levels: family, genus, and species. Simple totals are useful, but they favor regions with more observations because additional records create more opportunities to encounter a new taxon.

Table 16 and Figure 23 present the unadjusted counts. Figure 24 then places the regions on a common sampling scale. The last visualization, Figure 25, changes the question from "how many species?" to "how geographically concentrated is each common species?"

Region	Foliage	Families	Genera	Species	Observations
South	4	138	493	1640	727869
West	4	132	503	1835	535334
Northeast	4	85	202	653	381951
Midwest	4	84	199	660	230433

Table 16: Recorded taxonomic diversity and sample size by region

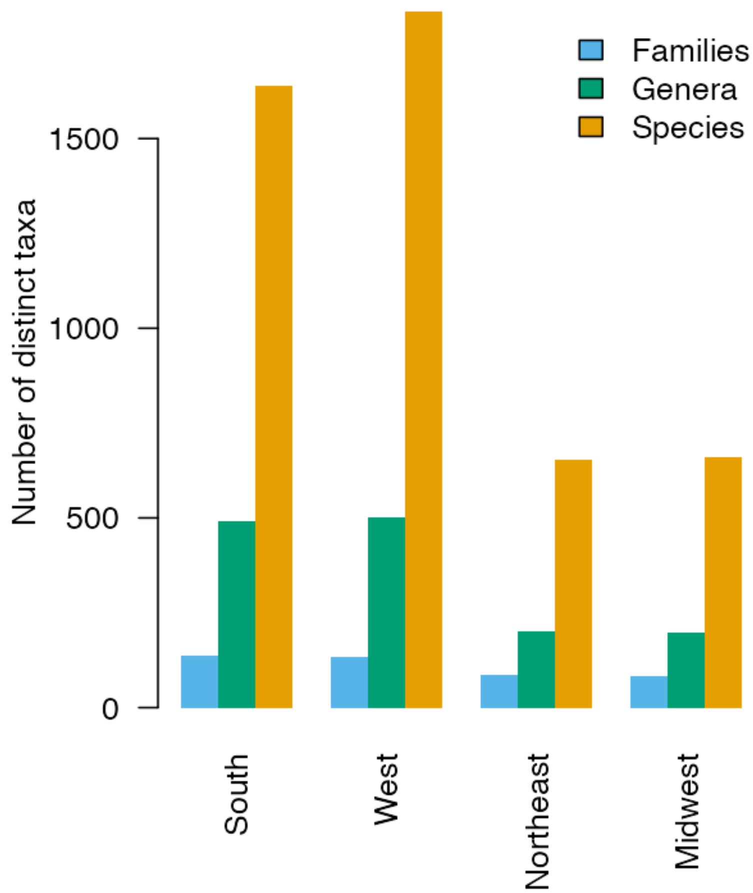


Figure 23: Unadjusted family, genus, and species counts by region

The South leads the raw counts in Table 16 and Figure 23, followed by the West. Their advantage is partly expected because these are also the two largest regional subsets. Despite having fewer observations, the Northeast and Midwest still include a notable share of the 3,001 species recorded for the United States.

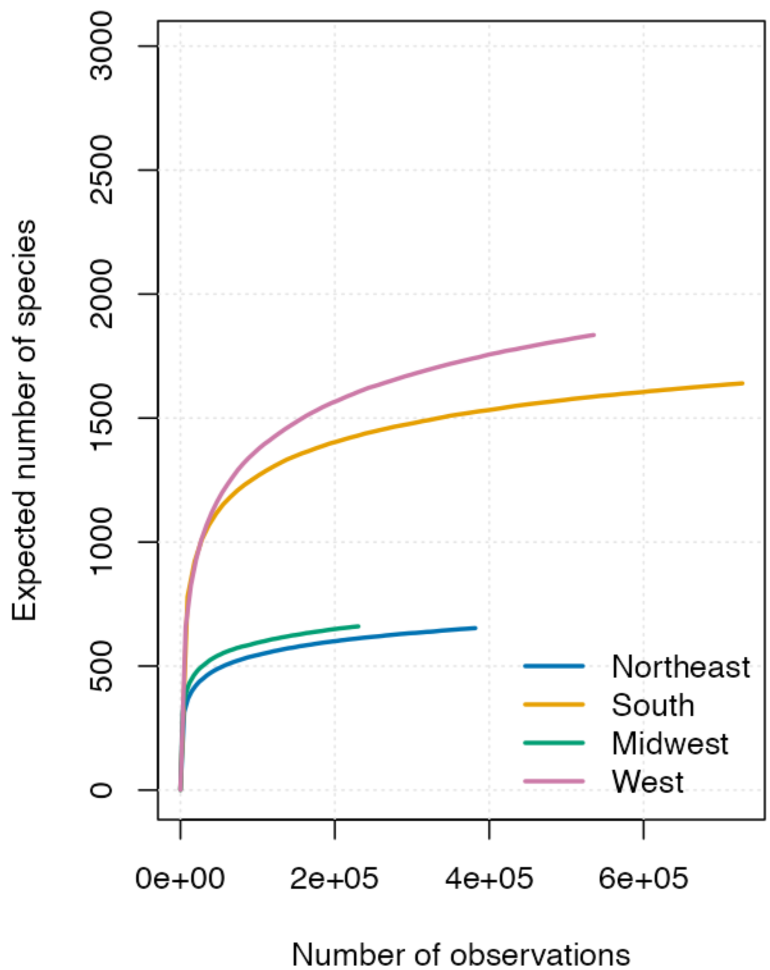


Figure 24: Growth of recorded species richness as regional sample size increases

In Figure 24, regions can be compared at the same horizontal position and therefore at the same number of sampled records. The curves reveal how quickly additional observations continue to add species in each region. Comparing them at a shared sample size separates part of the diversity pattern from the very different regional record totals.

Figure 25: Observed ranges of the 12 most frequent species in the contiguous states

There is no single answer to whether members of one species grow together. In Figure 25, several species are restricted to compact clusters on the Pacific coast, in Texas, Florida, or the Northeast, while others extend across several states or regions. Each species has its own observed range, but that range may be broad or discontinuous. Environmental requirements and dispersal history matter, and the gaps may also reflect missing observations.

11.5 Changes in the distribution over time

All previous figures combine ten collection years into one picture. This can hide changes in participation, especially for data coming mainly from citizen-science platforms.

The bars and line in Figure 26 compare annual volume with the number of represented states. Figure 27 identifies which parts of the country produce the increase, and Figure 28 checks whether later records spread into new territory or reinforce existing hotspots.

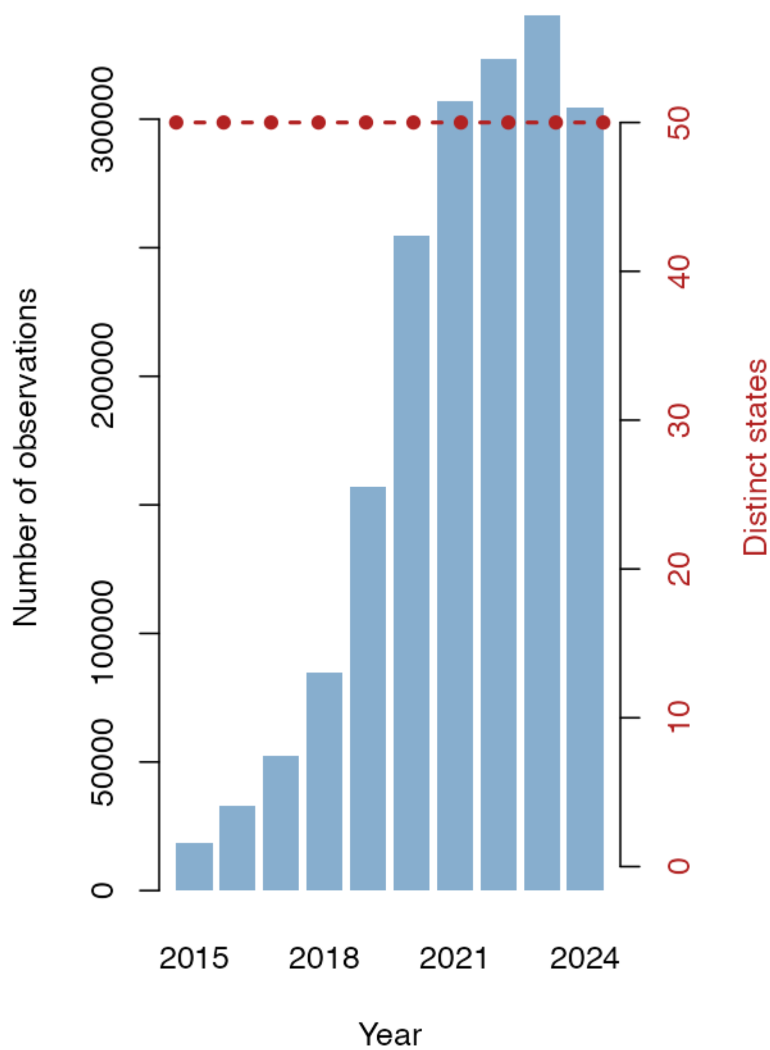


Figure 26: Annual U.S. record volume and number of represented states

The annual series in Figure 26 rises markedly after 2018 and reaches its highest level near the end of the period. State coverage expands earlier and then changes much less. In other words, recent growth mostly adds records within an already broad national footprint. It indicates greater data-collection activity, not rapid growth in the tree population.

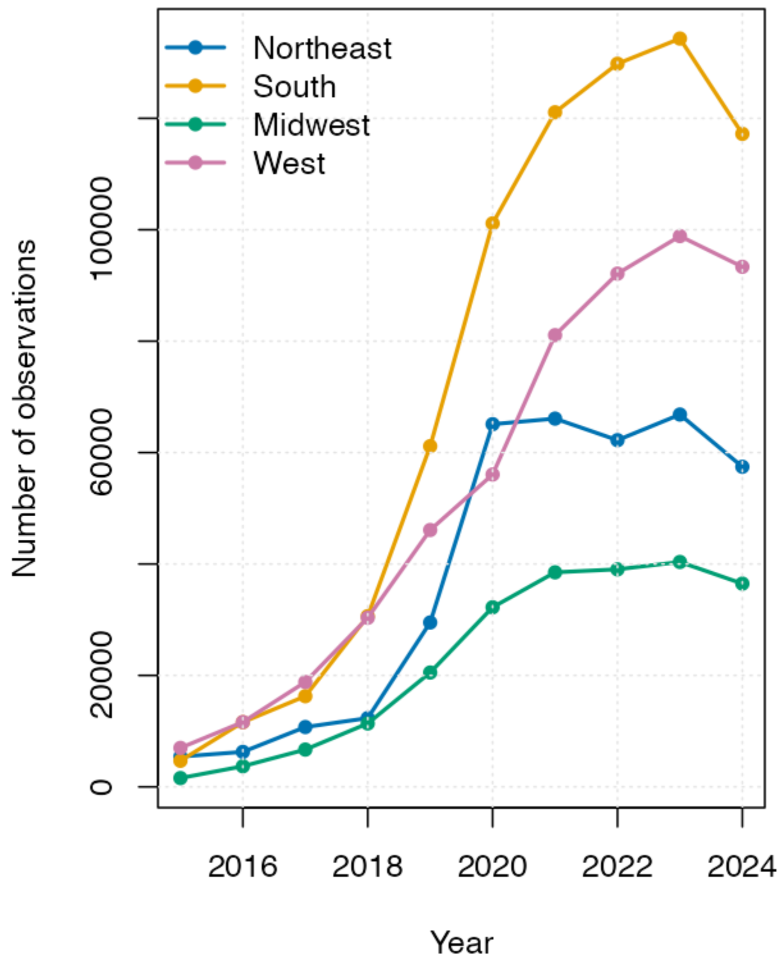


Figure 27: Yearly observation counts separated by U.S. region

The regional lines in Figure 27 do not grow in parallel. Most of the national increase comes from the South and West, whereas the Northeast and Midwest stay at lower levels. The regional imbalance is therefore repeated across several collection years rather than being produced by one exceptional year.

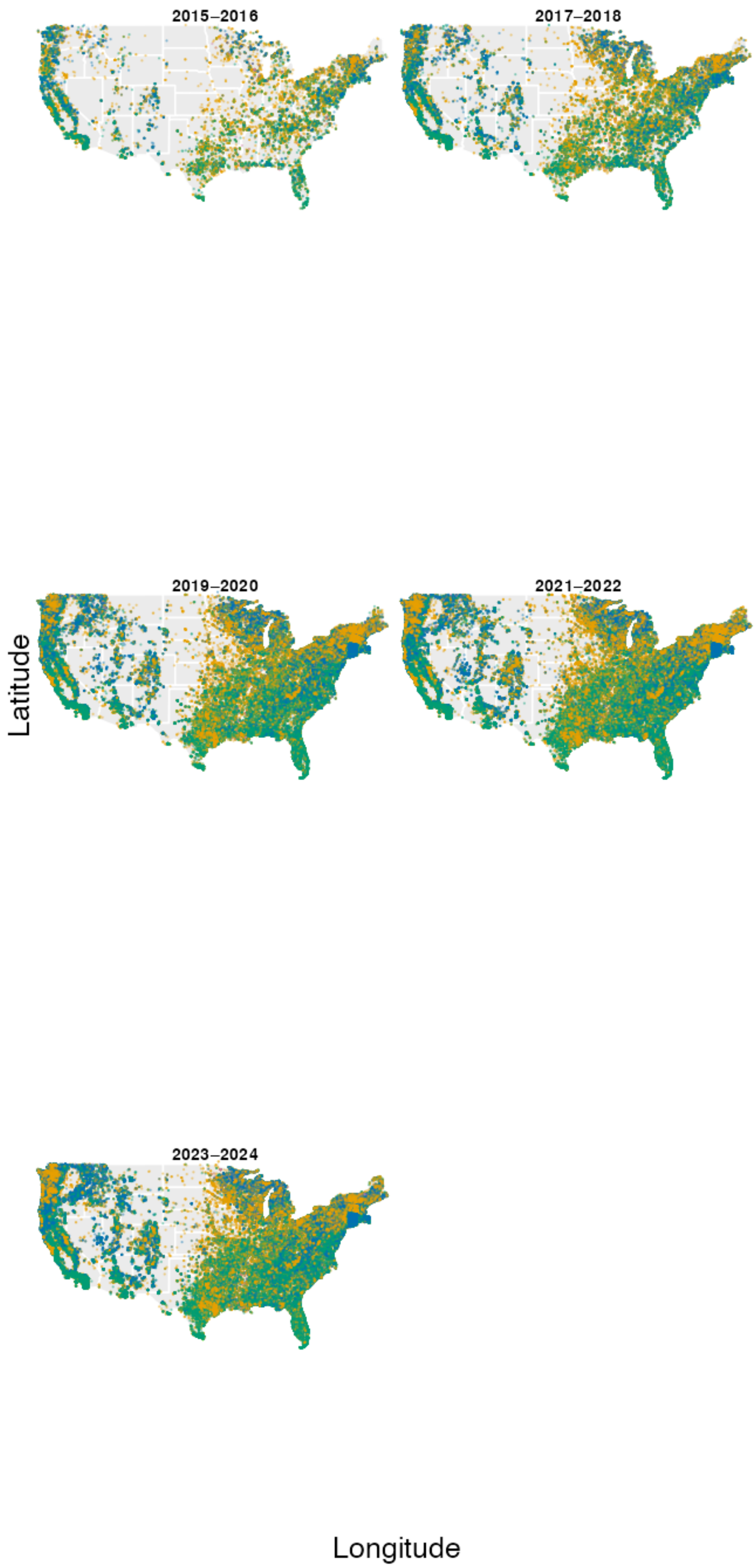


Figure 28: Spatial coverage of U.S. observations across five collection periods

The period maps in Figure 28 repeatedly show the same geographical cores, although the number of visible points increases. Coverage becomes denser more often than it becomes geographically broader. The dataset thus preserves a similar internal bias throughout the decade.

12 Overall Interpretation

This report has examined the GlobalGeoTree dataset from multiple angles — source bias, temporal trends, taxonomic composition, worldwide distribution, regional patterns in Germany, and a country-level analysis of the United States.

Source bias and temporal trends. The global analysis showed that citizen-science platforms — above all iNaturalist — dominate the dataset and strongly shape patterns by observation year and geography, making the combined dataset a composite of several geographically non-uniform "lenses" rather than a single coherent survey.

Taxonomic composition and worldwide distribution. The taxonomic analysis showed that Deciduous Broadleaf is the largest foliage category in the observations and that recorded species richness peaks in several temperate latitude bands. Sampling is heavily concentrated in North America and Europe, while regions such as central Africa, the Amazon interior, and Central Asia remain sparsely represented. This geographic bias is visible throughout 2015–2024 observation years, with later-dated records tending to intensify already represented areas rather than adding major new regions.

Germany and Berlin analysis. Within Germany, the same citizen-science dominance reproduces at the national level, with several platforms shaping partially different spatial footprints. At the Bundesland level, observation counts vary widely, reflecting differences in population density, platform coverage, and observer activity. Berlin is examined in detail, while Baden-Württemberg and NRW provide brief map-based context. The Berlin oak analysis demonstrates how recorded oak locations could be compared with official inventories and pest-surveillance data in a follow-up application.

United States. The US analysis — the most-represented country in the dataset — showed strong regional imbalance in recording intensity, with the South and West contributing the largest shares of observations. Foliage composition varies geographically: Deciduous Broadleaf dominates everywhere, while Evergreen Broadleaf is more southern/coastal and Evergreen Needleleaf is more western and northern. Taxonomic richness remains partly shaped by sample size, so regional diversity comparisons should be read alongside the rarefaction results.

Limitations and outlook. All visualisations in this report show **recording intensity** in the dataset, not a direct census of trees. High values can reflect true tree presence, but they equally reflect where observers and platforms are most active. For any analysis built on GlobalGeoTree, the source composition of the sample should be reported alongside geographic, temporal, or taxonomic results, and key findings should be checked with and without the largest citizen-science source. Future work could refine the source classification rules, particularly for non-English source names, and explore whether the patterns observed in this analysis hold when the dominant citizen-science sources are excluded as a sensitivity check.

Visualisation Techniques

The report intentionally uses multiple visual encodings, including complementary views of the same question. The table summarises the main techniques and what each contributes.

Technique	Main visual purpose
Bar and stacked-bar charts	Compare rankings, totals, and within-group composition
Line and area charts	Show changes across observation years and their source composition
Histograms and density curves	Contrast absolute frequency with distribution shape
Boxplots, heatmaps, and ridge plots	Compare distributions across taxa and latitude
Point and faceted maps	Show coverage, individual locations, and group-specific footprints
KDE maps and spatial bins	Emphasise relative hotspots and local concentration
Choropleth maps	Compare aggregated counts across administrative areas
Richness grids and rarefaction curves	Compare recorded taxonomic richness spatially and at equal observation counts
Spatial joins	Connect observation coordinates with continents and administrative boundaries

References

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